# Paediatric Urology Rapid Revision

Campbell-Walsh-Wein · Chapters 41, 49–53, 56, 57, 61, 63, 64

For MCh Urology theory & viva · NEET-SS / competitive MCQs

**How to use (total ≈ 6h 45min even at a slow, careful pace)**

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

**Conventions**: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 53* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.


| Chapter | Topic | Slow-reader time |
|---|---|---|
| 41 | Management Strategies for Vesicoureteral Reflux | 33 min |
| 49 | Anomalies of the Kidney and Collecting System | 40 min |
| 50 | Renal Dysgenesis and Cystic Disease of the Kidney | 41 min |
| 51 | Pathophysiology of Urinary Tract Obstruction | 18 min |
| 52 | Ectopic Ureter, Ureterocele and Ureteral Anomalies | 59 min |
| 53 | Surgery of the Ureter in Children | 40 min |
| 56 | Hypospadias | 34 min |
| 57 | Etiology, Diagnosis and Management of the Undescended Testis | 53 min |
| 61 | Urologic Considerations in Pediatric Renal Transplantation | 18 min |
| 63 | Paediatric Urologic Oncology: Renal and Adrenal | 35 min |
| 64 | Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia | 26 min |
| ★ | Putting it together: common paediatric decisions | 8 min |
| | **Total (28373 words, at 70 words/min)** | **6h 45min** |

# 41 Management Strategies for Vesicoureteral Reflux

*Reading time ≈ 33 min (slow pace)*

## 1 · DEFINITION AND CONCEPT

- **Vesicoureteral reflux (VUR)**: retrograde flow of urine from bladder to upper tract. Associated with **acquired scarring** and with **congenital renal dysplasia**.
- Now seen as a **spectrum of severity** needing risk-stratified management, not a yes/no disease.
- **Infection, not reflux, is the injury.** VUR does not cause urinary tract infection (UTI); it is an accelerant that carries infected bladder urine to the kidney. Sterile urine and good emptying are the targets of treatment.

## 2 · EMBRYOLOGY AND ANTIREFLUX MECHANISM

- Ureteric bud arises from the mesonephric (wolffian) duct at weeks 4–6; the common duct is absorbed into the urogenital sinus (UGS). **Mackie–Stephens**: bud position sets both orifice and kidney.
  - **Early bud** → over-rotation → orifice **high and lateral**, short tunnel → VUR.
  - **Late bud** → orifice **distal and medial** → ectopic, often obstructed ureter.
  - Abnormal bud–metanephros contact → dysplasia/hypoplasia. Hence high-grade VUR and dysplasia travel together and some "scars" are congenital.
- VUR = imbalance of three factors: ureteric peristalsis, ureterovesical junction (UVJ) anatomy, bladder dynamics.
- **Passive flap-valve**: intramural ureter compressed against detrusor backing during filling. **Paquin**: tunnel length : ureteric diameter **5:1** in non-refluxing vs 1.4:1 in refluxing UVJs; orifice configuration matters as much.
- Opening is active (longitudinal tunnel muscle shortens/widens the tunnel); closure is passive.

## 3 · PRIMARY VS SECONDARY VUR

| | Primary | Secondary |
|---|---|---|
| Fault | Congenital UVJ deficiency, normal low-pressure bladder | Normal-ish UVJ overwhelmed by pressure, or abnormal UVJ |
| Causes | Short tunnel, weak detrusor backing, dilated/stiff ureter | Outlet obstruction (posterior urethral valves (PUV), stricture); neurogenic dyssynergia; dysfunctional voiding; ureterocele, duplication, paraureteral diverticulum, iatrogenic |
| Treat | Wait for resolution / correct | Treat the cause first |

- Spina bifida: VUR in up to **30%** → early clean intermittent catheterisation (CIC), anticholinergic or β3-agonist, serial imaging. Look for occult dysraphism (sacral dimple, hairy patch, gluteal cleft anomaly, refractory constipation).
- **Infant male VUR**: high voiding pressures from poor sphincter relaxation; resolves as voiding matures — explains infant male predominance and high male resolution.
- Normal bladders void at ~100 cm $H_2O$ without reflux, so structural UVJ failure is key. **Reflux at low pressure / early filling = poor prognosis for resolution.**
- Detrusor overactivity is the commonest urodynamic finding; treating it (timed voiding, biofeedback, anticholinergics, α-blockers) improves resolution.

> **KEY POINT** Primary and secondary are not exclusive: "primary" reflux often has a treatable secondary component (overactivity, bladder and bowel dysfunction). Removing it leaves only the true primary VUR to manage.

## 4 · EPIDEMIOLOGY (HEADLINE)

- VUR in **~30%** of children with UTI; ~17% of those imaged for hydronephrosis. Younger child with UTI → likelier VUR (**38.6%** under 2 y vs 7.6% at 12–21 y).
- Girls ~2× overall, but **infant males predominate** (up to 3:1 at 0–6 months; uncircumcised boys have 10–12× more UTI). Less common in Black children.

## 5 · GRADING — INTERNATIONAL REFLUX STUDY (1981)

| Grade | Ureter | Pelvis and calyces |
|---|---|---|
| I | Into non-dilated ureter only | Not reached |
| II | Not dilated | Reached, no dilatation |
| III | Mild–moderate dilatation | Mild–moderate dilatation, minimal forniceal blunting |
| IV | Moderate tortuosity | Moderate dilatation |
| V | Gross dilatation and tortuosity | Gross dilatation; **papillary impressions lost** |

- I–II non-dilating; III–V dilating. Interobserver agreement poor (all three readers in RIVUR (Randomized Intervention for Children with Vesicoureteral Reflux) agreed in only ~60%, worst II–IV).
- Radionuclide cystography (RNC) has 3 grades: 1 = I; 2 = II–III; 3 = IV–V.
- **Ureteral diameter ratio (UDR)** = largest ureteric diameter in false pelvis ÷ L1–L3 vertebral distance. Predicts resolution and breakthrough UTI (BUTI) better than grade.

## 6 · DIAGNOSIS

### When to image after UTI

- Accurate urine collection and UTI confirmation first — major decisions (including surgery) rest on it.
- American Academy of Pediatrics (AAP) 2011, febrile UTI 2–24 months: **ultrasound (US) first**; voiding cystourethrogram (VCUG) only if ultrasound abnormal or **second febrile UTI**.
- National Institute for Health and Care Excellence (NICE, UK): US; VCUG mainly for atypical/recurrent UTI in younger children; none routinely >3 y. Misses much VUR including grade IV–V.
- **Top-down approach**: dimercaptosuccinic acid (DMSA) scan first; VCUG only if abnormal. Normal DMSA → **≤1%** chance of high-grade VUR; misses ~1/3 of (mostly low-grade) VUR.

### Cystographic methods

| Method | Strength | Weakness |
|---|---|---|
| **VCUG** (gold standard; gravity ≤70 cm $H_2O$, filling + voiding fluoroscopy) | Anatomy: grade, intrarenal reflux, diverticula, ureterocele, urethra, postvoid drainage | Catheter, radiation |
| **Direct RNC** (technetium-99m pertechnetate) | More sensitive for II–V; low dose; ideal for screening and follow-up | Misses grade I; little anatomy |
| Indirect RNC | No catheter | Needs toilet-trained child; sensitivity 70–90% |
| Contrast-enhanced voiding urosonography | No radiation; comparable to VCUG; detects intrarenal reflux | Grading not standardised |

- **Cyclic VCUG** (2–3 fill–void cycles) detects 12–20% more VUR.
- **Filling-phase reflux** = poor sign for resolution; voiding-only reflux is more favourable.
- Timing: once urine sterile and child improved, early VCUG (~8 days) gives same VUR prevalence as delayed.

- **Postvoid films**: retained pelvic contrast → suspect ureteropelvic junction obstruction (UPJO).
- Suspect coexisting **UVJ obstruction**: dilated proximal ureter, distal aperistaltic segment, dilute contrast, delayed drainage → **relative contraindication to endoscopic injection** (may worsen obstruction).
- Routine cystoscopy has no role (orifice shape correlates poorly with VUR); use just before open surgery.
- Non-invasive uroflowmetry + postvoid residual (PVR) ± electromyography (EMG) is useful; invasive urodynamics only for refractory cases.

**Upper tract**

- **Renal and bladder ultrasound (RBUS)**: initial and serial mainstay (size, growth, echogenicity). Cannot diagnose or exclude VUR.
- **DMSA**: gold standard for cortical defects (sensitivity 98%, specificity 92%). Acute photopenia may resolve; persistence = scar.

## 7 · CONGENITAL DYSPLASIA VS ACQUIRED SCAR

| | Reflux-associated dysplasia | Acquired scar |
|---|---|---|
| Cause | Abnormal bud–blastema interaction | Pyelonephritis: infected urine + VUR |
| Typical patient | Male infant, grade IV–V, antenatal hydronephrosis, no UTI | Girl, recurrent febrile UTI, bladder and bowel dysfunction |
| Imaging | Small kidney, globally reduced function | Focal photopenic defects after infection |
| Preventable? | No | Yes — keep urine sterile |

- Prenatally detected boys: damage confined to grades IV–V (nearly half of grade V severely damaged) before any infection.
- **Ransley–Risdon** (pig): reflux caused nephropathy only with infected urine. Sterile VUR damages only with high pressure (PUV, Hodson).
- **Compound (concave) papillae** at poles → intrarenal reflux → **polar scars**. Intrarenal reflux occurs at only 2 mm Hg in neonates.
- Scarring risk highest in first year. "Big bang" theory (most scarring with first pyelonephritis) is now challenged.
- Long-term: hypertension in **34–38%** of scarred children. **Correcting VUR does not lower blood pressure** (parenchymal mechanism); nephrectomy/segmental resection of a small kidney may.

## 8 · BLADDER AND BOWEL DYSFUNCTION (BBD)

- Symptoms: frequency, holding, urgency, incontinence, constipation; curtsying (girls), penile squeezing (boys). **Treat constipation first** — it causes co-contraction of anal and urinary sphincters.
- American Urological Association (AUA) guideline: BBD is one of the most critical modifiable variables. BBD → more UTI on prophylaxis and after surgery, less resolution, **lower endoscopic success, but no reduction in open surgical success**.
- Off prophylaxis, recurrent UTI: ~1/2 with BBD + VUR, ~1/5 VUR alone, ~1/3 BBD alone.
- **Phimosis**: in boys with VUR number needed to treat (NNT) by circumcision to prevent one UTI falls from ~111 to **~4**; topical steroid is equivalent.

## 9 · NATURAL HISTORY AND RESOLUTION

| Grade | Spontaneous resolution |
|---|---|
| I–II | Most; grade II 63–85% |
| III | ~50%; ~90% of resolutions within 4 years |

| Grade | Spontaneous resolution |
|---|---|
| IV–V, bilateral III | Few (but infants with IV–V resolve well in year 1) |

- Mechanism: UVJ remodelling (tunnel elongation) + stabilising bladder dynamics. Persistent BBD explains failure.
- Predictors of resolution: **low grade, young age, unilateral, no BBD, voiding-only reflux, higher bladder volume at onset, low UDR**. Age at presentation matters as much as grade (AUA 1997 nomograms).

## 10 · MEDICAL MANAGEMENT — CONTINUOUS ANTIBIOTIC PROPHYLAXIS (CAP)

- Principle: **manage infection risk, not VUR**; prevent UTI while awaiting resolution.
- Daily low-dose oral CAP, **at night** — longest bladder concentration during longest retention.

| Agent | Age | Note |
|---|---|---|
| Trimethoprim / amoxicillin | <2 months | Avoids sulfonamide in neonate |
| Trimethoprim-sulfamethoxazole (TMP-SMX) | >2 months | RIVUR agent; sulfonamide displaces bilirubin → neonatal jaundice; rare reversible leukopenia |
| Nitrofurantoin | >2 months | Least faecal-flora resistance; pulmonary fibrosis, hepatic toxicity (rare at low dose) |

- **Smellie's BUTI rule**: organism sensitive to the prophylactic → non-adherence/low dose; resistant → residual volume too high or dose too high.
- **Breakthrough febrile UTI on CAP despite good bladder and bowel habits → stop medical management and correct VUR** (confirm proper urine collection first).
- Stopping CAP after toilet training/BBD resolution: ~8–10% get febrile UTI → reassess, possibly correct.

### Risk stratification (Hidas)

| Factor | Low risk | High risk |
|---|---|---|
| Sex / age | Male; >5 y | Female; <1 y |
| Laterality / grade | Unilateral; I–II | Bilateral; III–V |
| BBD / scars | No | Yes |
| Presentation | Prenatal hydronephrosis | Multiple febrile UTIs |

- 2-year BUTI on CAP: **8.6% low, 26% intermediate, 62.5% high**. About two-thirds of children are low risk.

## 11 · PRACTICAL ALGORITHM (AUA-BASED, RISK-STRATIFIED)

**ALGORITHM**

History of number/severity of true febrile UTIs + RBUS → VCUG if multiple febrile UTIs, admissions or upper-tract findings (afebrile, doubtful UTI with normal US → conservative)

All: treat BBD (constipation first), hygiene, timed voiding

Low risk (grade I–II, no BBD, male, older) → observation ± CAP; treat febrile UTI promptly

High risk (grade ≥III, girl, BBD, febrile presentation, scars, infant) → CAP

Breakthrough febrile UTI, new scars, high grade with poor function, persistent VUR with low tolerance for infection → correction (endoscopic or reimplant)

- Evidence summary: little CAP benefit at grade ≤II without BBD; benefit at grade ≥III, at least in girls.
- Extensive scarring, globally reduced function or dysplasia lower tolerance for another infection → favour correction.

- Unresolved: asymptomatic bilateral high-grade VUR in an older boy; correction of persistent VUR in girls before pregnancy (unproven).

## 12 · SURGICAL CORRECTION

- Open correction succeeds in **>95%**; robotic correction is probably less morbid. No superiority of surgery over medical therapy for scarring or renal growth (International Reflux Study in Children (IRSC), Birmingham).
- After correction, almost no new scars despite asymptomatic bacteriuria — but adult UTI risk persists.
- **Endoscopic injection** of dextranomer/hyaluronic acid copolymer (Dx/HA): BBD lowers success; avoid with UVJ obstruction or obstructive refluxing megaureter (may obstruct). Mounds calcify in over half and **mimic ureteric stones** later.
- Surgical techniques — endoscopic injection, intravesical/extravesical reimplantation, complications → *Ch 53*.
- After cross-trigonal reimplant, **retrograde catheterisation and ureteroscopy are difficult** — know the surgical history.

## 13 · ASSOCIATED CONDITIONS

- **UPJO + VUR**: VCUG signs — dilated ureter with little pelvic filling, dilute pelvic contrast, ureter–pelvis discrepancy, pelvis not draining postvoid. **Correct obstruction first**; simultaneous open repair of both risks ureteric blood supply.
- **Duplex**: VUR usually into **lower pole** (Weigert–Meyer: lateral, proximal orifice, short tunnel). Duplication alone does not increase risk → conservative management reasonable.
- **Paraureteral diverticulum**: resolves like primary VUR. **Ureter entering diverticulum**: no muscle backing → will not resolve.
- Multicystic dysplastic kidney (MCDK): contralateral VUR ~1/4, usually resolves.
- **Megacystis-megaureter**: massive bilateral VUR recycles urine → large thin bladder; normal posterior urethra. Treat by reimplant (not outlet surgery).
- **Obstructive refluxing megaureter**: ureteric contrast more dilute than bladder; avoid injection; reimplant ± tapering.

## 14 · SIBLING SCREENING

- Sibling prevalence ~**30%** (up to 50%), higher in younger siblings; identical twins near-complete concordance.
- Target is **renal damage, not VUR**. AUA: **VCUG for siblings with cortical abnormality or renal asymmetry on US, or UTI history**; otherwise observe. Older sibling with normal kidneys and no BBD gains little — treat any febrile UTI promptly.

## 15 · FOLLOW-UP AND ADULTHOOD

- Lifelong follow-up if scarred: blood pressure, proteinuria, renal function; transition programmes.
- **Pregnancy**: VUR → pyelonephritis in ~1/3; renal scars → ~3× hypertension, **~8× pre-eclampsia**. **Scarring, not VUR presence, drives pregnancy morbidity.** Reimplanted women still at UTI risk but no excess miscarriage.

## 16 · LANDMARK TRIALS

| Trial | Design | Result | Take-home |
|---|---|---|---|
| IRSC | <9 y, high grade; CAP vs open surgery | UTI equal (38%); surgery fewer pyelonephritis; new scars equal; 10 y renal function equal | Surgery ≈ medical for scarring |

| Trial | Design | Result | Take-home |
|---|---|---|---|
| Birmingham Reflux Study | High grade; medical vs surgical, 5 y | New scars equal, all within 2 y | Equal; supports "big bang" |
| Swedish Reflux Trial | 1–2 y, grade III–IV; CAP vs endoscopic vs surveillance | Girls febrile UTI: 19% CAP, 23% endoscopic, **57%** surveillance; new scars mostly on surveillance, none on CAP; boys no difference | Treat girls with dilating VUR |
| RIVUR | 607 children, grade I–IV after first/second UTI; TMP-SMX vs placebo | Recurrent UTI **12.9% vs 23.6%**; scarring unchanged; benefit greatest with febrile index UTI/BBD | CAP prevents UTI, not scars |
| PRIVENT | 576 children with UTI; CAP vs placebo | Absolute UTI reduction 6% (19% → 13%) | Modest benefit |
| French / low-risk trials | Grade I–III; CAP vs none | No significant difference | No benefit at low risk |

**KEY POINT** Why do the trials disagree? Population risk. RIVUR was diluted by low-risk children (only ~8% grade IV); reanalysis gives NNT 5 in high-risk vs 18 in low-risk. Prophylaxis reduces infection in proportion to baseline risk; neither CAP nor surgery prevents scarring better than the other.

## 17 · MCQ ONE-LINERS

1. Early ureteric budding → lateral, high orifice → short tunnel → VUR.
2. Paquin ratio of tunnel length to ureteric diameter in non-refluxing UVJ: 5:1.
3. VUR prevalence in children with UTI: ~30%.
4. Grade V VUR: loss of papillary impressions.
5. First dilating grade on international classification: III.
6. Direct RNC misses grade I VUR.
7. Cyclic VCUG detects 12–20% more VUR.
8. Reflux during filling phase predicts poor spontaneous resolution.
9. Normal DMSA: <1% chance of high-grade VUR.
0. Compound papillae at poles allow intrarenal reflux and polar scars.
1. Ransley–Risdon pig model: scarring needs infected urine.
2. Congenital reflux nephropathy: male infant, grade IV–V, no UTI.
3. VUR correction does not cure reflux-related hypertension.
14. BBD lowers endoscopic but not open reimplant success.
15. Circumcision NNT to prevent one UTI in boys with VUR: ~4.
16. RIVUR agent: trimethoprim-sulfamethoxazole; halved recurrent UTI, no effect on scarring.
17. Swedish trial: surveillance girls had 57% febrile UTI recurrence.
18. Avoid sulfonamides under 2 months: bilirubin displacement → neonatal jaundice.
19. Breakthrough febrile UTI on CAP: indication for surgical correction.
20. Hidas high-risk group: 62.5% BUTI at 2 years.
21. Duplex VUR usually affects the lower pole.
22. Ureter entering diverticulum: VUR will not resolve.
23. UPJO with VUR: correct obstruction first.
24. Megacystis–megaureter: treat by ureteric reimplantation, not outlet surgery.
25. Calcified Dx/HA mound may mimic distal ureteric stone.
26. Pregnancy morbidity is driven by renal scarring, not VUR.

# 49 Anomalies of the Kidney and Collecting System

*Reading time ≈ 40 min (slow pace)*

## 1 · EMBRYOLOGY THAT EXPLAINS EVERYTHING

- Ureteric bud (UB) arises from the caudal wolffian duct (WD) at **5–7 wk**, invades the metanephric mesenchyme (MM) and branches into the collecting system; UB tips induce nephrons (**reciprocal induction**).
- Ascent with 90° ventromedial rotation: weeks 6–9; kidney ends below the adrenal, calyces lateral, pelvis medial.
- Renal and genital development are integrated → renal anomalies travel with WD (male) and müllerian duct (MD, female) anomalies. Adrenal (cortex from mesoderm medial to the ridge, medulla from neural crest) is independent of the kidney.

> **KEY POINT** Every anomaly of number is a failure of UB–MM induction: timing and extent set the phenotype. An early insult drags the WD/MD with it (genital anomalies). Arrested ascent leaves an unrotated kidney with anterior pelvis and fetal blood supply; fusion freezes kidneys where they meet.

## 2 · BILATERAL RENAL AGENESIS (BRA)

- **≈1 in 5000** births; **75% male**. First-degree relatives ≈1000× population risk → **ultrasound (US) parents and siblings** of any infant with unilateral renal agenesis (URA), BRA or dysgenesis.
- **Pathology**: kidneys absent; ureters atretic; bladder absent (≈½) or hypoplastic. Adrenal present but flattened — **"lying down" sign** on US. Bladder fails because UB/WD entry into the cloaca initiates bladder development (favoured over "lack of urine").
- **Potter facies** (sine qua non of absent amniotic fluid): senile look, semicircular fold below inner canthus, blunted nose, low-set-appearing ears; claw hands, bowed clubbed legs. These are **deformations**, not malformations (absent in a BRA twin sharing a normal sac).
- Associated: undescended testes 43%, testicular agenesis ≈10% (vas usually normal → insult acted after the WD formed); females — müllerian and ovarian anomalies; gastrointestinal anomalies ≈60%.
- **Pulmonary hypoplasia** kills. Fetal urine is >90% of amniotic fluid by the third trimester. **Peters two-step model**: an early kidney-derived "renal factor" (proline for bronchiolar collagen) acts before 16 wk, then amniotic fluid is needed in the canalicular phase (16–26 wk). Hence fluid only partly restores lung growth.
- **Diagnosis**: prenatal US — severe oligohydramnios, no kidneys, small chest. Neonate: **anuria >24 h without bladder distension** = BRA until proved otherwise. Postnatal US + power Doppler (renal arteries); equivocal → technetium-99m dimercaptosuccinic acid (DMSA) scan: no uptake in either fossa or any ectopic site confirms.
- **Prognosis**: ≈40% stillborn; rest die in 24–48 h of respiratory failure. Amnioinfusion is experimental (RAFT trial, §9): only under institutional review board oversight, with full disclosure and multidisciplinary counselling including palliative care.

> **KEY POINT** BRA is lethal through the lungs, not the kidneys. Amnioinfusion replaces only the amniotic step, so it converts pulmonary deaths into neonatal end-stage renal disease (ESRD).

## 3 · UNILATERAL RENAL AGENESIS AND THE SOLITARY FUNCTIONING KIDNEY (SFK)

- **1 in 1200**; male 1.8:1; left slightly more. Many apparent URAs are **involuted multicystic dysplastic kidneys (MCDK)** — clues: curvilinear calcification on computed tomography (CT); splenic or hepatic flexure in its normal place (a kidney once occupied the fossa). Renal aplasia

(rudimentary parenchyma + ureter) may be the commonest cause of a congenital solitary kidney. **Pitfall at the 20-week scan**: a flattened adrenal or the spleen mistaken for a kidney.

- Associations: VACTERL (vertebral, anal, cardiac, tracheo-oesophageal, renal, limb; ≈30% have URA), MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia), Turner, Kallmann, branchio-oto-renal, DiGeorge (22q11). **Maternal diabetes triples risk**. Isolated preauricular pit/tag or single umbilical artery → no routine renal US; ≥2 anomalies → US; recognised complex → image all systems incl. spine.
- **Ipsilateral ureter absent in ≈60%**. "Hemitrigone" is a misnomer: the trigone derives from the urogenital sinus and forms normally, merely looking indistinct without an intramural ureter.
- Primary defect is in the UB (RET mutations); the gonad (from adjacent mesenchyme) is rarely affected.
- **Contralateral tract**: vesicoureteric reflux (VUR) **≈30%**, pelviureteric junction (PUJ) obstruction ≈11%, vesicoureteric junction obstruction ≈7%.
- **Genital anomalies 20–40%**: females 25–50%, males 10–15% (although URA is commoner in males). Gonads usually normal; adrenal agenesis <10%.

| | Male (WD derivatives) | Female (MD derivatives) |
|---|---|---|
| Always present | Testis, head of epididymis | Fimbriated end of tube |
| Commonest | Absent vas/epididymal body-tail/seminal vesicle (≈50%). **79% of men with absent vas lack the ipsilateral kidney** | Unicornuate/bicornuate, didelphys or septate uterus; vaginal atresia or duplication; Gartner duct cyst with blind ectopic ureter; infertility in up to ⅓ with unicornuate uterus |
| Named syndrome | **Zinner**: seminal vesicle cyst (ejaculatory duct atresia) + ipsilateral agenesis | **OHVIRA** (obstructed hemivagina, ipsilateral renal anomaly; Herlyn–Werner–Wunderlich): pubertal haematocolpos, pelvic mass, pain |
| Trigger to image kidneys | Impalpable vas in infertility work-up, or at herniotomy/orchidopexy | Any uterine anomaly; pelvic US after thelarche, before menarche |

- **Magee classification** (URA + müllerian anomaly, by timing): I (before wk 4, whole ridge) → unicornuate uterus; II (early wk 4, WD + UB) → didelphys with obstructed hemivagina on agenesis side; III (after wk 4, UB only) → isolated URA, normal genitalia.
- **Mayer–Rokitansky–Küster–Hauser (MRKH)**: 46,XX, upper vaginal atresia to total müllerian agenesis (1 in 4500 females). Type I (typical): symmetrical buds, renal anomalies uncommon. **Type II (atypical, commoner): renal anomalies 40–60%** (URA, ectopia, horseshoe). MURCS = most severe end.
- **Work-up**: compensatory hypertrophy from 20 wk in ≈90%. US + colour Doppler; DMSA finds a missed pelvic or crossed ectopic kidney. Voiding cystourethrogram (VCUG) for 28% contralateral VUR is **not routine** — no reduction in febrile urinary tract infection (UTI); shared decision with parents.
- **Sport**: renal trauma comes more from motor vehicle accidents than sport. A solitary kidney does not preclude sport — individual assessment, protective padding.

## Prognosis and surveillance of the SFK

- **Brenner hyperfiltration**: reduced nephron mass → glomerular hypertension → albuminuria → glomerulosclerosis. Risk highest when the SFK itself has congenital anomalies of the kidney and urinary tract (CAKUT: agenesis, hypodysplasia, MCDK, PUJ obstruction, megaureter, posterior urethral valves, VUR).
- Normal nephron number varies eightfold; the low end carries more renal risk. Surrogates: low birth weight, prematurity, small kidney volume, diabetic mother.
- KIMONO: ≈⅓ show renal injury by ≈10 y; population data ≈4× ESRD risk. Verdict: lifelong chronic kidney disease (CKD) risk → lifelong follow-up.

| Situation | Blood pressure (BP) + first-morning urine albumin | Ultrasound | Creatinine |
|---|---|---|---|
| SFK without CAKUT | Annually | Every 5 y if glomerular filtration rate (GFR) >60 and no medication | Periodic, to confirm compensatory hypertrophy into adolescence |
| SFK with CAKUT | Twice yearly | As indicated | Periodic |
| GFR <60 or on antihypertensive/antiproteinuric | BP, albumin, creatinine 2–4×/year | | |

Warning signs: hypertension, microalbuminuria (>30 mg/24 h). Treat with angiotensin-converting enzyme (ACE) inhibitor, salt limitation, avoid excess protein. Renal hypoplasia and dysplasia → *Ch 50*.

## 4 · SUPERNUMERARY KIDNEY

- True accessory organ: **own capsule, own collecting system and own (separate) blood supply** — vs duplex kidney (one parenchyma, one capsule, shared supply). Rare; left; ≈60% caudal to the dominant kidney.
- Second UB; ureters converge (≈50%) or are separate (≈50%, Weigert–Meyer usually obeyed).
- ≈½ are hydronephrotic; symptoms (pain, fever, mass) in early adulthood from obstruction/infection/stone. Magnetic resonance urography (MRU) or retrograde pyelography for anatomy, renography for function; cystoscopy shows one or two ipsilateral orifices.

## 5 · ECTOPIA (ANOMALIES OF ASCENT)

### Simple ectopia

- **1 in 900** (autopsy); left slightly more; bilateral ≈10%. Types: **pelvic (commonest)**, lumbar/iliac, abdominal, cephalad, thoracic. "Vascular barrier" theory rejected — aberrant vessels are a consequence, not cause.
- Kidney small, fetally lobulated, malrotated (pelvis anterior). **Hydronephrosis common**: ½ obstruction (PUJ ≈70% > VUJ), ¼ VUR ≥ grade III, ¼ malrotation alone. **VUR 30–50%** (often into the orthotopic kidney).
- Why hydronephrosis and stones: anterior pelvis, high ureteric insertion, and anomalous vessels compressing a calyx or upper ureter.
- Blood supply anomalous (distal aorta, iliac, inferior mesenteric) → **map by CT or magnetic resonance angiography before surgery**.
- Genital anomalies 15–45%: females 20–66% (bicornuate/unicornuate uterus, vaginal duplication or agenesis); males hypospadias, undescended testis ≈5% each. Ectopic kidney in ≈14% of cloacal malformations. Contralateral kidney usually normal (may be absent, ectopic or refluxing); adrenal normally placed.
- Diagnosis: empty renal fossa; DMSA/MRU for small non-functioning kidney mistaken for URA. A girl with continuous incontinence → look for ectopic ureter of the ectopic kidney.
- Prognosis: normal except hydronephrosis, stones, blunt-trauma vulnerability and reduced differential function. **Never remove a "pelvic mass" without imaging** — solitary pelvic kidneys have been excised as tumours.

### Cephalad and thoracic kidney

- Cephalad (near T10): classic with **omphalocele**.
- Thoracic: ≈1 in 13,000. Lies in the **foramen of Bochdalek** within a thin membrane (not in pleural space); rotation complete; usually asymptomatic. Favoured cause: delayed mesonephric involution allowing ascent before diaphragmatic closure (only 0.25% of diaphragmatic hernias contain a kidney). Adjacent lower lobe may be hypoplastic. Chest X-ray: smooth rounded posterior mass on

the hemidiaphragm → CT/MRU (avoid thoracotomy for a "mediastinal tumour"). Operate only for respiratory distress; no late bowel herniation.

## 6 · FUSION ANOMALIES

### Crossed renal ectopia

- Kidney lies opposite its ureteric orifice; **90% fused** — second commonest fusion anomaly. Left-to-right 3:1. Crossed ureter always crosses midline at the pelvic brim; orifice usually orthotopic.
- **McDonald–McClellan types**: with fusion (90%), without fusion, solitary crossed (usually left kidney crossing right; imperforate anus ≈20%, skeletal and genital anomalies), bilateral crossed (rarest).

| Crossed fused form | Key feature |
|---|---|
| **Inferior ectopia** (≈⅔, commonest) | Crossed upper pole fused to normal lower pole; both pelves anterior (early fusion) |
| Sigmoid (S-shaped) | Second commonest; pelves face opposite ways (late fusion) |
| Lump (cake) | Lobulated pelvic mass, ureters leave below |
| L-shaped (tandem) | Crossed kidney transverse, anterior to L4 |
| Disc (pancake) | Medial fusion, reniform outline kept |
| Superior ectopia | Least common; crossed lower pole to normal upper pole |

- Pelvis orientation times fusion: **anterior = early** (before rotation); medial = late. No rotation after fusion; aortic bifurcation, IMA and mesenteric root stop ascent.
- VUR ≈20% (71% if bilateral crossed); ectopic orifice of the crossed unit ≈3%. A midline-fused kidney with a single ureter dividing into two pelves goes with imperforate anus and vertebral anomalies. Genital: cryptorchidism, hypospadias, absent vas, vaginal atresia.
- Symptoms (third–fourth decade): vague lower abdominal pain, UTI, haematuria, stones (abnormal position and vessels impede drainage); asymptomatic mass in ⅓. **Pain lateralisation**: stone in the crossed ureter hurts on the **anephric side** (the ureter's embryonic side); pyelonephritis or PUJ obstruction hurts where the kidneys lie.
- Imaging: US first; diuretic renography for drainage; MRU/MR angiography (MRA) before major surgery.

### Horseshoe kidney

- **Commonest fusion anomaly**: **1 in 400** (0.25%); male 2:1. **Lower-pole fusion in 95%**, before rotation.
- **Isthmus**: bulky parenchyma (own supply) or fibrous band at L3/L4, **just below the inferior mesenteric artery (IMA)**, which blocks ascent; usually anterior to aorta and inferior vena cava.
- Calyces normal in number but **point posteriorly**; lowest calyces point medially and may overlie the spine; pelvic axis vertical or outward; high lateral ureteric insertion; ureter bends over the isthmus.
- **Blood supply highly variable** (aorta above/below isthmus, IMA, iliac, sacral) → image (MRA/CT) before surgery.
- **Associations** (≈½ of children): **Turner syndrome — ≈60% have horseshoe** (commonest syndrome), caudal regression, neural tube defects (3%), anorectal malformations/VATER association (vertebral, anal, tracheo-oesophageal, renal/radial anomalies); hypospadias/undescended testis ≈4%.
- **PUJ obstruction 13–34%**: (1) high ureteric insertion, (2) ureter anterior to isthmus, (3) crossing vessels. Many hydronephroses are non-obstructed on renography. VUR 8–32% (managed as usual).
- **Stones ≈⅓ of adults**: stasis *plus* usual metabolic risk (hypercalciuria, hypocitraturia) → still do metabolic work-up.

- **Tumours**: **Wilms 1.76–7.9×** expected (favourable histology; preoperative chemotherapy preserved ≈75% of parenchyma — National Wilms Tumor Study); renal pelvic tumours increased (infection, stones); renal cell carcinoma (RCC) not increased.
- Presentation: ≥50% asymptomatic; commonest symptom vague abdominal pain radiating to the lower lumbar region; UTI ≈30%; palpable mass. Diagnosis: US (fibrous isthmus may be missed), DMSA (functioning band across midline), CT/MRU.
- Pregnancy unaffected; transplantable en bloc or divided.

> **KEY POINT** † PUJ obstruction in a horseshoe kidney: dismembered pyeloplasty with a dependent, anterior anastomosis is standard; division of the isthmus (symphysiotomy) is not routinely needed.

## 7 · MALROTATION AND VASCULAR ANOMALIES

- Malrotation: ≈1 in 939; male 2:1; frequent in Turner. **Weyrauch positions**: **ventral (no rotation, commonest)**; ventromedial (incomplete); dorsal (180°, rarest; vessels pass behind); lateral (hyper- or reverse rotation). **Vessel course is the only clue** to direction of rotation.
- Hilar fibrosis distorts the PUJ → intermittent obstruction, stones; function unaffected. Imaging: flattened elongated pelvis, stretched upper calyx, laterally displaced upper ureter; suspect when a stone lies in an unusual position. Bilateral malrotation mimics horseshoe (look for the isthmus and lower-pole outline).
- **Vessel terminology**: *multiple* = >1 aortic vessel (preferred term); *aberrant* = from vessels other than aorta/main renal; *accessory* = ≥2 branches to one segment. Single artery in 71–85%. Segmental arteries are **end-arteries**. The lower segment often has an accessory vessel. Polar or multiple arteries = incomplete degeneration of the embryonic vascular network.
- Crossing vessels occur in ≈¼ of normal unobstructed PUJs → a vessel alone does not prove obstruction.
- **Renal artery aneurysm** (≈0.1%; Abeshouse: saccular ≈93%, fusiform at branch points — congenital, dissecting, arteriovenous); ring ("wreath") calcification ≈30%; mostly silent; hypertension renin-mediated. Rupture only >3 cm. **Treat** (endovascular/open): rupture, rapid growth, pregnancy (or planned), **≥2 cm**, symptoms. Otherwise observe.
- **Arteriovenous fistula (AVF)**: congenital (cirsoid, <25%) vs acquired (biopsy, trauma, surgery — majority; may close within months → observe). Congenital AVF presents in the third–fourth decade: bruit ≈75%, haematuria >75%, renin-mediated hypertension and high-output cardiac failure ≈50%. Rarely resolves → **selective embolisation**, balloon occlusion, partial nephrectomy or nephrectomy.

## 8 · COLLECTING SYSTEM ANOMALIES

- **Bifid pelvis**: ≈10% of normal pelves; normal variant.
- † **Extrarenal pelvis**: a normal variant in which the pelvis lies outside the renal sinus; it looks dilated on US but drains normally — confirm with diuretic renography before calling it obstruction.
- **Calyceal diverticulum**: transitional-lined cavity joined to a calyx (type I, upper pole, small) or pelvis (type II, larger, symptomatic) via a narrow neck; ≈4.5/1000. Congenital or acquired. Trapped urine → enlargement, infection, milk of calcium, **stones in ≈40%**. Milk of calcium layers on US and shifts with position. Diagnosis: **delayed CT/MRU contrast pooling**. Asymptomatic → observe; pain, infection, stones → ureteroscopic neck dilatation/ablation, **percutaneous ablation + fulguration** (best long-term), or laparoscopic marsupialisation. Indications: enlargement with pain or infection, abscess, urosepsis, symptomatic stones. Ablating the cavity lowers recurrence.
- **Hydrocalycosis**: dilated major calyx with demonstrable connection; infundibular obstruction (vessel, cicatricial stenosis, achalasia of the infundibulopelvic muscle ring). Presents with flank pain, UTI, haematuria. Treat for symptoms.
- **Megacalycosis**: **non-obstructive** calyceal enlargement from malformed (falciform) papillae; calyces increased (12–20); pelvis and PUJ normal; only a mild concentrating defect; **normal**

**diuretic renogram**. Male 6:1, bilateral in males. Non-progressive → **no surgery**. May coexist with segmental distal megaureter (pitfall: mimics obstruction).

- **Infundibulopelvic stenosis**: usually bilateral, with VUR; spectrum toward cystic dysplasia. Function normal early, but **>⅓ develop renal insufficiency** (bilateral cases; hyperfiltration). Manage: baseline then yearly creatinine, estimated GFR and urinalysis; when function falls, ACE inhibitor or calcium-channel blocker and protein restriction; surgery only if obstruction harms function — endoscopic/percutaneous for increasing hydronephrosis, or open calicocalicostomy / lower-pole ureterocalicostomy planned on 3-D MRU.

> **KEY POINT** Collecting-system anomalies split on one question — is there obstruction? Dilatation without obstruction (bifid, extrarenal pelvis, megacalycosis) needs nothing; stasis (diverticulum, hydrocalycosis) is treated for symptoms; infundibulopelvic stenosis threatens function through bilateral dysplasia.

## 9 · LANDMARK TRIALS

| Trial | Design | Result | Take-home |
|---|---|---|---|
| RAFT (Renal Anhydramnios Fetal Therapy) | Non-randomised multicentre; isolated BRA, serial amnioinfusion before 26 wk | 94% live births; 82% survived ≥14 d with dialysis access; only 35% discharged on peritoneal dialysis; stopped early | Amnioinfusion prevents lethal lung hypoplasia but leaves neonatal ESRD — experimental, research setting only |

## 10 · MCQ ONE-LINERS

1. Flattened adrenal lying along the spine on US ("lying down" sign) = absent ipsilateral kidney.
2. Potter facies are deformations from oligohydramnios, not primary malformations.
3. URA incidence 1 in 1200; ipsilateral ureter absent in ≈60%.
4. Commonest associated urological anomaly in URA: contralateral VUR (≈30%).
5. 79% of men with absent vas have ipsilateral renal agenesis.
6. MRKH type II carries renal anomalies in 40–60%.
7. Maternal diabetes triples the risk of renal agenesis.
8. Supernumerary kidney requires its own capsule and separate blood supply.
9. Thoracic kidney lies in the foramen of Bochdalek, outside the pleural space.
0. Cephalad ectopia is classically seen with omphalocele.
1. 90% of crossed ectopic kidneys are fused; left-to-right is 3× commoner.
2. Commonest crossed fused form: inferior ectopia.
3. Anterior-facing pelvis in fused kidneys indicates fusion before rotation.
4. Horseshoe kidney: 1 in 400; isthmus lies below the IMA at L3/L4.
15. About 60% of Turner syndrome patients have a horseshoe kidney.
16. Horseshoe kidney raises Wilms tumour risk, not RCC risk.
17. PUJ obstruction in horseshoe kidney: 13–34%.
18. Commonest malrotation: ventral (no rotation).
19. Direction of malrotation is told by the course of the renal vessels.
20. Treat renal artery aneurysm >2 cm, enlarging, symptomatic or with pregnancy.
21. Congenital renal AVF may cause high-output cardiac failure.
22. Stone in a crossed ectopic ureter causes pain on the anephric side.
23. Zinner syndrome: seminal vesicle cyst with ipsilateral renal agenesis.
24. Bifid pelvis occurs in ≈10% of normal kidneys.
25. Calyceal diverticulum: stones in ≈40%; delayed contrast pooling confirms.
26. Megacalycosis is male-predominant (6:1); bilateral disease occurs almost only in males.
27. Megacalycosis: increased calyces, normal pelvis, normal diuretic renogram — no surgery.
28. Infundibulopelvic stenosis is usually bilateral and can progress to ESRD.

# 50 Renal Dysgenesis and Cystic Disease of the Kidney

*Reading time ≈ 41 min (slow pace)*

## 1 · GENETICS IN ONE TABLE

| Gene | Renal phenotype |
|---|---|
| PAX2 | Renal hypodysplasia or aplasia |
| EYA1 | Branchio-oto-renal (BOR) syndrome |
| HNF1B | Renal cysts and diabetes syndrome |
| PKD1, PKD2 (primary cilium) | Polycystic kidney disease |

## 2 · DYSGENESIS: DYSPLASIA VS HYPOPLASIA

- **Dysgenesis** = maldevelopment of size, shape or structure; subtypes dysplasia and hypoplasia. Agenesis → *Ch 49*.
- **Dysplasia** is a **histological** diagnosis: arrested metanephric differentiation from a wolffian duct–ureteric bud (UB)–blastema error. Hallmark (sine qua non): **primitive ducts** — tubules ringed by fibromuscular collars; also cartilage, immature glomeruli. Cysts optional: not all cysts are dysplastic, not all dysplasia is cystic.
- Cause: primary UB failure (genetic; syndromes — Fraser, BOR, renal–coloboma, Kallmann, Townes–Brocks, HDR [hypoparathyroidism, deafness, renal dysplasia]) or fetal outflow obstruction. **Dysplasia is the leading cause of childhood end-stage renal disease (ESRD).** Ultrasound (US): small, echogenic (brighter than liver) kidney, lost corticomedullary differentiation ± cysts.
- **Hypoplasia** = reduced renal mass from too few cells; simple hypoplasia has normal nephron number. Assess by US size, biochemistry, and radionuclide differential function if unilateral.

| Entity | Defining feature | Course / management |
|---|---|---|
| **Oligomeganephronia** | Few, hypertrophied nephrons; usually bilateral | After year 1: polyuria, polydipsia, short stature; **proteinuria is first sign** (precedes decline by years); function stable then falls rapidly in teens → ESRD in second decade. Fluids, correct salt loss/acidosis, protein 1.5 g/kg/d, angiotensin-converting enzyme (ACE) inhibitor; living related donor acceptable (not familial) |
| **Ask-Upmark** (segmental hypoplasia) | Mostly acquired reflux nephropathy; deep midzone "slit-scars", thyroid-like tubules | Girls ≥10 y with **severe hypertension**; partial/total nephrectomy cures blood pressure (BP) regardless of renin; if BP persists suspect a contralateral scar; bilateral → medical, dialysis/transplant. Reflux correction prevents further damage but does not cure hypertension |
| **Hypodysplasia** | With ectopic orifice, megaureter, ureterocele, posterior urethral valves (PUV), prune-belly | **Degree of dysplasia ∝ degree of ureteric ectopia**: lateral ectopia → reflux-associated; medial ectopia/ureterocele → obstruction-associated. PUV: mild form (small subcapsular cysts, near-normal function) vs severe (widespread cysts, cartilage, impaired function). Prune-belly: deformed kidneys, wide tortuous ureters, usually no outlet obstruction |

> **KEY POINT** The ureteric orifice predicts the kidney: a bud from the wrong place on the wolffian duct meets the wrong blastema, so ectopia, obstruction and dysplasia travel together.

## 3 · CYSTIC DISEASE: CONCEPTS AND CLASSIFICATION

- Cysts arise from tubules by three processes: **epithelial proliferation** (after a "second hit" on the normal allele), **fluid secretion** (net NaCl secretion when $Na^+$ absorption falls), and **abnormal extracellular matrix**. The **primary cilium** is central.

- **Multicystic** (multicystic dysplastic kidney, MCDK) = dysplastic, no normal nephrons; **polycystic** = normally developed kidney with nephrons throughout, cysts progressively replace them.

| Inheritable | Non-heritable |
|---|---|
| Autosomal recessive polycystic kidney disease (ARPKD) | Multicystic dysplastic kidney (MCDK) |
| Autosomal dominant polycystic kidney disease (ADPKD) | Benign multilocular cyst (cystic nephroma) |
| Juvenile nephronophthisis (recessive) / medullary cystic disease (dominant) | Simple cysts |
| Congenital nephrosis; familial hypoplastic glomerulocystic disease | Medullary sponge kidney; sporadic glomerulocystic disease |
| Syndromes: tuberous sclerosis, von Hippel–Lindau (VHL) | Acquired renal cystic disease; calyceal diverticulum |

**Extrarenal clue decides**: hepatic fibrosis → ARPKD; berry aneurysm → ADPKD; retinitis pigmentosa → nephronophthisis; angiofibroma → tuberous sclerosis; haemangioblastoma → VHL. Chromosome 16 carries both PKD1 and TSC2.

## 4 · ARPKD

- **PKHD1, chromosome 6** († protein fibrocystin/polyductin); collecting-duct ectasia; **all have congenital hepatic fibrosis**. 1 in 10,000–50,000. "Infantile" is a misnomer — the earlier it presents, the more severe.
- **Fetus/neonate**: huge, smooth, reniform, non-transilluminating flank masses; oligohydramnios → Potter phenotype, pulmonary hypoplasia. **30–50% die** neonatally (respiratory failure, uraemia); ≈½ of neonatal survivors alive at 10 y.
- **US**: bilaterally enlarged, **homogeneously hyperechoic** kidneys (innumerable microcysts creating interfaces); pyramids merge with cortex. Discrete cysts appear with age.
- Later-onset disease: slower renal decline; hepatic problems dominate. Older children: hypertension, chronic kidney disease (growth failure, anaemia, osteodystrophy) and liver — **portal hypertension, varices, splenomegaly with normal liver enzymes**. No neoplasm risk.
- Differential: ADPKD in a neonate (can be identical → **scan the parents**), bilateral hydronephrosis, mesoblastic nephroma, Wilms, renal vein thrombosis. Computed tomography (CT) if in doubt (detects tumour inhomogeneity).
- **Management**: supportive — ventilation, antihypertensives, portosystemic shunt/sclerotherapy, dialysis and transplantation.
- **Nephrectomy** (respiratory/feeding compromise, BP, space for peritoneal dialysis) — limited evidence. ARegPKD registry: **very early bilateral nephrectomy (<3 months)** and hypotensive episodes independently predicted severe neurological injury. Risks: loss of nephron mass (earlier dialysis) and hypotension after bilateral removal. The remaining kidney may grow rapidly after unilateral nephrectomy, forcing a second operation — so buy time.

## 5 · ADPKD — PAEDIATRIC FOCUS

- **Commonest inheritable cystic disease** (1 in 400–1000); 50% of offspring; ≈10% de novo; 7–15% of haemodialysis patients. Usually presents 30–50 y, but in utero/infantile onset is aggressive.

| | PKD1 | PKD2 |
|---|---|---|
| Locus / protein | 16p / polycystin-1 | 4q / polycystin-2 |
| Share | ≈85% | ≈15% |
| Cysts / ESRD | Most by 10 y, nearly all by 20; ESRD 50s | Later; ESRD 70s |

- **Contiguous gene syndrome**: deletion of PKD1 + adjacent TSC2 → tuberous sclerosis with severe early polycystic kidneys.

- In children ≈17% look unilateral/asymmetric; the other kidney follows.
- **Clinical**: pain (commonest adult symptom — haemorrhage, infection, stone), haematuria ≈50%, stones 20–30%, urinary tract infection (UTI). **Hypertension** (renin-mediated by cyst stretch) in ≈50% at 20–35 y with normal function — the key paediatric finding; ≈100% by ESRD. Tight control slows renal decline and lowers cardiac and aneurysm morbidity.
- **Early ESRD risk**: PKD1, male, haematuria <30 y, hypertension <35 y, hyperlipidaemia, sickle trait. Function falls in proportion to kidney/cyst volume.
- **Extrarenal**: liver cysts (commonest; rare in children, ≈all by 50, worse in females), **intracranial berry aneurysm 10–30%**, seminal vesicle cysts 40%, mitral valve prolapse, colonic diverticula.
- Renal cell carcinoma (RCC) **not increased** (but when present: younger, bilateral, sarcomatoid).

### Diagnosis

| Age (at 50% risk) | Ravine US criterion |
|---|---|
| <30 y | ≥2 cysts, uni- or bilateral |
| 30–59 y | ≥2 cysts in each kidney |
| ≥60 y | ≥4 cysts in each kidney |

- Negative US in a young person does **not** exclude PKD2 (sensitivity ≈⅔ <30 y) → genetic testing when certainty matters (e.g., living donor). Counsel before testing (insurance, psychological harms).
- **No family history**: bilateral renal cysts + ≥2 of bilateral enlargement, ≥3 hepatic cysts, cerebral aneurysm, cyst of arachnoid/pineal/pancreas/spleen. Take a three-generation history.
- CT/magnetic resonance imaging (MRI) beat US for extrarenal cysts. CT: haemorrhagic cyst 50–90 Hounsfield units (HU); MRI (no contrast, useful in impaired function): bright T1/dark T2.

### Management

- **BP**: ACE inhibitor/angiotensin receptor blocker (ARB) first line; target <130/80. HALT-PKD: tight BP slowed kidney growth.
- **Tolvaptan** (vasopressin V2 antagonist; ↓ renal cyclic adenosine monophosphate, cAMP) slows total kidney volume (TKV) growth and glomerular filtration rate (GFR) decline (TEMPO, REPRISE); first approved adult therapy (2018); costs — aquaresis, hepatotoxicity, discontinuation.
- **Urological issues**: pain → exclude infection, stone, tumour; avoid non-steroidal anti-inflammatory drugs; aspiration or **laparoscopic unroofing** (relieves pain only, no effect on renal decline). Infected cyst unresponsive to antibiotics → **lipophilic antibiotics** (trimethoprim–sulfamethoxazole, fluoroquinolones) → drainage. Exclude obstruction and abscess. Stones (uric acid and oxalate equally): alkalinisation, spontaneous passage, shock wave lithotripsy; cysts camouflage hydronephrosis. Nephrectomy for symptomatic kidneys in ESRD.
- Aneurysm screening (MR angiography) only for family history of aneurysm/subarachnoid haemorrhage, prior rupture, high-risk job, major surgery, anxiety. <7 mm on screening → observe; coil/clip >7–10 mm or symptomatic. Sudden severe headache → immediate evaluation. Adult management → *kidney book*.

> **KEY POINT** ADPKD is a volume disease: cyst growth stretches vessels (hypertension), compresses nephrons and hides stones and tumours. Every therapy is judged by whether it slows total kidney volume.

## 6 · JUVENILE NEPHRONOPHTHISIS (NPH) / MEDULLARY CYSTIC KIDNEY DISEASE (MCKD)

- Ciliopathies; shared triad: **tubular basement membrane thickening/disintegration, tubular atrophy with corticomedullary cysts, interstitial fibrosis**. Small-to-normal kidneys.
- Shared clinical: **polyuria/polydipsia >80%** (vasopressin-resistant, **salt wasting**); ESRD 5–10 y after presentation.

| | Juvenile NPH | MCKD |
|---|---|---|
| Inheritance | Autosomal recessive | Autosomal dominant |
| ESRD | Mean 13 y; **10–20% of childhood renal failure** | Third–fourth decade |
| Hypertension | Absent | Present |
| Extrarenal | 20%: **retinitis pigmentosa (Senior-Løken)**, Joubert, Cogan apraxia, hepatic fibrosis, Bardet–Biedl | None; type 2 — hyperuricaemia, gout in teenage males |

- US often misses cysts (echogenic small kidneys); CT/MRI better; absence of cysts does not exclude. Bardet–Biedl = obesity, intellectual disability, polydactyly, retinitis pigmentosa, hypogenitalism.
- Treatment: **sodium replacement; avoid salt restriction and diuretics**; dialysis/transplant — no recurrence in grafts (no anti-basement-membrane antibodies).

## 7 · SYNDROMES WITH RENAL CYSTS

### Tuberous sclerosis complex (TSC)

- Autosomal dominant; **TSC1 (chr 9), TSC2 (chr 16)**. Triad: epilepsy ≈80%, intellectual disability ≈60%, adenoma sebaceum ≈75%; earliest skin sign ash-leaf macule (Wood lamp). Diagnosis: 2 major or 1 major + 2 minor features (angiomyolipoma, AML = major; multiple renal cysts = minor).
- Kidney: cysts (even in utero; mimic ADPKD), **AML 40–80%** (bilateral, multiple, after 10 y; never metastasise; risk = haemorrhage), RCC ≈2% (younger, female, bilateral — suspect in a growing fat-free or calcified lesion).
- At diagnosis (International Consensus): abdominal MRI at any age, accurate BP, GFR. Imaging: AML echogenic on US, fat (≈−50 HU) on CT; **MRI preferred** (25–30% fat-poor AMLs missed on US). Fat-poor AML is benign — recognise it to avoid unnecessary surgery. AML elsewhere → TSC; cysts in other organs → ADPKD.
- **AML >3 cm and growing → mammalian target of rapamycin (mTOR) inhibitor (everolimus) first line**; embolisation then corticosteroids for acute bleed; kidney-sparing surgery second line; avoid nephrectomy. Why mTOR: tolerable long-term therapy beats renal loss from AML growth, embolisation or ablation. Imaging every 1–3 y (yearly if a lesion nears 3 cm or grows); BP, proteinuria and GFR yearly; ACE inhibitor/ARB for hypertension (may reduce AML formation). In advanced renal failure mTOR inhibitors still reduce bleeding; offer all renal replacement options.

### Von Hippel–Lindau disease

- Autosomal dominant, **VHL gene 3p25**; pVHL degrades hypoxia-inducible factor (HIF) → loss upregulates angiogenic factors (vascular endothelial growth factor foremost).
- Renal cysts 76% (earliest), **clear cell RCC ≈50%**, phaeochromocytoma 10–17%, cerebellar/retinal haemangioblastoma, pancreatic cysts, epididymal cystadenoma 10–26%. Metastatic RCC and haemangioblastoma are leading causes of death.
- Diagnosis: with family history — one haemangioblastoma, RCC or phaeochromocytoma; without — ≥2 haemangioblastomas, or one + another characteristic lesion. Molecular testing limits screening to gene carriers.
- Surveillance (VHL Alliance 2020): exam yearly from 1 y, BP from 2, eye from <1, metanephrines from 5, brain/spine MRI 2-yearly from 11, **abdominal MRI 2-yearly from 15**; stop routine imaging at 65; full check before conception.
- Goal: cancer control while preserving parenchyma — **nephron-sparing surgery** (accepts residual microscopic disease; local recurrence 75–85% but 10-year disease-specific survival 81–94%). Bilateral radical nephrectomy (dialysis, transplant) only for high-grade disease; ablation investigational. Laparoscopic partial adrenalectomy for small phaeochromocytomas avoids lifelong steroids.

## 8 · MULTICYSTIC DYSPLASTIC KIDNEY (MCDK)

- Multiple non-communicating cysts, **no normal parenchyma, no function**, atretic ureter ("bunch of grapes"). Types: infundibulopelvic (commoner, pelvis + ureter atretic) and hydronephrotic.
- **Commonest renal cystic disease of the newborn; second commonest neonatal abdominal mass**; 1 in 1000–4000; male 55–70%. Bilateral = lethal. Aetiology: extreme obstruction (pelvic/ureteric atresia) vs **UB-gene theory** (EYA1, PAX2, WT1, GDNF; favoured — explains contralateral anomalies). Rarely follows ureterocele obstruction, in a duplex upper pole or a horseshoe moiety.
- **Contralateral kidney**: compensatory hypertrophy; **vesicoureteric reflux (VUR) 18–43%, pelviureteric junction (PUJ) obstruction 3–12%**.
- **Natural history**: ≈40% involute (later imaged as "agenesis"); hypertension and malignancy (Wilms) not increased in large series.

| US feature | MCDK | Severe PUJ obstruction |
|---|---|---|
| Cysts | Haphazard, varied sizes, non-communicating | Calyces arranged around a central pelvis, communicating |
| Central large cyst | Absent | Present (pelvis) |
| Isotope uptake | None | Usually some |

**ALGORITHM**
Prenatal MCDK → US day 2–3 and at 1 month (contralateral kidney, bladder) → technetium-99m dimercaptosuccinic acid (DMSA) or mercaptoacetyltriglycine (MAG3) scan if doubt → voiding cystourethrogram (VCUG) only if UTI or abnormal contralateral kidney/bladder → primary care: BP, mass, UTI

- **Nephrectomy** only for mass effect (respiration, feeding), hypertension, cyst rupture, or **increasing solid tissue**. Routine US surveillance (3–12-monthly) has no proven benefit. Cystoscopy (absent orifice) or angiography rarely needed. Malignant transformation: extremely low, perhaps nil.

**KEY POINT** MCDK removes itself; management looks away from it to the solitary contralateral kidney and its reflux or obstruction. Mistaking a non-functioning PUJ kidney for MCDK costs little.

## 9 · BENIGN MULTILOCULAR CYST (CYSTIC NEPHROMA)

- Spectrum (no lesion converts): benign multilocular cyst → cystic partially differentiated Wilms tumour (CPDN) → multilocular cyst with Wilms nodules → cystic Wilms tumour. Adult counterpart: multilocular cystic RCC.
- **Bimodal: boys <4 y (M 2:1, flank mass) and women >30 y (F 8:1, haematuria from herniation into pelvis)**.
- US: echogenic septa, sonolucent loculi; CT: septa less dense than parenchyma. Imaging separates it from MCDK but not from cystic Wilms, mesoblastic nephroma or clear cell sarcoma.
- Imaging cannot exclude malignancy → **excision is diagnosis and treatment** (nephrectomy in children; partial if localised). Wilms nodules → treat per National Wilms Tumor Study staging (favourable outlook).

## 10 · SIMPLE CYSTS IN CHILDREN AND BOSNIAK

- **US criteria (all four)**: anechoic; thin smooth wall; posterior acoustic enhancement; round/oval. CT: −10 to +20 HU, no enhancement. Enhancement = vascular tissue = suspicion. Hyperdense (20–90 HU) cyst is still benign if non-enhancing, homogeneous, ≤3 cm and ≥¼ of wall extrarenal. Calcification: 2% of cysts (peripheral) vs 10% of RCC (central).

- Cyst puncture now rare: suspected infection, discordant US/CT, borderline lesion in a poor surgical candidate.
- **Children**: simple cyst <15 mm, asymptomatic → repeat US at 2 y, discharge if stable. Continue surveillance if ≥15 mm, complex, family history of cystic disease, or genitourinary anomalies. Many bilateral cysts without liver cysts — do not label ADPKD without family history/genetics.

| Bosniak | Features | Management |
|---|---|---|
| I | Simple | None |
| II | Fine septa, minimal calcification, hyperdense ≤3 cm | None |
| IIF | More septa/calcification, no measurable enhancement | Follow-up (≈11% progress) |
| III | Thick/irregular enhancing walls or septa | Excision (paediatric III–IV: >90% intermediate/malignant) |
| IV | Enhancing solid components | Excision |

Asymptomatic benign cyst → no surgery. Symptomatic (pain, obstruction, rare hypertension): aspiration ± sclerosant (ethanol), or laparoscopic unroofing; infected → drainage + antibiotics. Paediatric: Bosniak I–II periodic US; III–IV excise. Bosniak 2019 adds MRI and down-classifies to curb over-treatment.

## 11 · MEDULLARY SPONGE KIDNEY, GLOMERULOCYSTIC AND ACQUIRED CYSTIC DISEASE

- **Medullary sponge kidney**: ectatic medullary collecting ducts ("brush bristles", "bouquet of flowers" on urography — triad: enlarged kidneys ± papillary calcification, contrast-filled papillary tubules, persistent medullary blush); non-heritable; presents >20 y with colic 50–60%, UTI, haematuria. Hypercalciuria ⅓–½, incomplete distal renal tubular acidosis 30–40% (hypocitraturia). Prevention: **potassium citrate first**; thiazide (even without hypercalciuria); inorganic phosphate if thiazide unsuitable — not with urease infection (struvite); fluids, low sodium.
- **Sporadic glomerulocystic disease**: bilateral enlarged kidneys, Bowman-space cysts; diagnosis of exclusion (no family history).
- **Acquired renal cystic disease (ARCD)**: ≥3 cysts per kidney in ESRD; caused by **uraemia**, not dialysis; 10% pre-dialysis → >90% at 10 y; children on dialysis 20–45%. Small kidneys (vs large in ADPKD). Haemorrhage commonest; adenomas common; RCC risk ↑ (aggressive after transplantation, mostly in native kidneys) → screen long-term dialysis and transplant patients.
- ARCD management: bleeding → rest, then embolisation or nephrectomy (nephrectomy if retroperitoneal bleed, as occult RCC is likely); heparin-related haematuria → switch to peritoneal dialysis. Tumour >3 cm → excision; <3 cm → nephrectomy if fit or annual CT. Avoid iodinated contrast pre-dialysis; gadolinium risks nephrogenic systemic fibrosis.
- **Parapelvic/renal sinus cysts** (lymphatic) mimic hydronephrosis on US — delayed contrast CT shows an undilated system displaced by cysts; conservative.

## 12 · LANDMARK TRIALS

| Trial | Design | Result | Take-home |
|---|---|---|---|
| HALT-PKD | Randomised: lisinopril + telmisartan vs lisinopril; rigorous vs standard BP | Dual blockade no benefit; low BP slowed TKV growth, ↓ left ventricular mass and albuminuria, GFR unchanged | Single renin–angiotensin blocker; tight BP early |
| TEMPO / REPRISE | Randomised, tolvaptan vs placebo | Slower TKV growth and GFR decline | First disease-modifying drug; aquaresis, hepatotoxicity |

| Trial | Design | Result | Take-home |
|---|---|---|---|
| ARegPKD registry | Observational ARPKD cohort | Bilateral nephrectomy <3 months → worst neurological outcome | Avoid early bilateral nephrectomy |

## 13 · MCQ ONE-LINERS

1. Sine qua non of renal dysplasia: primitive ducts with fibromuscular collars.
2. Leading cause of childhood ESRD: renal dysplasia.
3. Degree of hypodysplasia correlates with degree of ureteric orifice ectopia.
4. Oligomeganephronia: fewer, hypertrophied nephrons; proteinuria first sign.
5. Ask-Upmark kidney: segmental hypoplasia, severe hypertension, cured by nephrectomy.
6. ARPKD gene: PKHD1 on chromosome 6 (fibrocystin).
7. ARPKD is always accompanied by congenital hepatic fibrosis.
8. Neonatal ARPKD US: large homogeneously hyperechoic kidneys.
9. Neonatal ADPKD may mimic ARPKD — scan the parents.
0. PKD1 (chr 16) ≈85%; PKD2 (chr 4) milder, later ESRD.
1. PKD1–TSC2 contiguous deletion gives TSC with polycystic kidneys.
2. Commonest extrarenal ADPKD manifestation: hepatic cysts.
3. Berry aneurysm in ADPKD: 10–30%.
4. Tolvaptan is a vasopressin V2 antagonist lowering renal cAMP.
5. Infected ADPKD cyst: use lipophilic antibiotics.
6. NPH: salt wasting, no hypertension, ESRD around 13 years.
17. Senior–Løken syndrome: nephronophthisis plus retinitis pigmentosa.
18. TSC AML >3 cm and growing: everolimus first line.
19. Fat-poor AML is missed on US in 25–30%; MRI preferred.
20. VHL gene: chromosome 3p25; RCC is clear cell type.
21. Commonest renal cystic disease in newborns: MCDK.
22. Contralateral VUR in MCDK: 18–43%.
23. About 40% of MCDKs involute spontaneously.
24. MCDK nephrectomy indication: increasing solid tissue or mass effect.
25. Multilocular cyst: boys <4 y and women >30 y.
26. Bosniak IIF: follow-up; III and IV: excision.
27. Paediatric simple cyst <15 mm: repeat US at 2 years, then discharge.
28. Medullary sponge kidney stone prevention: potassium citrate first.
29. Acquired cystic disease is caused by uraemia, not dialysis.
30. Ask-Upmark hypertension is not cured by reflux correction.
31. Simple cyst calcification is peripheral; tumour calcification is central.
32. VHL: nephron-sparing surgery aims at cancer control, not cure.
33. Nephronophthisis allografts do not develop recurrent disease.

# 51 Pathophysiology of Urinary Tract Obstruction

*Reading time ≈ 18 min (slow pace)*

## 1 · DEFINITION AND CLASSIFICATION

- **Obstruction**: any anatomic or functional abnormality impeding urine flow, anywhere from calyces to meatus, that injures the kidney or lower tract. Classify: **acute/chronic · partial/complete · intrinsic/extrinsic · unilateral/bilateral**.
- Defined by effect on **flow and function, not dilation**: a dilated system may be unobstructed and resolve alone, so the partial vs high-grade distinction, not the image, decides treatment.
- Incidence ≈ **1 in 500** live births. High-grade obstruction: lifetime end-stage renal disease (ESRD) risk up to **30%**.

| | Lower urinary tract obstruction (LUTO) | Upper tract obstruction |
|---|---|---|
| Commonest cause | Posterior urethral valves (PUV) and urethral atresia ≈ 50% | Ureteropelvic junction obstruction (UPJO), 1 in 1500; M:F 2:1 |
| Others | Anterior urethral valves, megalourethra, prolapsing ureterocele, cloaca, megacystis–microcolon–intestinal hypoperistalsis syndrome, trisomy 18/21 | Ureterovesical junction obstruction (UVJO), ectopic ureter, ureterocele, fibroepithelial polyp, stones, crossing vessel (extrinsic) |
| Outcome driver | ESRD + pulmonary hypoplasia (bilateral, whole-fetus disease) | Progressive unilateral function loss; normal contralateral kidney buffers |

**Presentation** depends on site, severity, age at onset and duration: prenatal hydronephrosis on anomaly scan (most at 18–20 weeks); postnatally mass, fever/recurrent infection, flank pain, polyuria or retention, failure to thrive, electrolyte imbalance, or respiratory failure from pulmonary hypoplasia.

> **KEY POINT** Trap: **detrusor hypertrophy** is a lower-tract cause of upper-tract (UVJ) obstruction — the thickened wall narrows the intramural ureter. Neurogenic bladder obstructs without an anatomical lesion.

## 2 · HAEMODYNAMIC CHANGES

Urine production continues against the block → proximal pressure rises → net filtration pressure falls → glomerular filtration rate (GFR) and renal blood flow (RBF) fall → ischaemia. Ischaemic injury (inflammation, tubular atrophy, interstitial fibrosis) begins within **2 h** of complete obstruction. Bilateral obstruction or a solitary kidney → more profound disease.

| † Phase (complete obstruction) | Unilateral | Bilateral / solitary kidney |
|---|---|---|
| I (0–1.5 h) | RBF ↑, ureteric pressure ↑ (afferent vasodilatation: prostaglandin E2, nitric oxide) | Modest RBF ↑ |
| II (1.5–5 h) | RBF ↓, ureteric pressure still ↑ (efferent constriction) | RBF ↓, pressure ↑ |
| III (> 5 h) | RBF ↓, ureteric pressure ↓ (afferent constriction: angiotensin II, thromboxane) | Pressure stays high (retained solute, atrial natriuretic peptide) — underlies post-obstructive diuresis |

- **Hypertension**: ↑ intrarenal pressure → ↓ perfusion → renin–angiotensin–aldosterone system activation → vasoconstriction + Na retention. Bilateral ≈ **77%** hypertensive; unilateral ≈ **20%** (renin lateralises to affected kidney).

## 3 · TUBULAR FUNCTION

Obstruction **downregulates aquaporins (AQP1–4)** and Na transporters along the nephron → concentrating defect (hyposthenuria), salt wasting, polyuria; impaired $H^+$ transport → **distal renal tubular acidosis**.

| Early obstruction — wasting | Prolonged obstruction — retention |
|---|---|
| Na, K, $HCO_3$, Mg, Ca, phosphate | K, $H^+$, Cl, ammonium |

- K is **wasted early, retained late**.
- Fractional excretion of sodium (FENa): low (< 1%) in acute obstruction (filtration falls, tubular avidity kept); high (> 1%) with established tubular injury (salt wasting).

## 4 · PATHOLOGY AND FIBROSIS MEDIATORS

| Immature kidney | Mature kidney |
|---|---|
| Apoptosis (pressure, ischaemia); impaired nephrogenesis and abnormal differentiation — nephron number permanently capped | Compensatory tubular proliferation → if persistent, collagen deposition → scarring |

Renal fibrosis (hallmark ↑ α-smooth muscle actin and interstitial extracellular matrix [ECM]) is the final common path to ESRD.

| Pathway | Mechanism |
|---|---|
| Mechanotransduction | Stretch-sensitive receptors convert stretch into fibrotic signals (initiator) |
| Inflammation | Tumour necrosis factor-α (TNF-α), interleukins → immune cell recruitment, ECM deposition |
| **Transforming growth factor-β (TGF-β)** | TGF-β superfamily → **Smad** signalling → fibrogenesis; **angiotensin II** contributes (amplifier) |
| MicroRNAs | miR-34a ↑ → epithelial-to-mesenchymal transition; possible therapeutic target |

**Bladder** (outlet obstruction): hypertrophy + hyperplasia → compensation → decompensation. Interleukin-6, TNF-α, monocyte chemoattractant protein-1 → collagen/fibronectin → poor compliance. Storage pressure > **40 cm $H_2O$** transmits injury to kidneys.

## 5 · RENAL RECOVERY AND COMPENSATORY GROWTH

- Prolonged obstruction → scarring and atony → renal failure over days–months. Relief can reverse damage, but **incomplete recovery is common**.
- Acute obstruction on chronic kidney disease (CKD) → rapid acute kidney injury; obstruction erodes **renal reserve**.
- **Compensatory growth** of the contralateral/remaining nephrons: insulin-like growth factor-1, TGF-β1, hepatocyte growth factor → glomerular enlargement and tubular hypertrophy (**glomerulomegaly, not new nephrons**).

## 6 · POST-OBSTRUCTIVE DIURESIS (POD)

- Polyuric phase after relief; unilateral ≈ **2%**, bilateral ≈ **50%** (no normal kidney to regulate volume).
- **Mechanism**: aquaporin downregulation, loss of medullary tonicity, tubular transporter dysfunction (apoptosis); physiological excretion of retained salt and water.
- **Risk after valve ablation**: oligohydramnios, preterm birth (low AQP2), raised creatinine.

**ALGORITHM**
Monitor intake/output, serum and urine electrolytes → replace losses, correct electrolytes → **do not over-replace** (perpetuates POD) → usually resolves in **24 h**, seldom > 48 h

## 7 · FETUS AND CHILD

- Fetal kidneys visible from 9 weeks; bladder by 12 weeks (not seen by 15 weeks = abnormal); visible fetal ureter = dilated.
- Urinary tract dilation classification; anteroposterior diameter (APD) most reliable: obstruction considered at **≥ 7 mm** (16–27 weeks), **≥ 10 mm** (≥ 28 weeks).
- LUTO: raised pressure impairs glomerular development; anhydramnios → pulmonary hypoplasia. Second-trimester oligo/anhydramnios → mortality **60–80%**.
- **Fetal urine** (vesicocentesis, three serial samples; last reflects current function) favourable if: Na < 100 mEq/L, Cl < 90, Ca < 8 mg/dL, osmolality < 200 mOsm/L, β2-microglobulin < 6 mg/L — the healthy tubule makes hypotonic urine.
- Fetal therapy: vesicoamniotic shunt/fetal cystoscopy (LUTO, preserved function) treats obstruction; amnioinfusion only replaces fluid.
- Postnatal: ultrasound (US) first; mercaptoacetyltriglycine (MAG3) for drainage; dimercaptosuccinic acid (DMSA) for scar; voiding cystourethrogram (VCUG) for outlet and vesicoureteral reflux (VUR). Whitaker test: obstructed if pelvic pressure > **12 cm $H_2O$** or continuous rise.
- Cystatin C–based GFR is the most accurate paediatric estimate.

## 8 · TREATMENT PRINCIPLES

Logic is **temporise, then define**: drainage buys renal function and stability; definitive reconstruction follows; complications of relief (POD, electrolytes) are managed as carefully as the obstruction.

| Option | Indication / key point |
|---|---|
| Clean intermittent catheterisation (CIC) | Neurogenic bladder; ↓ urinary tract infection (UTI), stones, retention |
| Continuous overnight catheter drainage | Valve bladder with nocturnal polyuria and progressive CKD; delays progression |
| Vesicostomy | Infant with complex LUTO, unstable, or unable to do CIC; temporising |
| Percutaneous nephrostomy | Upper tract obstruction; needs some hydronephrosis |
| Ureteral stent / cutaneous ureterostomy | Temporising or with definitive repair; stent excellent for extrinsic obstruction |
| Pyeloplasty / reimplantation | Correct all intrinsic and extrinsic causes; aberrant vessel → anastomose anterior to it; non-refluxing reimplant in children |

- Pain: non-steroidal anti-inflammatory drugs only if renal function is preserved.
- Oliguria/acute kidney injury: treat cause, optimise fluids and perfusion. CKD: blood pressure, diet; ESRD → dialysis or transplant → *Ch 61*.

## 9 · MCQ ONE-LINERS

1. Commonest cause of LUTO: posterior urethral valves.
2. Commonest upper tract obstruction: UPJO; male:female 2:1.
3. Ischaemic injury starts within 2 h of complete obstruction.
4. Unilateral obstruction late phase: RBF ↓ and ureteric pressure ↓.
5. Bilateral obstruction: ureteric pressure stays high.
6. Characteristic acidification defect: distal renal tubular acidosis.
7. Potassium: wasted early, retained in prolonged obstruction.
8. Aquaporins downregulated in obstruction: AQP1–4.
9. Key fibrotic signalling: TGF-β → Smad.
10. MicroRNA linked to obstructive fibrosis: miR-34a.
11. Compensatory growth = glomerulomegaly, not nephrogenesis.

2. POD: unilateral ≈ 2%, bilateral ≈ 50%.
3. POD usually resolves within 24 h; over-replacement perpetuates it.
4. Preterm infants: low AQP2 → POD risk after valve ablation.
5. Hypertension: bilateral ≈ 77%, unilateral ≈ 20%.
6. Dangerous bladder storage pressure: > 40 cm $H_2O$.
7. Prenatal APD threshold: ≥ 7 mm (16–27 wk), ≥ 10 mm (≥ 28 wk).
18. Favourable fetal urine Na < 100 mEq/L; β2-microglobulin < 6 mg/L.
19. Fetal bladder not seen by 15 weeks: abnormal.
20. Whitaker test positive: pelvic pressure > 12 cm $H_2O$.
21. Megacystis: fetal bladder failing to empty over 40 min at ≥ 18 weeks.
22. FENa < 1% in acute obstruction; > 1% with tubular injury.

# 52 Ectopic Ureter, Ureterocele and Ureteral Anomalies

*Reading time ≈ 59 min (slow pace)*

## 1 · DEFINITIONS AND CLASSIFICATION

- Ectopic ureter and ureterocele are one developmental defect of **ureterotrigonal connection** expressed along a continuum (intermediate cases exist); appearance changes the technique, not the strategy.
- **Ectopic ureter**: any ureter, single or duplex, that does not enter the trigone. Ectopia requires entry **at or distal to the bladder neck**; a ureter merely near the bladder neck is not ectopic.
- **Ureterocele**: ectopic-ureter variant with **cystic dilation of the distal ureter**, lying within the bladder or spanning bladder neck and urethra. Never forms entirely within the urethra; never attaches to wolffian structures — the points that separate it from ectopic ureter.
- In a **duplex system, both lesions always involve the upper pole** ureter.

| | **Ectopic ureter** | **Ureterocele** |
|---|---|---|
| Distal ureter | Dilated or non-dilated, opening ectopically | Cystic dilation |
| Position | At/distal to bladder neck; perineum, vagina, rectum (female) or wolffian ducts (male) | In bladder or spanning bladder neck and urethra |
| Wolffian attachment | Yes (male) | Never |
| Incontinence (untreated) | Yes, in females | No |

### Ureterocele classification

| **System** | **Basis** | **Categories** | **Clinical value** |
|---|---|---|---|
| **Glassberg** (most useful) | Position vs bladder neck | **Intravesical** (entirely above bladder neck) vs **ectopic/extravesical** (part permanently at bladder neck or urethra; orifice in bladder, bladder neck or urethra) | Predicts outcome of endoscopic incision |
| Stephens | Morphology | Stenotic, sphincteric, sphincterostenotic, **cecoureterocele**, blind, non-obstructed | Identifies the surgically dangerous cecoureterocele |
| Churchill | Functional impact | Upper pole only → whole ipsilateral kidney → contralateral too (reflux or bladder outlet obstruction (BOO)) — last group ~1/4 | Extent of threat; outcome prediction unreliable |

- **Cecoureterocele**: orifice inside the bladder (large, incompetent), but the cavity tunnels submucosally beyond the bladder neck into the urethra as a long tongue. Often missed preoperatively; troublesome at incision; risk factor for incontinence.
- Prefer "single intravesical ureterocele" to the adult "simple" ureterocele — intravesical ureteroceles can still obstruct severely.
- Distinguish ectopic ureterocele from an intravesical ureterocele that **prolapses** into the urethra only during voiding.
- **Ureterocele disproportion** (non-obstructive): small dysplastic, non-dilated upper pole invisible on most imaging, no ureteric dilation above the bladder — a typical ureterocele with an apparently normal kidney.

## 2 · EMBRYOLOGY ESSENTIALS

- Ureteric bud arises from the mesonephric (wolffian) duct. In a duplex, the upper pole bud arises **later and more cephalad**, so it is incorporated into the urogenital sinus later and its orifice is carried **caudally** (ectopic).

- **Weigert–Meyer rule**: inverse relationship of duplex orifices — the **upper pole orifice is caudal (and medial) to the lower pole orifice**. Upper pole → ectopia/ureterocele (obstruction); lower pole → short tunnel → vesicoureteral reflux (VUR).
- **Chwalle membrane**: failure of rupture of a membrane at the distal ureteric orifice → ureterocele (classic theory).
- Classic model: the common nephric duct (CND) forms the trigone. Revised model: the CND regresses by apoptosis and the trigone forms from endoderm, like the bladder.
- Renal dysplasia is **downstream** of the connection defect, not a parallel event.
- Bud derives from the wolffian duct, so an ectopic ureter never inserts directly into müllerian structures; it reaches vagina/cervix/uterus via the **Gartner duct** (wolffian remnant alongside the vagina). A cystic Gartner duct rupturing into the vagina is the anatomical basis of female incontinence.

## 3 · ECTOPIC URETER — SITES AND SEX ASYMMETRY

| | Female | Male |
|---|---|---|
| Sites | Bladder neck → perineum; vagina, uterus, rectum (via Gartner duct) | Always **above the external sphincter**: bladder neck, prostatic urethra, vas, seminal vesicle, ejaculatory duct |
| Continence | **Continuous incontinence** (classic) | **Never incontinent** |
| Presentation | Continuous dribbling, perineal (purulent) discharge, urinary tract infection (UTI) | Epididymitis/orchitis, UTI, hydronephrosis, flank pain |
| Association | Gartner duct cyst | **Zinner syndrome**: ipsilateral renal agenesis + seminal vesicle cyst (retrograde injection shows ureter–seminal vesicle–vas communication) |

> **KEY POINT** Why does only the girl leak? Everything follows from where the ureter opens relative to the external sphincter. The female ureter may open distal to it and leaks continuously; the male ureter, confined to wolffian derivatives above it, cannot leak and presents as infection or obstruction.

- **Single-system ectopic ureter** may appear as an "absent kidney" (small, poorly functioning unit missed on ultrasound) → computed tomography (CT) with contrast or renal function study.
- Single or duplex systems may show severe hydroureteronephrosis from distal obstruction, sometimes enough to stop renal development (non-function).
- **Bilateral single-system ectopic ureters** (rare): hypoplastic bladder (no fetal bladder filling), poorly developed trigone and bladder neck, bilateral dysplasia.

## 4 · CLINICAL PRESENTATION

### Prenatal

- Most detected on prenatal ultrasound as hydronephrosis; specific diagnosis usually postnatal. A duplex is hard to see unless one moiety is dilated. Fetal upper pole "cyst" = upper pole hydronephrosis until proven otherwise.
- **Diagnosis is made at the bladder**: upper tract appearances are identical. A large ureterocele can fill and mask the bladder; await filling.
- Record upper pole parenchymal thickness and echogenicity; check other units — ipsilateral lower pole or contralateral dilation suggests reflux (less often obstruction). Ureterocele BOO can rarely cause oligohydramnios.
- Whole-kidney-and-ureter dilation suggests a single system; may be indistinguishable from megaureter or severe reflux until after birth.

### Incidental and infection

- Hydronephrosis found on imaging for trauma or pain. A "presumed ovarian cyst" may be a dilated ureter — include it in the differential of any cystic pelvic mass.
- Urosepsis → ultrasound in all, even with a normal late prenatal scan. Responding child → no urgency; unresponsive sepsis → **urgent transurethral incision (TUI)** of ureterocele, or percutaneous/open drainage (usually end ureterostomy) of ectopic ureter.
- Ectopic ureter: subacute low-grade fever with spikes. **Negative urine culture does not exclude infection** — the infected system may not drain into the bladder. Young girls: purulent perineal discharge → ultrasound. An unobstructed ectopic ureter rarely infects but may cause incontinence.
- Males: epididymitis (often recurrent before diagnosis). **Any young male with bacterial epididymitis → upper tract ultrasound.**

### Incontinence

- Toilet-trained girl with verified **continuous low-volume dampness day and night**, despite normal voiding, sometimes worse with activity → evaluate for ectopic ureter. Untreated ureteroceles do not cause incontinence.
- Discriminating question: **can she be dry for 30–60 minutes?** With ectopic ureter — no. Caution: parents of children wetting for other reasons also say "always wet".
- Before training, parents may notice persistent dribbling at changing. Rarely intermittent (leak through a Gartner duct membrane).
- The leaking moiety may be non-dilated and missed on imaging; suspicion is history-driven. Look for voiding postponement, posturing and constipation before blaming dysfunctional voiding; wetting may be misattributed to laziness or abuse.

### Ureterocele prolapse

- **Smooth, congested, mucosa-covered interlabial mass** from the urethra, distinct from the vagina, often with voiding difficulty.
- Lobulated, grape-like mass suggests **sarcoma botryoides** instead.
- Bladder ultrasound confirms; VCUG shows prolapse into the urethra, lower pole reflux and the "drooping lily".

### Late presentation

- Teenager/adult: infection or pain, rarely incontinence. Adult single-system non-obstructing ureterocele, frequently containing a **stone**.
- Affected moiety usually has little function → upper pole heminephrectomy (mainstay) or upper-to-lower ureteroureterostomy.

## 5 · EVALUATION

### Physical examination

- Search the perineum in every continuously wet girl: an ectopic orifice lies **between urethral meatus and vagina** and can be cannulated with an angiocatheter for contrast study. Rarely a Gartner duct cyst is seen.
- Dilated upper pole may be palpable in a relaxed infant.

### Imaging

| Test | Answers | Note |
|---|---|---|
| **Ultrasound** | Anatomy; inferred function; bladder | First in every case; cannot separate ectopic ureter from ureterocele above the bladder |
| **Voiding cystourethrogram (VCUG)** | Reflux (lower pole, contralateral, into ureterocele); bladder base, urethra | Optional if obstruction alone dictates incision; repeat after incision |

| Test | Answers | Note |
|---|---|---|
| **Dimercaptosuccinic acid (DMSA)** | Differential function — gold standard | Only "some vs none"; no normal upper pole criteria |
| Mercaptoacetyltriglycine (MAG3) diuretic renogram | Function + drainage | For observation candidates; generally after 2 months |
| Magnetic resonance (MR) urography | Most detailed anatomy + function | Complex/undefined anatomy only; needs sedation |
| Intravenous pyelogram (IVP) | Anatomy + qualitative function | Non-dilated ectopic ureter with incontinence |

- **Ureterocele on ultrasound**: thin-walled cyst within the bladder, not beyond its wall. **Pseudoureterocele**: dilated ectopic ureter indenting the bladder — **bilaminar, thicker wall, lumen extending outside the bladder**.
- Pitfalls: a large ureterocele may fill the bladder (no urine seen); a full bladder may efface it so it looks like an ectopic ureter; a lobulated one may look like two cysts; cecoureterocele extension is hard to see (non-visualisation is not exclusion). Repeat observation over filling and emptying avoids errors.
- Bladder ultrasound over a fill–void cycle gives emptying efficiency and wall thickness (outlet obstruction).
- VCUG: reflux most often into the lower pole, recognised by calyceal orientation; the **"drooping lily"** = lower pole displaced down/out by dilated upper pole. With BOO or severe bilateral obstruction, puncture can proceed on ultrasound alone; obtain a post-puncture VCUG.
- MR urography: duplex with non-dilated upper pole, massive dilation obscuring duplication, or grossly distorted anatomy.
- † On excretory urography a single-system intravesical ureterocele gives the **"cobra head" (spring-onion)** appearance — contrast-filled dilated distal ureter with a radiolucent halo (ureterocele wall).

## Function

- Nuclear imaging assesses upper pole and all other moieties (especially with lower pole reflux). No criteria define "normal" upper pole function or the amount worth saving; **preoperative function does not predict function after decompression**.
- Bladder: ureterocele BOO; poor bladder neck function with either lesion.

## VCUG terms

| Term | Meaning | Significance |
|---|---|---|
| Effacement | Ureterocele flattens with filling | Mimics ectopic ureter on full-bladder imaging |
| Prolapse | Ureterocele moves into urethra | Outlet obstruction; predicts secondary surgery |
| **Eversion** | Diverticulum-like protrusion of bladder base with filling/voiding | **Weak trigonal floor → likely needs open repair** (not a true diverticulum) |

## Endoscopy

- A **site-selection** exercise before incision: start with little filling and increase slowly — the lowest portion of the ureterocele (best incision site) is seen with limited filling.
- Document urethra, bladder neck, trigone and other orifices; seek the affected orifice (examine the urethra if not in the bladder). Detect urethral extension (cecoureterocele) so that incision does not leave an obstructing segment.
- Retrograde injection can confirm ureterocele disproportion (narrow proximal ureter, no upper pole filling) or ureter–vas–seminal vesicle communication.
- Ectopic orifice at the bladder neck is often patulous and hard to find. An ectopic ureter separated from the urethra by only a thin layer looks like a ureterocele, goes with bladder neck defects, and supports wider excision with bladder neck reconstruction.

- Patulous bladder neck / cecoureterocele → risk of later incontinence; consider bladder neck tightening at definitive surgery (videourodynamics may help).

## 6 · MANAGEMENT PRINCIPLES

- Goals: **preserve renal function; eliminate infection, obstruction and reflux; maintain continence; minimise total procedural morbidity**.
- Early daily prophylactic antibiotic in significant hydronephrosis (agreed), even without reflux.
- Relieving obstruction may **create** reflux or allow existing lower pole reflux to resolve — the central trade-off. Correct reflux with the drainage procedure where possible.
- Function cannot be predicted before or after decompression; upper pole contributes little and recovery after early decompression is limited (though some puncture series found unsuspected function). Decisions are empirical and individual.
- **Fetal intervention**: none for ectopic ureter; for ureterocele only if large and obstructing all units (needle aspiration or laser ablation — experimental).

### Neonatal algorithm

**ALGORITHM**

Prenatal hydronephrosis → postnatal ultrasound → antibiotic prophylaxis if significant hydroureteronephrosis → VCUG (optional if it will not change decompression) → renal scan after 2 months

Small ureterocele, minimal hydronephrosis, no obstruction → observe

Significant obstruction, parenchyma at risk → TUI (or cutaneous distal ureterostomy if endoscopy impossible, until old enough for reimplant)

Sepsis not responding → TUI (ureterocele) / nephrostomy or end ureterostomy (ectopic ureter); after stabilisation an elective drainage procedure is still needed, as the system has proven infection-prone

- Optimal age for elective decompression undefined (soon after birth to avoid infection vs several months to lower anaesthetic risk).
- Newer infant option for ectopic ureter: side-to-side non-dismembered refluxing anastomosis to the bladder, intended as definitive — long-term data lacking.

### Observation (selected ureteroceles)

- Criteria: no obstruction of ipsilateral lower pole or contralateral kidney, lower pole reflux ≤ grade III; non-functioning upper pole (e.g. multicystic dysplastic upper segment without ureteric dilation); or no obstruction on diuretic renogram.
- Upper pole dilation and lower pole reflux may resolve. Long-term risk undetermined; counsel about unpredictable acute presentation and educate parents.

## 7 · OPERATIVE OPTIONS

### Transurethral incision (TUI) of ureterocele

- Simplest decompression; relieves obstruction in **78–97%**. Procedure of choice in **sepsis** (nephrostomy an alternative) and in the **infant**.
- Technique: **low transverse full-thickness incision** with cutting current, **as distal and as close to the bladder floor as possible** to lessen reflux. Angled-tip wire preferred to Bugbee (finer, easier to manipulate); Collins knife via resectoscope in older children; laser equivalent. Incise deeply — walls may be thick.
- Adequacy: jet of urine or view of urothelium inside.
- Ectopic ureterocele: longitudinal incision from intravesical down into urethral portion, or separate intravesical and urethral punctures; retrograde incision of the urethral part also described (avoid leaving an obstructing urethral flap).

- **"Watering-can"** technique: multiple holmium laser punctures — equal decompression, less de novo reflux.
- Aftercare: uninfected well child — no catheter, outpatient. Ultrasound and VCUG at **4–6 weeks** (decompression, lower pole and new ureterocele reflux).
- New reflux after TUI **0–50%**. **Intravesical**: decompression without reflux in 70–80%. **Extravesical**: secondary surgery up to 70%. Reoperation relative risk 1.74 with lower pole reflux, 2.78 if extravesical (not additive). Single-system ureteroceles do better than duplex.
- Persisting reflux into either pole → typically ureterocele excision and reimplantation (some resolve; observation depends on infection and parental preference). Individual response cannot be predicted.

> **KEY POINT** Why offer TUI first even if half need more surgery? It is an outpatient procedure that may be definitive; the second operation can be deferred to an older child; and decompressing the ureter often avoids tapering at later reimplantation. Less appealing in the older child with a massive non-functioning upper pole, in whom definitive surgery at diagnosis is preferable.

### Upper pole partial nephrectomy (heminephrectomy)

- For a clearly non-functioning upper pole or doubtful drainage because of massive dilation. Open (flank — best vessel exposure; or dorsal lumbotomy — less morbid, harder in older children), laparoscopic or robotic — same intrarenal steps. Primary concern: **do not damage the viable lower pole**.
- Steps: gentle kidney retraction (traction causes vascular injury and spasm) → transect upper pole ureter, traction stitch as handle → dissect it above and below the vessels → pass it **behind the renal vessels** → ligate upper pole vessels (usually 2–3) → demarcation → resect upper pole → strip capsule in continuity → close defect over a fat pedicle (mattress sutures, 3-0 polydioxanone (PDS)/Vicryl).
- Adjuncts: atraumatic pedicle clamp with intravenous mannitol before and after (prevents acute tubular necrosis); papaverine for vasospasm. Vascular control usually not essential.
- Laparoscopic/robotic: clips, energy or suture ties (ties avoid clip avulsion); intravenous methylene blue checks for collecting system leak; fibrin glue + Gerota cover is an alternative closure. Advantages: less pain, quicker discharge, magnification, no second incision.
- **Dissect on the wall of the upper pole ureter**; leave its back wall attached to the lower pole ureter near the bladder — the shared blood supply runs between the two ureters.
- Distal stump: ligate and leave if small (little trouble); large or obstructed-and-refluxing stump → excise as low as possible without injuring bladder neck (second Gibson incision in open surgery; protect vas). Penrose drain to fossa.
- Postoperative Doppler ultrasound: anatomy, urinoma, lower pole flow; nuclear scan if lower pole viability doubtful.
- Ureteral clipping alone (no excision) is tentative — pyonephrosis reported; monitor long term for infection and hypertension.
- Outcomes: ectopic ureter — usually curative (stump rarely problematic). Ureterocele — pre-existing lower pole reflux resolves in up to 20%; new reflux **15–50%**; secondary surgery **40–50%** (depends on whether any reflux or only clinical reflux triggers surgery).

### Ureteroureterostomy / pyeloureterostomy

- Upper pole drained into lower pole system (end-to-side). For functioning upper poles and, increasingly, non-functioning ones (hypertension concern unsubstantiated); herniorrhaphy-sized incision.
- **Distal** (inguinal): **stent the lower pole ureter cystoscopically first** to identify the recipient. **Proximal** preferred for dilated upper pole/size mismatch (less stasis; "yo-yo" significance uncertain).
- Minimal (especially medial) dissection; generous recipient ureterotomy/pyelotomy; aspirate the distal upper pole ureter with a feeding tube, then resect it as low as possible staying on its wall.

Stent the recipient only, never across the anastomosis.

- Distal stump: no reflux → resect low, may leave open; refluxing ectopic ureter → take close to bladder neck; ureterocele → excision avoided if well decompressed.
- Laparoscopic/robotic at iliac vessel level; robotic-assisted laparoscopic ureteroureterostomy (RALUU) ≈ open with slightly shorter stay. Size mismatch does not prevent a good anastomosis.

### Lower tract reconstruction

- **Ureterocele excision + common sheath reimplantation**: relieves obstruction and corrects reflux in one operation; also the fallback for persistent reflux after other procedures. Risks: bladder neck, sphincter, vagina; complexity.
- Steps: transverse incision between stay sutures → plane between ureterocele and bladder wall → dissect to its junction with the lower pole ureter → mobilise **both ureters as a unit** (never separate them — shared longitudinal blood supply) → taper upper pole ureter if needed → excise distal ureterocele to bladder neck → **plicate the detrusor defect** → submucosal reimplant → mucosal flaps.
- Below the bladder neck, take extreme care of the sphincter; if the whole ureterocele cannot be excised, fulgurate the remnant and close in two layers.
- Cecoureterocele remnant can act as an obstructing **urethral windsock** flap-valve: gentle retraction and resection, two-layer closure, or fulguration; watch postoperatively for progressive voiding dysfunction.
- **Marsupialisation**: excise thin intravesical roof and oversew edges, without back-wall dissection; avoids vaginal injury; for small ureteroceles without detrusor defect (risk of diverticulum). Complete excision with trigonal reconstruction is preferred.
- Persisting reflux **5–10%**, more with tapering.
- **Total reconstruction** (heminephrectomy + excision + lower pole reimplant): definitive, two incisions (laparoscopic partial nephrectomy + Pfannenstiel); best for **older child with massive ureterocele, non-functioning upper pole and significant lower pole reflux**.

### End ureterostomy for ectopic ureter

- Neonate with sepsis or massive dilation (alternatives: endoscopic incision with stent, nephrostomy). Decompresses and lets function be assessed later.
- **Do not trim redundant ureter** — it shortens with time and length is needed for reimplantation. Stoma at the lateral end of a future Pfannenstiel incision.
- DMSA at ~4 months or after 6 months → salvageable → reimplant/ureteroureterostomy; no function → partial/total nephrectomy with ureter to the stoma.

## 8 · DECISION SUMMARY

### Ectopic ureter (duplex)

**ALGORITHM**

Keep upper pole? → Yes + lower pole reflux → common sheath reimplant, or lower pole reimplant + distal ureteroureterostomy

Yes + no lower pole reflux → proximal or distal ureteroureterostomy

Function unclear / massive dilation / sepsis → temporary end ureterostomy → DMSA → reconstruct or remove

Remove (marked dilation, preference to excise dysplastic tissue) → upper pole nephrectomy with ureterectomy

- If lower tract surgery is needed anyway for reflux, upper tract surgery may be avoided altogether.
- Single-system ectopic ureter: preserve vs remove by function and preference — no outcome data favour either.

**Ureterocele (Table 52.1)**

| Procedure | Ideal indication | Advantage | Limitation |
|---|---|---|---|
| TUI | Infant; large ureterocele with VUR | Outpatient; effective decompression; occasionally definitive | Reflux into ureterocele moiety → bladder surgery |
| Upper pole nephrectomy | Older; large non-functioning upper pole; no VUR | Removes pathology; avoids bladder | Lower pole risk; may still need bladder surgery |
| Ureteroureterostomy / pyelostomy | Older; functioning upper pole; no VUR | Drains with little risk | Ureterocele left; may reflux |
| Excision + common sheath reimplant | VUR; functioning upper pole without major dilation | Eliminates obstruction and reflux; no renal risk | Complex; vagina/bladder neck risk; tapering |

> **KEY POINT** VUR is the pivot: no VUR → upper tract solution; VUR → bladder-level solution — except in the infant, where TUI is chosen precisely to defer bladder surgery. Upper pole function then chooses nephrectomy vs ureteroureterostomy (though ureteroureterostomy is now also used for non-functioning poles). TUI is not a lesser version of the others: its failure costs little and may simplify the next operation.

## 9 · COMPLICATIONS AND THEIR MANAGEMENT

- **Loss of lower pole function** after heminephrectomy (most significant; fever, pain, haematuria in first week; may be recognised late) — prevented by preserving the inter-ureteric blood supply and gentle traction; same risk on every platform.
- **Urinoma**: up to 20% after laparoscopic/robotic cases, mostly when the defect is not closed; asymptomatic non-expanding → observe.
- **De novo reflux** after TUI or heminephrectomy → observe if no infection; excision and reimplantation if persistent/infecting.
- **Voiding dysfunction after repair**: infrequent voiding is common, incontinence uncommon; partly intrinsic bladder abnormality. Inadequate trigonal support → weak, ballooning trigone behind the bladder neck → obstruction, poor emptying, infection, upper tract dilation, recurrent VUR; also after TUI of cecoureterocele.
- Assess with VCUG **including lateral voiding views** ± videourodynamics or antegrade (suprapubic) cystoscopy; treat by trigonal/bladder neck reconstruction, clean intermittent catheterisation (CIC) or bulking injection for bladder neck incompetence.

## 10 · OTHER URETERAL ANOMALIES

### Duplication

- ~**1 in 125** (0.8%) at autopsy; F:M 1.6:1; unilateral 6× bilateral. **Look for contralateral duplication** — an occult second ectopic ureter explains persistent wetting.
- **Bifid pelvis** = single ureter; **bifid ureter** = partial duplication joining below the ureteropelvic junction (UPJ). Duplication with orthotopic orifices or partial duplication is compatible with normal function.
- **Lower pole gets single-system disease** (UPJ obstruction, VUR); **upper pole gets ureteral-formation disease** (ectopia, ureterocele).
- Lower pole UPJ obstruction: manage as single system (may resolve); renal scan reveals the normal upper pole hidden on ultrasound; option of pyeloureterostomy to the normal upper pole ureter. Upper pole UPJ obstruction is exceptionally rare.
- Lower pole VUR: often high grade with reduced (mostly congenital) function; slower resolution; common sheath reimplant, endoscopic injection, or low lower-to-upper ureteroureterostomy.

### Triplication and quadruplication

- Smith: type 1 three separate ureters (**35%**, commonest); 2 incomplete, two orifices (21%); 3 trifid, one orifice (31%); 4 two ureters, three orifices (13%). Presents with infection or wetting; associated ureterocele, ectopia, UPJ obstruction, VUR. Weigert–Meyer usually holds; violated in quadruplication.

### Congenital ureteral valves

- † Transverse mucosal folds containing smooth muscle that obstruct the ureter. Diagnosis needs proximal dilation, a normal ureter below, and no other cause; treat by excision of the valve segment with ureteroureterostomy (or reimplant if distal).

### Ureteral diverticulum

- † Usually a **blind-ending bifid ureter** (a duplication branch that never reached the kidney), rarely a true congenital diverticulum. Causes stasis, infection, stones or pain; excise the blind limb if symptomatic.

### Fibroepithelial polyp

- Flank pain, haematuria, hydronephrosis; commonest at **UPJ**; **89% male, 78% left**; only 22% seen preoperatively (filling defect, mistaken for clot). A filling defect on retrograde study before pyeloplasty → look for a polyp.
- At UPJ → pyeloplasty; elsewhere → **ureteroscopic removal**. Sleeve resection/nephrectomy unjustified.

### Preureteral vena cava (retrocaval/circumcaval ureter)

- A **venous** anomaly: lumbar **subcardinal vein persists** as the cava (normally the cava forms from the right supracardinal vein), trapping the ureter dorsally. Always **right**-sided, even with bilateral cavae. ~1 in 1500; male 3:1; presents 3rd–4th decade.
- Ureter passes behind the cava at L3–L4 and crosses in front medial-to-lateral. **Type I** (commoner): low crossing, **fishhook/J** pelvis, obstruction at iliopsoas edge. **Type II**: high crossing, smooth horizontal curve, little/no obstruction.
- Excretory urography often fails to show the ureter beyond the J hook; retrograde pyelogram: **S curve**; **CT urography** is test of choice; MRI shows the cava; diuretic renogram grades obstruction only. Classic mimic of UPJ obstruction — a pyeloplasty will fail.
- Treatment: divide ureter, relocate anterior to cava, reanastomose with **excision/bypass of the aperistaltic retrocaval segment**, respecting blood supply (renal/aortic above, iliac below); laparoscopic/robotic possible. Move the ureter, not the vessel. Associated: horseshoe kidney, left renal anomalies.

### Other position anomalies

- **Preureteral iliac artery**: ventral root of umbilical artery persists; either side; obstruction at L5–S1; associated vasal/mesonephric duct anomalies and ureteral ectopia.
- Ureteral herniation (inguinal, scrotal, femoral): mostly paraperitoneal, ureteric loop **medial** to the sac.

## 11 · MCQ ONE-LINERS

1. Ectopic ureter in a duplex system always drains the upper pole.
2. Weigert–Meyer: upper pole orifice lies caudal and medial to the lower pole orifice.
3. Female ectopic ureter: continuous dribbling with normal voiding.
4. Male ectopic ureter never causes incontinence; always above external sphincter.
5. Commonest male presentation of ectopic ureter: epididymitis or UTI.
6. Zinner syndrome: ectopic ureter, ipsilateral renal agenesis, seminal vesicle cyst.
7. Female ectopic ureter reaches vagina via the Gartner duct.
8. Most useful ureterocele classification: intravesical versus ectopic (extravesical).

9. Cecoureterocele: intravesical orifice, submucosal cavity extending into urethra.
0. Chwalle membrane persistence: classic theory of ureterocele formation.
1. Smooth interlabial mass: prolapsed ureterocele; grape-like: sarcoma botryoides.
2. "Drooping lily": lower pole displaced by dilated upper pole.
3. Pseudoureterocele: bilaminar thick wall, lumen outside bladder.
4. Eversion on VCUG: weak trigone, likely needs open repair.
5. Negative urine culture does not exclude infected ectopic system.
6. Ectopic perineal orifice lies between urethral meatus and vagina.
7. Preoperative upper pole function does not predict post-decompression function.
8. TUI incision: low, transverse, close to bladder floor.
9. Watering-can puncture: equal decompression, less reflux.
0. Intravesical ureterocele after TUI: 70–80% decompressed without reflux.
21. Extravesical ureterocele: secondary surgery after TUI up to 70%.
22. Secondary surgery after upper pole nephrectomy for ureterocele: 40–50%.
23. Major heminephrectomy risk: loss of lower pole function.
24. Never separate duplex ureters intravesically: shared blood supply.
25. Distal ureteroureterostomy: stent lower pole ureter first.
26. End ureterostomy for ectopic ureter: do not trim redundant ureter.
27. Post-repair voiding dysfunction: inadequate trigonal support; lateral voiding views.
28. Duplication incidence: ~1 in 125 (0.8%).
29. Lower pole: UPJ obstruction and VUR; upper pole: ectopia and ureterocele.
30. Commonest Smith triplex type: type 1.
31. Fibroepithelial polyp: male, left, at UPJ.
32. Retrocaval ureter: persistent right subcardinal vein; fishhook deformity at L3–L4.
33. Retroiliac ureter obstructs at L5–S1.

# 53 Surgery of the Ureter in Children

*Reading time ≈ 40 min (slow pace)*

## 1 · PELVIURETERIC JUNCTION (PUJ) OBSTRUCTION — DEFINITION AND AETIOLOGY

- Paediatric hydronephrosis: renal pelvic anteroposterior diameter (APD) **12 mm**; hydronephrosis is **not synonymous with obstruction**.
- **Koff's definition**: obstruction is "any restriction to urinary outflow that, if left untreated, will lead to progressive kidney damage." It is retrospective — so every surgical indication is an attempt to predict harm before function is lost.
- Male and left predominance; bilateral in **10–40%**.

| | Intrinsic PUJ obstruction | Extrinsic (crossing vessel) |
|---|---|---|
| Age | Newborns, infants (commonest at this age) | Older children, adolescents |
| Lesion | Narrow segment with interrupted circular muscle → functional discontinuity of peristalsis → poor pelvic emptying | Accessory lower-pole vessel crossing anteriorly; ureter kinks over it |
| Presentation | Antenatal or incidental dilatation | Intermittent loin pain, nausea, vomiting |

- After mobilisation off a crossing vessel there is often no intrinsic narrowing, yet **dismembering and transposing the ureter anterior to the vessel prevents recurrence**.
- **Secondary PUJ obstruction**: high-grade vesicoureteral reflux (VUR) with tortuous ureter kinks the PUJ. Avoid simultaneous reimplant + dismembered pyeloplasty (devascularises ureter from both ends) → conservative or **two-stage** repair.
- **Lower-moiety PUJ obstruction (duplex)**: incomplete duplication → upper-pole ureter to lower-pole pelvis (ureteropyelostomy); complete → standard pyeloplasty; moiety ≤ **5–10%** function → partial nephrectomy.
- **Associated anomalies** in up to 50%: contralateral PUJ obstruction (commonest, 10–40%) → renal dysplasia → multicystic dysplastic kidney → agenesis. VUR in up to 40% (high spontaneous resolution). Horseshoe kidney and VATER association.

## 2 · DIAGNOSIS

- Antenatal ultrasound (US) at ≈ 20 weeks; postnatal US (incidental or after urinary tract infection [UTI]). Febrile UTI usually responds to antibiotics; few need nephrostomy.
- † **Society for Fetal Urology (SFU) grading**: 0 none; 1 renal pelvis only; 2 pelvis + few calyces; 3 all calyces dilated, parenchyma normal; 4 as 3 + parenchymal thinning.
- † **Mercaptoacetyltriglycine (MAG3) diuretic renogram**: drainage half-time (T½) < 10 min unobstructed, 10–20 min equivocal, > 20 min obstructed; differential renal function (DRF) < 40% or falling is the more reliable marker.
- Neonatal washout curves are confounded by renal immaturity, hydration, diuretic timing, posture and pelvic volume/compliance — false positives up to **44%**, so decide on DRF trend and dilatation, not T½ alone.

## 3 · INDICATIONS FOR PYELOPLASTY

About **one-third** need surgery; most resolve.

- Increasing APD on US
- Low or falling DRF
- Breakthrough infection on prophylaxis

- Symptoms: pain (older child); failure to thrive, feeding difficulty in infants; haematuria (ruptured mucosal vessels); rarely hypertension (renin–angiotensin activation)

## 4 · ANDERSON–HYNES DISMEMBERED PYELOPLASTY

**Gold standard** whatever the access, success ≈ **95%**; originally devised for retrocaval ureter. Dismembering preserves a crossing vessel (ureter transposed anterior), excises the adynamic segment and allows pelvic reduction.

**ALGORITHM**
Open Gerota fascia → expose PUJ → free lower pole (do not miss an accessory vessel) → sparing bipolar diathermy (protect pelvic/ureteric blood supply) → stay sutures on pelvis and ureter → decompress tense pelvis with 21-G needle → dismember ± reduce pelvis → **spatulate laterally** well into healthy ureter, stenotic segment kept as handle ("no-touch") → apex of spatulation to inferior pelvis → tension-free, watertight 6-0/5-0 absorbable anastomosis → excise stenotic segment just before completion → irrigate clot → cover with perinephric fat

- Long gap: full renal mobilisation; horizontal lower-pelvis incision ("bypass" side-to-side configuration).
- Pitfall: excess stay-suture tension kinks the ureter when released. Gradual taper; drain usually unnecessary.

| Alternative | Best indication | Point |
|---|---|---|
| Foley Y-V-plasty | High insertion, no vessel | Base of V on dependent medial pelvis, apex at PUJ; stem down lateral ureter |
| Spiral flap | Long proximal stricture, dependent PUJ | Length:width ≤ 3:1 |
| Vertical flap | Dependent PUJ, large box-shaped extrarenal pelvis | Flap length must match gap |
| Vascular hitch (Hellström) | Anterior lower-pole vessel, **no intrinsic stenosis**: moderate hydronephrosis, no calyceal dilatation, good cortex, preserved split function; normal PUJ with peristalsis at surgery | Vessel moved cranially, fixed in pelvic wall; no hypertension seen |
| Uretero-calicostomy | Failed repair, short scarred ureter, dilated lower calyx with thin cortex | Lower-pole nephrectomy to expose calyx widely → spatulated ureter sutured over stent ± nephrostomy; capsule not closed near anastomosis (compression); cover with fat/omentum |

Flaps are reserved for severe hydronephrosis with a long stenotic segment where tension-free primary anastomosis is impossible. Non-dismembered (Heineke–Mikulicz) repair is inferior.

**KEY POINT** Every variant answers two questions: is there a crossing vessel (dismember), and is the gap too long for tension-free anastomosis (flap, mobilisation)?

**KEY POINT** Repair has one fixed principle and many routes: dismember, excise, spatulate, anastomose without tension. Access, stent and robot are choices of convenience; failure is managed by repeating the same principle.

## 5 · ACCESS: OPEN VS LAPAROSCOPIC/ROBOTIC

- **Infants**: open preferred (incision barely larger than ports). **Posterior lumbotomy** (prone, lumbodorsal fascia) only in infants/toddlers; flank approach at 11th–12th rib tip at any age; open route extraperitoneal.
- Laparoscopic outcome = open (meta-analysis); robotic: shorter learning curve (3D, dexterity), faster resolution of hydronephrosis vs open.

| | Transperitoneal | Retroperitoneal |
|---|---|---|
| For | Easier, familiar landmarks; left side transmesocolic | No bowel injury; leak stays extraperitoneal; shorter stay; mirrors open |
| Against | Bowel risk; urinary ascites → ileus | Learning curve; small space; orientation |
| Note | Preferred for horseshoe kidney, retrocaval ureter | One randomised trial favoured it (time, stay, feeding); 80–90° flank, table **not flexed**, balloon space 200–300 mL |

- **Endopyelotomy**/balloon: poor long-term in < 18 years; niche is **failed pyeloplasty** (anatomy known, no vessel) with double-J for 6 weeks.
- Stones found at pyeloplasty → flexible scope through a port.
- Lap/robotic set-up: transperitoneal 45–60° flank, Hasson camera port, instrument ports more medial for large pelvis or small infant; stay suture on PUJ (transabdominal hitch stitch) for traction. Retroperitoneal: gel roll under contralateral iliac crest, extra 5-mm assistant port for suction and sutures.

## 6 · STENTS AND DRAINAGE

| Stent | Removal | Point |
|---|---|---|
| Internal double-J | Second anaesthetic unless string/magnet | Bladder catheter 24–48 h (stent transmits reflux pressure); string → more febrile UTI (girls) |
| Externalised nephro-ureteric | Pulled at 7–14 days, no anaesthesia | Fewer complications and shorter stay in infants; can be unknotted to drain as nephrostomy |
| Stentless | — | Needs flawless technique |

- Stent duration 1–4 weeks does not change outcome. Antegrade double-J may not pass the narrow infant ureterovesical junction (UVJ) — confirm bladder position with methylene blue in bladder.
- Emergency diversion before repair: **nephrostomy, not retrograde stent** (stent inflames the PUJ).

## 7 · OUTCOMES AND FAILED PYELOPLASTY

- Success 90–100%, but "success" variably defined (strictest: no reintervention). Commonest reason for conversion: bleeding.
- **Early** failure (infection, pain, ↑ dilatation): usually clot/oedema → settles, ± temporary nephrostomy.
- **Late** failure (falling function, ↑ dilatation): balloon dilatation/endopyelotomy with stent 4–6 weeks, or — usually best — **redo pyeloplasty, transperitoneal** (retroperitoneum scarred); short scarred ureter → uretero-calicostomy.

## 8 · PRIMARY MEGAURETER

- Descriptive: dilated ureter **≥ 7 mm** (normal ≤ 5–6.5 mm to 16 years). Primary (intrinsic) vs secondary (neurogenic bladder, outlet obstruction).
- **Four types**: obstructed · refluxing · non-obstructed non-refluxing · refluxing + obstructed. Hard part: separating obstructed from non-obstructed non-refluxing — dilatation does not prove obstruction.
- **Primary obstructive megaureter (POM)**: adynamic juxtavesical segment. Ureteric muscle matures **craniocaudally**, distal segment last (after birth) → transient dysfunction and spontaneous resolution. Also excess collagen, abnormal interstitial cells of Cajal.
- 10–23% of antenatal upper-tract dilatation; males, left; bilateral 25%. Symptoms: UTI, pain, haematuria, stones.

| Ureter diameter | < 10 mm | ≥ 10 mm |
|---|---|---|
| Course | Few febrile UTIs; usually resolves | Febrile UTI, stones, pain; minority resolve |

| Ureter diameter | < 10 mm | ≥ 10 mm |
|---|---|---|
| Surgery | Virtually none | Substantial minority |

**Indications**: recurrent febrile UTI · progressive dilatation · DRF **< 40%** · DRF fall **≥ 5%** on comparable scans. Follow into adulthood (some worsen at puberty).

> **KEY POINT** Why not rely on the renogram? Neonatal washout depends on immature function, hydration and the capacity of a huge ureter — so serial DRF, diameter and infection history drive surgery. Megaureter is the PUJ problem moved to the bottom of the ureter: an immature adynamic segment that usually matures on its own.

### Temporising in infancy

Small bladder and massive ureter are incongruent — a safe non-refluxing tunnel is impossible, so definitive repair is delayed.

| Option | Point |
|---|---|
| Distal (loop) cutaneous ureterostomy | Oblique inguinal incision, ureter brought out as loop, fixed to fascia, opened and everted to skin; rapid decompression of infected system; stomal stenosis rare (wide ureter) |
| Refluxing reimplant | Converts dangerous obstruction into the "lesser evil" of reflux |
| Endoscopic balloon dilatation + double-J 2–6 months | Dilate with high-pressure balloon under fluoroscopy until the waist disappears; failure → re-stent or reimplant. Least invasive; success ≈ 70–80% (weak, mostly retrospective evidence); complications up to one-third (migration, infection, VUR, stones) |
| Nephrostomy | Hard to maintain in infants |

### Definitive repair

**ALGORITHM**
Excise stenotic distal segment → straighten → taper → non-refluxing reimplant at **5:1** tunnel length:diameter → stent (feeding tube 7 days or double-J 4 weeks)

| | Starr plication | Kalicinski folding | Hendren excisional tapering |
|---|---|---|---|
| Principle | Imbricate lateral wall inward | Exclude lateral lumen, fold over | Excise lateral wall over 8–10 Fr catheter |
| Blood supply | Preserved — less ischaemia | | At risk → ischaemic stricture |
| Use | Moderate dilatation (< 1.75 cm); bulky if larger | | Massively dilated ureter |

- A refinement of Hendren excises only mucosa and muscle, leaving vascular adventitia.
- Hendren: running locking suture proximal two-thirds, interrupted distal third (shortened at reimplant). Taper gradually — an abrupt calibre change is itself obstructive.
- Approach: Pfannenstiel, mobilise as for intravesical reimplant (Cohen or Politano–Leadbetter tunnel).
- Very large ureter: extravesical approach avoids a huge trigonal hiatus. Laparoscopic/robotic results comparable short–medium term.
- Success > **90%**; POM better than refluxing megaureter (stiffer collagen-rich UVJ). Worse with valves/neurogenic bladder.
- Complications: obstruction → stent first (oedema); ischaemic stricture → redo reimplant. Reflux → observe/subureteric injection → revision. Transureteroureterostomy for unilateral severe scarring.

## 9 · SURGERY FOR VESICOURETERAL REFLUX

Aim: one-way, unobstructed antegrade flow by reimplanting the ureter in an antireflux tunnel. Open, laparoscopic/robotic and endoscopic options.

**KEY POINT** **Principles**: exclude or treat secondary (bladder) causes first · handle the ureter gently (blood supply) and the bladder gently (haematuria, spasm) · mobilise enough ureter for a tension-free repair · tunnel length:diameter **5:1** (Paquin) · keep the course straight at both empty and full bladder (kinking).

- Classified by access (open/laparoscopic), approach (intravesical/extravesical), and tunnel relative to the original hiatus (suprahiatal/infrahiatal).
- Pfannenstiel 2 cm above symphysis; vertical midline for redo (more cranial ureteric mobilisation).

| Technique | Approach / tunnel | Strength | Weakness |
|---|---|---|---|
| Politano–Leadbetter | Intravesical; suprahiatal — new superomedial hiatus, vertical tunnel | Long tunnel for high grade | Blind neo-hiatus (bowel, vas, vagina); kink if too lateral/anterior |
| Glenn–Anderson | Intravesical; infrahiatal, towards bladder neck | Least kinking | Short tunnel |
| **Cohen** | Intravesical; transhiatal cross-trigonal | Commonest; success > **98%**; thick trabeculated bladder; with bladder-neck reconstruction | Difficult later retrograde catheterisation |
| **Lich–Gregoir** | Extravesical detrusorotomy, ≈ 5 cm | No haematuria or spasm; short catheter time | Transient voiding dysfunction/retention up to **10%**, especially bilateral |
| Robotic extravesical (RALUR-EV) | Y-shaped detrusorotomy 4–5 cm | Less pain, spasm, stay | Early series: leak, obstruction, ureteric injury |

## Cohen — key steps

**ALGORITHM**
Open bladder vertically to 2 cm from neck → pack dome, ring retractor; stay sutures on lateral walls → 5-Fr feeding tube in orifice as handle → circumscribe orifice 1–2 mm with fine diathermy → free ureter in correct plane (preserve distal blood supply) until it reaches the contralateral wall without tension → submucosal tunnel across trigone → new orifice just above contralateral orifice → mucosa-to-mucosa 6-0, detrusor hitch at 6 o'clock

Later access to a cross-trigonal orifice: flexible ureteroscope, or suprapubic needle/cannula aimed at the orifice to pass a guidewire.

## Lich–Gregoir — key steps

**ALGORITHM**
Bladder one-third full → divide obliterated umbilical artery → free ureter **hugging it**, only down to the UVJ → mark 5-cm tunnel along natural course with bladder moderately distended → detrusorotomy to intact mucosa (uniform bulge) → raise flaps wide enough not to constrict → lay ureter in → close detrusor caudocranially, catching ureteric adventitia at apex (prevents slippage) → Foley 12–48 h

**KEY POINT** Why voiding dysfunction? Pelvic plexus lies ≈ 1.5–2 cm dorsomedial to the UVJ, with branches along the medial ureter. Stay on the ureter, avoid the dorsomedial UVJ, stop at the UVJ. The Zaontz modification (detrusorotomy around the UVJ) encroaches on these nerves and is now avoided.

- Open benchmark: American Urological Association (AUA) meta-analysis — success **95.9%**. Minimally invasive results converging with experience.
- **Follow-up**: US at 6–12 weeks for all. Voiding cystourethrogram (VCUG) may be omitted after low-grade reflux if US normal and no voiding dysfunction or UTI; do it for high grade, new hydronephrosis, UTI or dysfunctional bladder.

### Complications of reimplantation

| | Early | Long-term |
|---|---|---|
| Reflux | Ipsilateral low grade (bladder inflammation) — usually resolves by 1 year. Contralateral ≈ 1 in 10 after unilateral repair, mostly resolves → observe; prophylaxis < 4–5 years; intervene for pyelonephritis. Routine bilateral repair not justified | Rare after low grade; due to high grade, short tunnel/untapered ureter, or **unrecognised secondary reflux** (neurogenic or valve bladder) → treat bladder first (anticholinergic + clean intermittent catheterisation) |
| Obstruction | Oedema, clot, submucosal haematoma, kink; within 2 weeks with pain, vomiting → double-J or nephrostomy (internalise early); most resolve | Distal ischaemia, angulation (hiatus too lateral/anterior), tight tunnel, orifice stenosis → balloon + stent or unroofing → re-reimplant |

- Mild early hydronephrosis is common; it should not exceed the maximal dilatation the reflux caused.
- Contralateral reflux risk: age < 6 years, capacity < 50% expected, refluxing duplex; technique irrelevant.
- **Reoperation**: combined intra/extravesical mobilisation; excise ischaemic ureter to bleeding, peristaltic tissue; short ureter → **psoas hitch** (non-absorbable, before tunnelling), Boari flap, nipple valve (dilated ureter), or reconfigured (Monti) bowel ureter — no tapering, no metabolic effects.

> **KEY POINT** Early complications are tissue reactions — drain and wait. Late ones are design errors (blood supply, hiatus position, tunnel ratio) or an untreated bladder. Reimplanting into an abnormal bladder only worsens dilatation.

### Endoscopic injection

- Dextranomer/hyaluronic acid (Dx/HA, Deflux) is the standard agent; ≈ **90%** initial resolution of low-grade reflux after one injection. Teflon (STING, subureteric Teflon injection) abandoned — particle migration, never approved by the US Food and Drug Administration.
- Technique: cystoscopy first (abandon if inflamed) → needle bevel up at 6 o'clock → mound should appear within **0.1–0.2 mL**, else reposition → aim for a **"volcano"** mound with slit/crescent orifice. Bleeding: press with scope, **do not cauterise** (extrusion). Day case, no catheter.

| Technique | Needle entry | Result |
|---|---|---|
| STING | 2–3 mm below orifice, 4–5 mm submucosal | Baseline |
| HIT (hydrodistension implantation) | Inside hydrodistended orifice, intramural ureter | Better than STING for all grades |
| Double HIT | Two intraureteric punctures (proximal + distal) | ≈ 90%, durable |

- Follow-up: antibiotics 3 months; US at 2–6 weeks (acute obstruction); US ± VCUG at 3 months. Failure → 1–2 repeat injections → reimplant (not harder after Deflux; implant encapsulated).
- **Late obstruction** reported (Dx/HA, especially polyacrylate polyalcohol [Vantris]) → long-term US. Deflux loses ≈ 23% volume; Swedish Reflux Trial: **20%** recurrence at 2 years.
- Migration: particles < 50 µm pass lungs to systemic circulation; < 80 µm phagocytosed. Deflux 80–250 µm.

## 10 · OTHER URETERIC LESIONS

- † **Retrocaval ureter**: right side, ureter passes behind inferior vena cava (persistent posterior cardinal vein); "fish-hook/S" deformity → dismember and re-anastomose anterior to the cava.
- **Congenital mid-ureteric stricture**: misread as megaureter; excise (1–3 cm) + ureteroureterostomy.
- **Fibroepithelial polyp**: upper third, male, left; pain + haematuria in later childhood; confirm by retrograde pyelography or ureteroscopy; endoscopic polypectomy, or resection with

pyeloplasty/ureteroureterostomy.

- **Retroperitoneal fibrosis**: steroids → azathioprine → double-J → ureterolysis.
- **Short ureter in reoperation**: psoas hitch (one side only), Boari flap, transureteroureterostomy, reconfigured (Monti) bowel ureter.

## 11 · MCQ ONE-LINERS

1. Gold-standard PUJ repair: Anderson–Hynes dismembered pyeloplasty (≈ 95%).
2. Anderson–Hynes was originally devised for retrocaval ureter.
3. Commonest associated anomaly with PUJ obstruction: contralateral PUJ obstruction.
4. Crossing vessel PUJ obstruction typically presents in older children with intermittent pain.
5. Foley Y-V-plasty is contraindicated with a crossing vessel.
6. Spiral flap length:width ratio ≤ 3:1.
7. Vascular hitch needs no intrinsic stenosis and visible peristalsis.
8. Posterior lumbotomy: infants and toddlers only.
9. Retroperitoneoscopic positioning: table not flexed.
10. Emergency drainage before pyeloplasty: nephrostomy, not stent.
11. Stent duration 1–4 weeks does not affect pyeloplasty outcome.
12. Redo pyeloplasty: transperitoneal route preferred.
13. Endopyelotomy niche: failed primary pyeloplasty.
14. Diuretic renogram false-positive rate in neonates up to 44%.
15. Megaureter: diameter ≥ 7 mm.
16. Ureteric muscle matures craniocaudally; juxtavesical segment last.
17. Megaureter < 10 mm virtually never needs surgery.
18. Megaureter surgery if DRF < 40% or falls ≥ 5%.
19. Tunnel length:diameter for reimplant: 5:1 (Paquin).
20. Folding/plication limit: ureter < 1.75 cm.
21. Hendren excisional tapering: risk of ischaemic stricture.
22. POM outcomes better than refluxing megaureter.
23. Balloon dilatation of POM: double-J 2–6 months, complications up to one-third.
24. Duplex lower-moiety with ≤ 5–10% function: partial nephrectomy.
25. Fibroepithelial polyp: upper ureter, left, males.
26. Psoas hitch: unilateral only; second short ureter → transureteroureterostomy.
27. Externalised nephro-ureteric stent: removed at 7–14 days without anaesthesia.
28. Secondary PUJ obstruction from high-grade VUR: two-stage repair, not simultaneous.
29. Commonest reason for conversion of minimally invasive pyeloplasty: bleeding.
30. Commonest intravesical reimplant: Cohen cross-trigonal; drawback is difficult retrograde access.
31. Glenn–Anderson infrahiatal tunnel: least kinking, length limited by bladder neck.
32. Lich–Gregoir voiding dysfunction: pelvic plexus injury dorsomedial to the UVJ.
33. Open reimplant success (AUA meta-analysis): 95.9%.
34. Endoscopic injection: mound within 0.1–0.2 mL; volcano shape predicts success.
35. Injection-site bleeding: compress, never cauterise.

# 56 Hypospadias

*Reading time ≈ 34 min (slow pace)*

## 1 · DEFINITION AND DIAGNOSIS

- **Classic triad**: proximal (ventral) ectopic meatus, ventral penile curvature, dorsally hooded foreskin — all from arrested penile development. Curvature and hood are not always present.
- Incidence ≈ **1 in 150–300** male births; one of the commonest congenital anomalies. Diagnosis is clinical.
- **Megameatus with intact prepuce (MIP)**, ≈ 5%: wide distal meatus, normal foreskin, no severe curvature — hidden until the foreskin is retracted. Retract before every newborn circumcision.
- Avoid circumcision with curvature or ventral skin deficiency (foreskin may be needed); curvature with orthotopic meatus → defer circumcision until curvature is repaired.

## 2 · EMBRYOLOGY

- Critical window **weeks 8–16**; urethral closure proceeds from the anus towards the glans tip.
- Sex-determining region Y (SRY) at 7–8 weeks → testis → Leydig testosterone → **5α-reductase type 2** → dihydrotestosterone (DHT) locally. **DHT at 8–12 weeks** is the key mediator. Sonic hedgehog drives tubercle and plate formation.
- **Endodermal "two-zipper" theory** (Kurzrock, current): plate continuous through the glans; "opening zipper" canalises plate into a groove (both sexes), then androgen-dependent "closing zipper" (male only). Older **ectodermal ingrowth theory** (Glenister): glanular urethra from ectodermal ingrowth fusing at the corona. MCQ
- Prepuce fuses dorsally first; ventral fusion awaits urethral fold fusion (≈ 20 weeks), so ectopic meatus → incomplete foreskin in **95%**.

> **KEY POINT** Hypospadias is an arrest of the androgen-dependent closing zipper: the meatus stops where closure stopped, the prepuce cannot close ventrally over an open urethra, and unfused dysplastic ventral tissue tethers the shaft into curvature.

## 3 · AETIOLOGY

- **Multifactorial**; every thread converges on insufficient androgen action in weeks 8–16.
- **Fetal/placental**: prematurity, small for gestational age, intrauterine growth restriction, placental insufficiency (≈ 2× risk); assisted reproduction (independent risk).
- **Genetic**: twin concordance 9–27%; first-degree relative 9–15%. Chromosomal abnormality in up to 30%; no single gene. Gene groups: penile development (HOX, fibroblast growth factor), gonadal (WT1, SOX9, DMRT1, GATA4), androgen cascade (LH receptor, 5α-reductase, androgen receptor). ≈ 200 syndromes: WAGR syndrome (Wilms, aniridia, genitourinary anomalies, intellectual disability) (Wilms tumour, aniridia, genitourinary anomalies, developmental delay), Denys–Drash, Smith–Lemli–Opitz.
- **Endocrine-disrupting chemicals (EDCs)**: pesticides, oestrogens/progestins, diethylstilbestrol, phthalates, phytoestrogens — compete at the androgen receptor; proven in animals, unproven in humans.
- **Testicular dysgenesis syndrome**: hypospadias, cryptorchidism, poor semen quality and testis cancer share one fetal androgen-disrupting exposure.
- **Anogenital distance (AGD)** (anus to penoscrotal junction): marker of androgen action; shorter AGD = more severe hypospadias.

## 4 · CLASSIFICATION AND SEVERITY

- By meatal position: distal (glanular, coronal, distal shaft) **70–85%**; proximal (proximal shaft, penoscrotal, scrotal, perineal) 10–25%.

- Meatus is the **least reliable** marker: a "distal" meatus may overlie absent spongiosum to the penoscrotal junction with severe curvature. **Classify after degloving**: meatus + curvature + plate.

| Score | Glans (G) | Meatus (M) | Shaft curvature (S) |
|---|---|---|---|
| 1 | Good size, deep grooved plate | Glanular | None |
| 2 | Adequate, grooved | Coronal sulcus | <30° |
| 3 | Small; narrow/flat plate | Mid or distal shaft | 30–60° |
| 4 | Very small; indistinct plate | Proximal shaft/penoscrotal | >60° |

**GMS score** (Merriman 2013): total 3–12, scored in theatre; higher → more complications. A coronal meatus with a small flat glans and 45° curvature scores worse than its position suggests.

**Plate objective scoring tool (POST)**: ratio of neomeatus length (AB) to glans fusion length (BC); low ratio = poor plate → favour inlay graft over plain tubularised incised plate (TIP) repair.

## 5 · ASSOCIATED ANOMALIES AND DSD WORK-UP

- Inguinal hernia/hydrocele 9–16%; undescended testis (UDT) ≈ 7%, ≈ 10% in proximal cases.
- **Disorder of sex development (DSD) work-up** when: UDT (uni- or bilateral) + proximal hypospadias; perineal hypospadias; atypical genitalia. Since 2006 severe hypospadias counts as 46,XY DSD. DSD risk ≈ **50%** with non-palpable vs 15% palpable testis; 64% proximal vs 7% distal.
- Work-up: ultrasound (US) for Müllerian remnants, karyotype, testosterone, follicle-stimulating hormone (FSH), luteinising hormone (LH), anti-Müllerian hormone (AMH). A 46,XY karyotype does not exclude 5α-reductase deficiency, androgen insensitivity, testicular dysgenesis or ovotesticular DSD → *Ch 59*.
- **Prostatic utricle**: commoner in proximal cases; large ones → stasis, urinary tract infection (UTI), difficult catheterisation. Most asymptomatic; excision (endoscopic/robotic) rare because the vas is at risk.
- No routine voiding cystourethrogram (VCUG) or renal US; image only with other systemic anomalies.

> **KEY POINT** Hypospadias + cryptorchidism signals a systemic androgen problem, not a local one — hence karyotype before surgery.

## 6 · PRE-OPERATIVE ASSESSMENT

- **Goals**: straight erection, urethra to the tip, acceptable cosmesis — fewest procedures, durable into adulthood. A straight penis voiding standing may gain little; counsel honestly.
- History: previous surgery, UTI, erectile curvature (parents' photographs), bleeding disorders. Examine: stretched penile length (exclude micropenis), both groins, AGD, penoscrotal transposition, torsion, scrotal cleft, meatal stenosis ("laser-beam" stream — may justify surgery even in distal cases).
- **Timing**: genital awareness ≈ 18 months. American Academy of Pediatrics (AAP, 1996): 6–12 months; authors start at 6 months within **6–18 months**. Systematic review (>16,000) → fewer complications when younger; repair by 18 months. One short anaesthetic appears low-risk for neurodevelopment.
- **Androgen stimulation**: glans width **<14 mm** → more complications. Testosterone, DHT or human chorionic gonadotrophin (hCG) enlarge glans, reducing glansplasty tension. Authors: testosterone cypionate **2 mg/kg intramuscular (IM) at 5 and 2 weeks** pre-op if glans <15 mm (≈ 4 mm gain). Two randomised trials → fewer complications; meta-analyses inconclusive. Side effects (aggression, erections, pigmentation) resolve within 6 months; concerns: bleeding, healing; severe cases may be partly androgen-insensitive.

## 7 · INTRA-OPERATIVE PRINCIPLES

- Caudal block (most popular; fistula link disputed), penile or pudendal block; ketorolac 0.5 mg/kg once. IV antibiotics at induction. Tourniquet (release every 30 min) or epinephrine.
- Cystoscopy only for severe proximal/redo (utricle may block catheter).
- Sutures: rapidly absorbed (chromic, polyglactin) beat polydioxanone; 6-0–8-0 in infants. Slow sutures leave epithelialised tracks → fistula.
- Magnification, fine handling, **tension-free** urethra/glans/skin (tension → ischaemia → fibrosis, breakdown), non-overlapping suture lines, haemostasis (haematoma kills flaps).
- **Never use hair-bearing skin** for urethroplasty: postpubertal hair → stones, recurrent UTI.
- Higher surgeon volume → fewer complications.

## 8 · CURVATURE: ASSESSMENT AND CORRECTION

Curvature, not meatal position, is the main determinant of **one-stage vs staged repair**. Causes: tethered ventral skin, dysplastic dartos/Buck's fascia, short urethra, corporal disproportion.

**ALGORITHM**
Circumscribe → deglove to penoscrotal junction, excise dysplastic dartos → **artificial erection** (Gittes–McLaughlin tourniquet + saline; or corporal compression; or prostaglandin) → straight → urethroplasty · residual <30° → dorsal plication · >30° → divide urethral plate → still >30° → ventral corporal lengthening (usually staged)

Grading: mild <30°, moderate 30–45°, severe ≥45°. Eyeballing underestimates; a tourniquet can hide basal curvature (compress corpora against pubis instead).

| Feature | Dorsal plication (shortening) | Ventral corporotomy ± graft (lengthening) |
|---|---|---|
| Technique | Nesbit ellipses; Baskin single midline 12 o'clock plication (nerve-free zone); Heineke–Mikulicz dorsal incision | Transverse ungrafted corporotomies (staged tubularised autograft, STAG) or single 9-to-3 o'clock corporotomy + graft (dermis 20% larger, tunica vaginalis, small intestinal submucosa) |
| Indication | ≤30° (authors ≤15°) | Moderate/severe after degloving and plate division |
| Risks | Neurovascular injury off midline; recurrence at puberty; shortens | Complex; limits later ventral skin; graft contraction |

Recurrence: plication 28% vs grafting 9% in one series; not dividing the plate is an independent risk. Correct fully first time — under-correction generates repeated surgery through puberty.

## 9 · URETHROPLASTY TECHNIQUES

Tenets: correct curvature → advance urethra or urethroplasty → glansplasty → skin cover ± preputioplasty. TIP is used for >90% of distal repairs.

| Technique | Indication | Key point / pitfall |
|---|---|---|
| **MAGPI** (meatal advancement and glanuloplasty; Duckett 1981) | Glanular meatus, no curvature, mobile meatus | No tubularisation; used more proximally → retrusive meatus, flat glans |
| Urethromeatoplasty | Stenotic distal meatus, blind pit | Heineke–Mikulicz |
| GAP (glans approximation) | Wide deep groove, fish-mouth meatus (MIP) | Deep glans wings, two-layer neourethra |
| Thiersch–Duplay | Wide plate, adequate glans | U-incision, tubularise over 12 Fr, dartos flap |
| **TIP / Snodgrass** (1994) | Distal and midshaft; plate too narrow to tubularise; mild curvature | Deep midline plate relaxing incision; stop 2–3 mm short → slit meatus; risk meatal stenosis; add inlay graft if poor plate |

| Technique | Indication | Key point / pitfall |
|---|---|---|
| **Mathieu** (flip-flap) | Distal, shallow/deficient plate | Parameatal flap; narrow base → ischaemia, fistula; fish-mouth meatus |
| **Onlay island flap** | Plate kept but too narrow/thin; curvature corrected by degloving | Inner preputial island flap; diverticulum if flap too wide; less stenosis |
| **Duckett tube** (transverse preputial island flap) | Proximal; plate divided for curvature | Single-stage tube; higher complications; anchor flap edge to avoid diverticulum |
| **Two-stage Bracka** (free graft) | Proximal with severe curvature, redo, poor plate | Stage 1: correct curvature, divide plate, quilted inner preputial/buccal graft; stage 2 (≥6 months): tubularise |
| **Two-stage Byars** (pedicle flaps) | Proximal with severe curvature | Split dorsal prepuce transposed ventrally; highest staged complication rates (fistula, diverticulum) |

- Variants: **STAG** (ungrafted corporotomies + graft; lowest reported proximal complications, ≈ 23%); STAC (no graft at stage 1, graft stage 2, tubularise stage 3 — authors' choice); distal-first/Ulaanbaatar.
- **Why stage**: a urethroplasty laid over a fresh corporotomy/graft fails more often; stage 1 creates supple ventral tissue. Always repeat artificial erection at stage 2.
- Proximal TIP abandoned: plate mobilisation devascularises and more recurrent curvature.

> **KEY POINT** Why proximal repairs fail — Poiseuille: resistance ∝ length / radius⁴. A long, rigid, spongiosum-free skin tube ending in a fixed non-distensible glans raises proximal pressure → fistula, diverticulum, stricture. 25–75% need extra procedures after planned two-stage repair.

### Glansplasty and coverage

- Deep glans wing mobilisation, tension-free **two-layer glans closure**, mucosal collar. Glans tension → dehiscence.
- **Barrier layer** of vascularised non-epithelialised tissue over the neourethra reduces fistula: de-epithelialised dorsal preputial dartos (Belman flap), ventral dartos, **tunica vaginalis** (mobilise to external ring to avoid tethering), spongiosum. Offset suture lines; must not compress glans.

## 10 · DRESSING, STENTS AND POST-OPERATIVE CARE

- Compressive dressing (circumferential Tegaderm including glans, or "sandwich" dressing) for ≈ **72 h**; then petroleum jelly.
- Diversion **3–21 days** by complexity; soft dripping stent; suprapubic tube for older/redo. Short distal repairs can be unstented.
- Oxybutynin until stent removal (bladder spasm → voiding around stent). Stiff stents/laterally pulled Foley → glans dehiscence.
- Paracetamol + ibuprofen regularly (non-steroidal anti-inflammatory drugs do not increase bleeding). Antibiotic prophylaxis: evidence mixed.

## 11 · COMPLICATIONS AND MANAGEMENT

| Complication | Standard definition |
|---|---|
| Fistula | Urethral leak anywhere below meatus |
| Glans dehiscence | Wing separation → coronal/proximal meatus; glans fusion <2 mm |
| Meatal stenosis | Obstructive symptoms + calibration **<8 Fr** prepubertal / <12 Fr postpubertal |
| Diverticulum | Segmental sacculation on voiding |
| Recurrent curvature | **>30°** on erection |

- **Urethrocutaneous fistula** — commonest (just under 10%), usually at original meatus or coronal margin. Causes: ischaemia, haematoma, infection, overlapping suture lines and especially **distal obstruction**. Rarely closes spontaneously. Repair ≥ **6–12 months** after surgery: calibrate (bougie à boule) ± urethroscopy, stress test with saline/methylene blue → excise tract, close in offset layers with dartos/tunica vaginalis interposition. Coronal fistula → redo urethroplasty and glans closure, not simple excision. Recurrence ≈ 20%.
- **Meatal stenosis**: dilatation ± topical steroid → meatotomy or redo urethroplasty. Commoner after TIP than Mathieu.
- **Glans dehiscence**: reoperate only for voiding symptoms; inlay graft avoids glans tension.
- **Urethral stricture** (often late): cystoscopy; dilatation/urethrotomy rarely durable → revision urethroplasty with buccal mucosa, often staged.
- **Diverticulum** (after flaps, staged, proximal): excise redundant epithelium, reduce to uniform calibre, keep dartos as extra layer; always exclude distal obstruction.
- **Balanitis xerotica obliterans (BXO)/lichen sclerosus**: white scarring of meatus/neourethra. Topical clobetasol/betamethasone usually fails → excise all involved tissue (histology), staged buccal or inner preputial graft. Recurrence up to **40%**; BXO can invade buccal grafts.
- **Recurrent curvature**: under-correction, graft contraction, skin tethering, pubertal disproportion. Full degloving + artificial erection; staged ventral lengthening, buccal graft, Cecil modification if skin short.
- **Hairy urethra**: hair + narrowing → stones, UTI → laser epilation, endoscopic stone removal, or resect and replace with buccal graft.

> **KEY POINT** Nearly every urethroplasty complication traces to distal obstruction, ischaemia or tension — so fistula/diverticulum repair begins by calibrating the distal urethra.

### Reoperation

- Each redo degrades tissue; Devine: avoid "small operations" — wide scar excision and tissue transfer instead. Always cystoscope and do artificial erection.
- **Buccal mucosa** = graft of choice (hairless, wet-adapted, little contraction); **lip mucosa** preferred within the glans (thinner). Bladder mucosa abandoned (meatal prolapse). Graft 10% larger, quilted, on preserved dartos; urethroplasty ≥6 months later. Dorsal inlay beats onlay or tube in redo.
- **Cecil–Culp**: bury repaired penis in scrotum to parasitise its blood supply; modified release at 9–12 months.
- Adjuncts (unproven): topical nitroglycerin, hyperbaric oxygen, tissue expanders.

## 12 · OUTCOMES AND FOLLOW-UP

- TIP meta-analysis (Pfistermuller 2015, 4,675 boys): complications 10.6%, reoperation 4.5%; fistula > stenosis > stricture; dartos layer improves results.
- Distal success 83–95%; proximal complications 23–68%.
- Only ≈ 50% of complications appear in year 1, ≈ 75% by 3 years; fistula early, stricture/curvature late → follow proximal cases **through puberty** (stream after toilet training; curvature, sexual function at puberty).
- Recurrent UTI after repair (≈ 2%) → image for diverticulum, stricture, utricle.
- Instruments: Hypospadias Objective Scoring Evaluation (HOSE; surgeon), Penile Perception Score (patient/parent), Hypospadias Objective Penile Evaluation (HOPE; photo panels; cosmetic only).

### Adult issues

- Untreated mild distal hypospadias: minimal sexual, fertility or voiding problems.
- Proximal repairs: worse satisfaction, sensation and fertility; ejaculatory problems; concern about penile length (chief adult complaint); sexual inhibition and lower self-esteem but no major psychopathology. Even 20–30° curvature can impair intercourse.

## 13 · MCQ ONE-LINERS

1. Hypospadias triad: ventral ectopic meatus, ventral curvature, dorsal hooded foreskin.
2. Critical window of urethral development: weeks 8–16; DHT at 8–12 weeks is key.
3. Current embryological model: endodermal "two-zipper" theory (Kurzrock).
4. Megameatus intact prepuce: ≈ 5%; found on foreskin retraction.
5. Distal hypospadias forms 70–85% of cases.
6. GMS score ranges 3–12; higher predicts complications.
7. Hypospadias + undescended testis → karyotype; DSD ≈ 50% if testis non-palpable.
8. Glans width <14 mm increases complication risk.
9. Pre-op testosterone cypionate 2 mg/kg at 5 and 2 weeks before surgery.
0. Recommended repair age: 6–18 months.
1. Residual curvature <30° after degloving → dorsal plication.
2. Curvature >30° after plate division → ventral corporotomy ± graft, usually staged.
3. MAGPI suits a glanular meatus with no curvature.
14. TIP: midline plate incision; commonest late problem meatal stenosis.
15. Mathieu flip-flap: fewer meatal stenoses than TIP, similar fistula rate.
16. Onlay island flap complication typical: diverticulum.
17. Duckett tube = transverse preputial island flap tube.
18. STAG has the lowest reported proximal complication rate (≈ 23%).
19. Byars two-stage flaps: highest complication rate among staged options.
20. Poiseuille: resistance ∝ length / radius⁴ — explains proximal failures.
21. Commonest complication: urethrocutaneous fistula.
22. Coronal fistula → redo urethroplasty, not simple closure.
23. Meatal stenosis: symptoms + <8 Fr prepubertal.
24. Recurrent curvature defined as >30°.
25. BXO recurrence after excision and buccal graft: up to 40%.
26. Graft of choice for redo: buccal mucosa; lip mucosa within glans.
27. Only half of complications appear within 1 year.

# 57 Etiology, Diagnosis and Management of the Undescended Testis

*Reading time ≈ 53 min (slow pace)*

## 1 · DEFINITIONS

| Term | Definition / key point |
|---|---|
| **Undescended testis (UDT) / cryptorchidism** | Testis not in normal scrotal position (midpoint at or below mid-scrotum); clinically palpable vs non-palpable; non-palpable is either undescended or absent |
| High scrotal testis | Stable testis above scrotal midpoint — a mixed group (stable, retractile, "gliding"); not routinely called undescended |
| Ectopic testis | Off the line of descent; commonest site **superficial inguinal pouch (SIP)**; others perirenal, perineal, femoral, prepubic, penile, transverse/contralateral scrotal |
| **Retractile testis** | Retracts but can be brought into the scrotum and stays there at least temporarily; one that rarely stays down or rests high may progress to acquired UDT |
| Acquired (ascending) | Previously documented scrotal; usually diagnosed at 8–11 years |
| Recurrent | Undescended at birth → descended spontaneously → re-ascended (10–22%, at 1–5 years) |
| Secondary | Suprascrotal after inguinal hernia repair (primary maldescent or scarring) |
| Testicular retraction | Suprascrotal testis as a complication of orchidopexy (iatrogenic) |
| Vanishing testis | Formed, then lost (perinatal torsion/vascular accident): monorchia; bilateral = anorchia |
| Agenesis | Never formed; associated with ipsilateral Müllerian persistence (extremely rare) |

## 2 · EMBRYOLOGY OF DESCENT

- SRY (sex-determining region Y) + steroidogenic factor 1 (SF1) → SOX9 → Sertoli cells → testis cords. Fetal Leydig cells active from 6–7 weeks, involute after 24 weeks. Each step licenses the next: only a hormonally competent testis can drive its own descent.
- Two Leydig hormones control descent: **insulin-like 3 (INSL3)** via its receptor RXFP2 (relaxin family peptide receptor 2), peak 15–17 weeks, and **androgens** via the androgen receptor, testosterone peak 14–16 weeks (placental human chorionic gonadotrophin [hCG]-driven, then fetal luteinising hormone). Anti-Müllerian hormone (AMH) from Sertoli cells regresses Müllerian ducts at 9–10 weeks. Sertoli cells lack androgen receptor early — otherwise androgen would suppress AMH and trigger premature spermatogenesis.
- The gubernaculum attaches to the epididymis; it does not pull the testis. Humans have no cranial gonadal ligament (unlike rodents). The scrotum is formed by dihydrotestosterone at 10–12 weeks, long before the testis arrives.

| Phase | Timing | Event | Control |
|---|---|---|---|
| Transabdominal (relative) | 10–15 weeks | Testis stays near internal ring (from 10 weeks) while gubernaculum swells from 12 weeks (outgrowth); cranial mesonephric ligament regresses by 13 weeks | Classically INSL3 ("non-androgenic factor"); now INSL3 + androgen synergistic |

| Phase | Timing | Event | Control |
|---|---|---|---|
| Inguinoscrotal | **20–28 weeks** | Gubernacular swelling + cremaster development widen the canal, processus vaginalis forms; distal gubernaculum released → rapid transinguinal passage (rarely before 22 weeks) → gubernaculum regresses; most scrotal after 27 weeks | Androgens acting via **genitofemoral nerve (GFN)**, released **calcitonin gene-related peptide (CGRP)**; cremaster myogenesis (Wnt pathways) |

> **KEY POINT** Outgrowth creates the canal; regression lets the testis follow — so a defect in either produces the same phenotype. INSL3 and androgen act on one target (gubernacular/cremaster myogenesis), hence one clinical picture whichever fails.

**Minipuberty** (hypothalamic–pituitary–gonadal [HPG] axis surge after maternal hormone withdrawal, peak 1–3 months, earlier and stronger in preterm boys): luteinising hormone (LH), follicle-stimulating hormone (FSH), testosterone, AMH and inhibin B (INHB) rise; Sertoli cells proliferate (sets final testis size); gonocytes migrate to the basement membrane and transform into **adult dark (Ad) spermatogonia** (stem-cell pool). Failed transformation leaves gonocytes prone to apoptosis or malignant change. This is the window in which a positional defect becomes a germ cell deficit.

## 3 · EPIDEMIOLOGY, AETIOLOGY AND ASSOCIATIONS

- Prevalence **2–4% at birth** (term) → **1–2% at 3 months**; up to 45% in preterm. Spontaneous descent mainly in first 3–6 months; rarely after 6 months (none in referred boys presenting after 6 months). High scrotal testes descend far more often (≈ 88%) than extrascrotal ones (≈ half).
- 80–85% isolated (non-syndromic). Risk factors: **prematurity / low birth weight**, family history (twins > brothers > sons; maternal half-brothers > paternal — implies maternal environment or X-linked factors), maternal diabetes, smoking; diethylstilbestrol (DES) exposure doubles risk (early, high dose); first-trimester corticosteroids in assisted reproduction; endocrine-disrupting chemicals (EDCs) — mixture effects in susceptible fetuses, unproven singly.
- Genetics: polygenic threshold trait; rare INSL3, RXFP2, NR5A1 variants (homozygous variants with bilateral disease suggest recessive inheritance).
- Testicular dysgenesis syndrome: cryptorchidism + hypospadias + infertility + germ cell tumour from impaired fetal testis function. Reduced anogenital distance marks low prenatal androgen; low placental weight also associated.
- Hormones: in non-syndromic UDT, primary gonadal (Sertoli/Leydig) dysfunction — raised LH/testosterone ratio ("compensated Leydig function"), low AMH/INHB in bilateral cases — is better supported than central hypogonadism. No hormone predicts germ cell number, so testing has no routine role outside disorders of sex development (DSD) and anorchia.

| Associated condition | Cryptorchidism | Implication |
|---|---|---|
| Prune-belly syndrome | All (bilateral abdominal) | Abdominal orchidopexy |
| Spigelian hernia | 53–80% | Absent gubernaculum/canal complicates orchidopexy |
| Cerebral palsy | 25–54% | Often bilateral (central/global defect) |
| Gastroschisis | 5–25% | **50–55% descend spontaneously — observe** |
| Omphalocele, imperforate anus, myelomeningocele, posterior urethral valves | 12–33% | Caudal-field anomalies ipsilateral in 90% |
| Congenital diaphragmatic hernia | 7–30% | Always ipsilateral |
| Klinefelter (47,XXY) | > 50% of prepubertal Klinefelter boys | < 2% of non-syndromic UDT; more with hypospadias |

| Associated condition | Cryptorchidism | Implication |
|---|---|---|
| Persistent Müllerian duct syndrome (PMDS) | Bilateral 59%; hernia uteri inguinalis; transverse testicular ectopia 22% | AMH or AMHR2 mutation; adult infertility ≥ 80%, malignancy risk |

### Congenital vs acquired

| Feature | Congenital | Acquired |
|---|---|---|
| Diagnosis | Birth/neonatal | Mean 8–11 years |
| Position | Often higher | Lower; often SIP |
| Processus vaginalis | Usually patent (87% unilateral) | Usually closed (patent 45–50%) |
| Epididymal anomalies | 37–60% | 11–31% |

Acquired UDT is probably a never-fully-descended testis unmasked by somatic growth: it shows reduced Ad spermatogonia and semen outcomes like congenital cases. Retractile testes share pathology (small volume, patent processus, epididymal anomalies) and carry a risk of ascent (prospective ≈ 2–3%; higher in referral series) → **annual examination through childhood**, especially after spontaneous descent or with retractile testes. Comprehensive screening (Japan) reduces late surgery.

## 4 · HISTOLOGY, ASSOCIATED ANOMALIES AND FERTILITY

- Germ cell loss begins **after infancy** and progresses; contralateral descended testis also affected to a lesser degree. The UDT is smaller at birth and grows less even after spontaneous descent — an intrinsic defect, worsened by position (the only part the surgeon controls).
- Normal spermatogonia per tubule (S/T): 2.5 at birth, nadir 1.2 at 2–3 years — needs age-specific reference.
- Delayed gonocyte disappearance and **Ad spermatogonia** formation; Ad absent in 7–85%; primary spermatocytes (normally at 4–5 years) delayed. Sertoli cell number falls after 4 months; reduced type IV collagen in both testes.
- Similar pathology whether ectopic, SIP, abdominal or ascending (some studies worse if abdominal). Higher germ cell counts with younger age at surgery, palpable testis, less fibrosis.
- Associated anomalies: **patent processus** in 87% unilateral / 71% bilateral UDT; aberrant lateral gubernacular attachment 66–75%; epididymal non-fusion > elongation/looping > atresia (track with severity). Complete testis–epididymis non-union is rare, mainly abdominal. UDT is at higher torsion risk — can mimic an incarcerated hernia and occur during infant observation.

| Anomaly | Key features | Management |
|---|---|---|
| Polyorchidism | Supernumerary testis, usually left, scrotal 75%; Kumar classification by ductal drainage (A with vas, B without) | Observe normal scrotal testis; orchidopexy if undescended with intact drainage |
| Splenogonadal fusion | Spleen–gonad union, left; continuous form with limb defects, micrognathia | Recognise at surgery — avoid needless orchidectomy |
| Transverse testicular ectopia | Both testes to one side; hernia + contralateral non-palpable testis; PMDS in 20–50% | Laparoscopy; trans-septal placement if vasa fused |

| Trial | Design | Result | Take-home |
|---|---|---|---|
| **Kollin et al.** | Randomised controlled trial: orchidopexy at 9 months vs 3 years | Early group: higher Sertoli and germ cell counts, better testicular growth (to 4 years) | Operate in first year; age at surgery, not hormone levels, predicts germ cells |

**Fertility outcome**

| Outcome | Former unilateral | Former bilateral |
|---|---|---|
| Oligospermia / azoospermia | 31% / 14% | 42% / 31% |
| Paternity | **≈ 90% of those attempting — same as controls** | ≈ 65% of those attempting (≈ 1/3 overall) |

- Cryptorchidism accounts for 10–17% of men evaluated for infertility.
- Unilateral: the normal testis carries fertility; semen overstates infertility; paternity after orchidectomy of the affected side equals controls. Bilateral: semen reflects true infertility.
- Ad spermatogonia at biopsy predict adult counts (present in both → normal count in 84%; absent → subnormal in all); Ad/tubule > 0.2 predicts fertility better than S/T. In bilateral cases INHB/FSH and AMH/FSH ratios help counselling.
- Even after first-year surgery, 20–25% remain at infertility risk (intrinsic defect).
- Population study: surgery **before 18 months** reduced need for assisted reproduction; each 6-month delay increased it by 5%.
- Orchidopexy before 19 years improves testicular sperm retrieval in azoospermia; retrieval is better than in idiopathic non-obstructive azoospermia.
- Uncomplicated retractile testis: insufficient evidence of infertility. Acquired UDT: outcomes like congenital.

## 5 · MALIGNANCY

- Testicular germ cell tumour (TGCT) risk **2–5×** (pooled odds ratio ≈ 4); ≈ 1% of cryptorchid men; 5–10% of TGCT patients have a history of UDT. Contralateral testis: odds ratio ≈ 1.5.
- **Swedish population study**: cryptorchidism odds ratio 3.2 — ≈ 2.8 if diagnosed before puberty vs ≈ 5.0 after. † Swedish orchidopexy cohort (Pettersson 2007): relative risk 2.23 if operated before 13 years vs 5.40 at ≥ 13 years. Meta-analysis: orchidopexy after 10–11 years → relative risk **5.8**.
- Risk rises with age at orchidopexy (untreated adults odds ratio ≈ 14) and with each 6-month delay beyond 18 months. Early surgery reduces the positional component; the intrinsic (contralateral) excess remains.
- Persistently cryptorchid → **seminoma** (74%); after orchidopexy → mostly non-seminoma (63%).
- Precursor germ cell neoplasia in situ (GCNIS, formerly intratubular germ cell neoplasia/carcinoma in situ) arises from gonocytes that fail to differentiate: 2–3% of formerly cryptorchid men, ≤ 0.4% of non-syndromic children. Higher risk with DSD/chromosomal defects — biopsy these.
- Does biopsy at orchidopexy raise TGCT risk? A cohort suggesting so was confounded by biopsy indication; a Scandinavian cohort with universal biopsy found no excess — biopsy risk is theoretical (no rise in microlithiasis or antisperm antibodies).
- Contralateral GCNIS occurs in 4–8% of men with TGCT; predictors are atrophy, cryptorchidism and age < 40.
- Postpubertal UDT with normal contralateral testis: **orchidectomy** preferred (up to age 50) — poor spermatogenesis, GCNIS and torsion risk; prosthesis at same sitting.
- **Testicular microlithiasis**: 3–6% of boys, similar after cryptorchidism, not clearly an independent TGCT risk factor → routine ultrasound (US) surveillance not recommended.
- Counsel on infertility and cancer risk and teach testicular self-examination (cost–benefit favours it in this group).

## 6 · DIAGNOSIS

### Examination — the gold standard

**ALGORITHM**
Warm room, warm hands, distraction; supine (then cross-legged, squatting or standing if needed); thigh abduction inhibits cremaster; lubricant/liquid soap on fingers → sweep from anterior superior iliac spine along the canal towards the scrotum with one hand → palpate with the other for the lowest attainable position → while compressing the canal, invaginate scrotum to feel SIP/distal testis → sustained gentle traction fatigues cremaster: **stays in scrotum = retractile; springs back = undescended**.

- Document palpability, position, size, mobility, lie, hernia/hydrocele, penile size, meatus. Scrotal asymmetry supports unilateral UDT.
- ≈ **80% palpable**; 60–70% unilateral. Non-palpable at surgery: abdominal/peeping 25–50%, vanished 15–40%, extra-abdominal missed 10–30%.
- Contralateral hypertrophy (volume > 2 mL: specificity 100%; length ≥ 1.8–2 cm) and absence of scrotal appendage tissue suggest monorchia — useful for counselling, not a reason to skip exploration. Vanished testis = torsion/vascular accident after masculinisation (haemosiderin in nubbins).
- Interobserver agreement is poor; preoperative position correlates poorly with operative findings; examination under anaesthesia (EUA) is the final arbiter.

| AUA algorithm step | Action |
|---|---|
| Bilateral non-palpable | Specialist evaluation for DSD |
| Otherwise, < 6 months corrected age | Re-examine at 6 months |
| Still undescended at 6 months | Refer to surgical specialist |
| EUA: retractile / palpable UDT | Yearly review / scrotal or inguinal orchidopexy |
| EUA: non-palpable, nubbin ± contralateral hypertrophy | Consider inguinal/scrotal exploration |
| EUA: non-palpable otherwise | Abdominal exploration (laparoscopic or open) |

### Imaging — not recommended

American Urological Association (AUA) 2014: **US or magnetic resonance imaging (MRI) is not recommended**. US sensitivity ≈ 45%, specificity 78%; MRI 65%/100% (needs sedation). A positive scan does not avoid surgery and a negative scan does not exclude a testis — the only effect is a delay of ≈ 3 months. Possible MRI use: ectopic abdominal testis not found at laparoscopy.

### Bilateral non-palpable or UDT with hypospadias → DSD work-up

- Bilateral non-palpable newborn (especially with abnormal phallus): **urgent karyotype and hormones** to exclude **congenital adrenal hyperplasia (CAH)** (undiagnosed salt-wasting); pelvic US for Müllerian structures; defer circumcision.
- Hypospadias coexists in 12–24%; proximal hypospadias + UDT → chromosomal abnormality in 32–47% → *Ch 56*.
- Micropenis + bilateral UDT: hypogonadotrophic hypogonadism or anorchia (micropenis in nearly half of anorchia) — measure LH, FSH, testosterone during minipuberty → *Ch 55*.

| Marker | Anorchia | Bilateral abdominal UDT |
|---|---|---|
| FSH, LH | Raised | Normal |
| **AMH, INHB** (best prepubertal Sertoli markers) | Undetectable/low | Present |
| Basal testosterone | Low in first year only | Normal |
| Testosterone after hCG stimulation | No rise | Rises |

hCG stimulation is poorly standardised and is no longer the test of choice; gonadotrophins may be normal in mid-childhood anorchia (quiescent axis). If hormones prove testicular tissue, it still has to be found: **laparoscopy remains the confirmatory test**. In a genetic male with bilateral non-palpable testes, ≥ 95% are abdominal.

## 7 · MANAGEMENT PRINCIPLES AND TIMING

- Aims: optimise fertility, reduce/allow detection of malignancy, cosmesis, prevent torsion and hernia.
- **Timing**: observe to **6 months** (corrected age) → refer → orchidopexy **6–18 months**, preferably by 12 months. Rationale: spontaneous descent unlikely after 6 months; better testicular growth after surgery at 9 months than 3 years; abdominal orchidopexy easier soon after minipuberty. Preterm: observe to 6 months beyond due date (1 year if position marginal).
- Most children worldwide are still operated after 2 years (late referral, acquired cases, undetected re-ascent, confusion with retractile testes).
- Retractile testis: no surgery; examine yearly.
- Acquired UDT: operate. Observation series show spontaneous pubertal descent in many (mostly high scrotal/retractile testes), but without paternity or cancer data observation cannot be recommended, especially bilateral.
- Pubertal/postpubertal with normal contralateral testis: orchidectomy (laparoscopic if abdominal) ± prosthesis.

### Hormonal therapy — not recommended (AUA)

- Luteinising hormone-releasing hormone (LHRH) intranasal: descent ≈ 20%; hCG: limited, never placebo-tested. Neither reliably distinguishes retractile from undescended or removes the need for re-examination.
- hCG may increase germ cell apoptosis and reduce adult volume (infertility relative risk 4.7 after hCG in one paternity series).
- Adjuvant LHRH (buserelin 10 µg intranasally alternate days × 6 months; Hadziselimovic) for bilateral UDT with absent Ad spermatogonia, on the theory of "defective minipuberty": pilot randomised data only; candidates need biopsy; not widely used.

> **KEY POINT** Hormones fail on both claims for one reason: position, not endocrinology, is the modifiable variable — descent is not androgen-limited after fetal life, and germ cells track age at scrotal placement.

## 8 · SURGERY FOR THE PALPABLE TESTIS

### Inguinal orchidopexy — key steps

**ALGORITHM**

EUA confirms palpability and lowest position → skin-crease incision superolateral to pubic tubercle → look for testis in SIP during subcutaneous dissection → open external oblique from external ring, sparing ilioinguinal nerve → free testis and divide gubernaculum distal to the sac (protect a long-looping vas) → incise internal spermatic fascia, **separate the patent processus from vas and vessels without skeletonising them and high-ligate at internal ring** (the main length-gaining step) → if needed: divide lateral spermatic fascial bands → incise transversalis fascia for retroperitoneal dissection → **Prentiss manoeuvre** (route medial to inferior epigastric vessels) → transverse scrotal incision, **subdartos pouch**, tunnel anterior to pubis → pass testis without twisting the cord → excise appendages, inspect epididymis, record volumes → close dartos around cord incorporating tunica vaginalis edge (transalbugineal sutures unnecessary — 98.7% success without) → local infiltration, ilioinguinal or caudal block.

- Cannot reach scrotum → staged procedure; orchidectomy only for atrophic/abnormal testis, postpubertal patient, or insufficient vas.

- Biopsy: not routine (does not change management); indicated in DSD/ambiguity or dysgenesis; consider > 12–13 years at orchidopexy and in bilateral UDT (Ad spermatogonia).

### Scrotal (Bianchi) orchidopexy

Single high scrotal-border incision (Bianchi–Squire; low transverse or midline for testes drawable into scrotum) → deliver testis → dissect sac and cremaster high above the canal, ligate processus through the scrotal wound → fixation in subdartos pouch. Inguinal conversion 3.5–4.4% (for sac ligation or cord length); atrophy 0.3–0.6%; less pain than inguinal; efficacy similar — increasingly first choice for palpable (especially low) testes. Same goal as inguinal: only the starting point differs.

## 9 · THE NON-PALPABLE TESTIS

**ALGORITHM**

EUA — testis now palpable? → inguinal/scrotal orchidopexy. **Still non-palpable → diagnostic laparoscopy** — inspect internal ring patency, vessels, vas, testis size and position:

• **Blind-ending spermatic vessels** (± vas) above a closed ring = vanished testis → stop (only unequivocal endpoint).

• **Vessels + vas entering a closed or open ring** → inguinal/scrotal exploration; excise the nubbin (viable tubules in 5–15%).

• **Intra-abdominal testis** → laparoscopic orchidopexy: primary (vessels long enough) or Fowler–Stephens (one or two stage); orchidectomy if postpubertal or dysgenetic.

• Peeping testis at ring → laparoscopic or inguinal orchidopexy.

• Neither vas nor vessels seen → search perivesical area and retroperitoneum to the kidney (agenesis extremely rare; perirenal testis may accompany renal dysplasia/agenesis).

- Pitfalls: an open ring/hernia often (not always) means a viable distal testis; apparently blind-ending vessels may coexist with a distal or ectopic testis — if the view is equivocal, keep exploring (add ports).
- Palpable scrotal nubbin + contralateral hypertrophy (≥ 1.8 cm) → scrotal exploration first is acceptable (laparoscopy then needed in ≈ 1/4); futile if the vanished testis is intra-abdominal, and the "nubbin" may be a dissociated epididymis.
- Contralateral fixation of a solitary testis: debated — fixation removes the risk of losing the only testis (bell-clapper possible), but prenatal torsion does not imply contralateral risk; AUA algorithm does not require it.

### Laparoscopic orchidopexy

**ALGORITHM**

Empty bladder and stomach → open (Hasson) umbilical 5-mm port, carbon dioxide ($CO_2$) 8–12 mmHg → inspect both rings → 2–3-mm working ports triangulated to the ring → divide gubernaculum distal to the ring (use it as a handle, minimal cautery) → incise peritoneum lateral to vessels and distal to vas → mobilise vessels proximally **preserving the peritoneal triangle between vas and vessels** (keeps Fowler–Stephens possible) → new hiatus medial to inferior epigastric vessels and lateral to medial umbilical ligament (or via internal ring) → trans-scrotal clamp/port brings testis to subdartos pouch; avoid tension that avulses vessels.

- Reaching the opposite internal ring is an unreliable test of adequate length. Ring closure unnecessary. Contralateral patent processus in ≈ 9% — repair optional.
- Primary success > 90–97%. Laparoscopy also suits bilateral cases, abdominal wall defects, polyorchidism, splenogonadal fusion and transverse ectopia. Robot-assisted: feasible, no clear advantage.

### Fowler–Stephens (FS)

- Principle: divide/clip the testicular vessels; testis survives on collaterals from the **deferential (vasal) artery** and cremasteric/gubernacular vessels. For testes not within 2–4 cm of the internal

ring; a long-looping vas helps but is not required.

- **Single-stage**: clip/coagulate and bring down at once. **Two-stage**: clip vessels high (1.5–3 cm above testis; as proximal as possible) → wait ≈ **6 months** for collaterals → mobilise on a wide peritoneal pedicle. Two-stage slightly favoured (better position, similar atrophy).
- Keep peritoneum over vasal vessels intact; **spare the gubernaculum** (supplies collaterals in 60%) — lowers atrophy.
- Success: FS 60–97%; primary procedures more reliable — avoid vessel division when possible (it reduces germ cells, especially in young boys). Retrovesical testis with short vas may need orchidectomy.
- Alternatives: **Shehata traction** (testis fixed to abdominal wall near contralateral anterior superior iliac spine; respiration lengthens vessels; orchidopexy after 12 weeks; success 84–85%, atrophy 10%) keeps the artery; Dessanti traction (PTFE [polytetrafluoroethylene]-wrapped cord on scrotal traction); **microvascular autotransplantation** (success ≈ 96%; for the solitary abdominal testis; needs microsurgical skill and time).
- Bilateral high testes: stage sides ≈ 6 months apart to avoid bilateral atrophy.
- Open transabdominal (Jones/Bagley): muscle-splitting approach at internal ring, open peritoneum, free vas and vessels, tunnel to scrotum.

| Approach | Success |
|---|---|
| Primary laparoscopic | > 90–97% |
| Primary open (Jones/Bagley transabdominal) | 89–95% |
| One-stage / two-stage FS | 63% / 68% (series range 60–97%) |
| Microvascular autotransplant | ≈ 96% |

> **KEY POINT** Every abdominal technique answers one question: how to gain length without losing blood supply. FS buys length by sacrificing the testicular artery; traction buys it over time while keeping it — so vessel division is the last resort.

## 10 · COMPLICATIONS AND FOLLOW-UP

| Complication | Cause / prevention | Management |
|---|---|---|
| **Atrophy** (≈ 2% in modern series; atrophy/failure 24% for abdominal/peeping historically vs 10% distal) | Vessel injury, skeletonisation, FS, excess tension | No further surgery; offer **prosthesis** ≥ 6 months after scrotal surgery or after puberty, inguinal route, fixed to dartos with purse-string fascial closure |
| **Retraction / non-scrotal position** (≈ 2–10%) | Inadequate retroperitoneal mobilisation, unligated processus | Redo orchidopexy: prescrotal → scrotal approach; otherwise approach cord laterally through unscarred plane (Redman) or keep a patch of external spermatic fascia on the cord (Cartwright); ligate processus, retroperitoneal mobilisation. Results match primary surgery |
| **Vas injury** | Long-looping vas cut with gubernaculum; dissecting sac off vas | Divide gubernaculum distal to sac; avoid skeletonising |
| Torsion after orchidopexy | Rare | Prevented by extravaginal subdartos fixation |
| Laparoscopic | Bladder/vessel injury, hypercapnia, bowel obstruction | Open (Hasson) entry, bladder drainage |

**Follow-up**: examine at ≥ **6 months** for position and size (US by some after FS); long-term counselling on fertility and cancer; teach self-examination from puberty; semen analysis when relevant.

## 11 · MCQ ONE-LINERS

1. Cryptorchidism prevalence: 2–4% at term birth, 1–2% at 3 months.
2. Commonest ectopic site: superficial inguinal pouch.
3. Retractile testis stays in scrotum after cremaster fatigue; observe yearly.
4. INSL3 (Leydig cell) acts on gubernacular RXFP2 receptor.
5. Inguinoscrotal descent: 20–28 weeks, androgen via genitofemoral nerve and CGRP.
6. Gubernaculum attaches to the epididymis, not the testis.
7. Gonocyte → Ad spermatogonia transformation occurs during minipuberty (1–3 months).
8. Kollin randomised trial: orchidopexy at 9 months beats 3 years for testicular growth.
9. About 80% of UDTs are palpable.
10. Patent processus vaginalis in 87% of unilateral UDT.
11. Imaging (US/MRI) not recommended for UDT (AUA 2014).
12. Bilateral non-palpable testes in newborn: exclude CAH urgently.
13. Anorchia: high FSH/LH, undetectable AMH and INHB, no hCG response.
14. Contralateral testis > 2 mL suggests monorchia but exploration still needed.
15. Orchidopexy timing: 6–18 months, ideally by 12 months.
16. Hormonal therapy for descent: not recommended; LHRH success ≈ 20%.
17. Laparoscopy blind-ending vessels: stop, vanished testis confirmed.
18. Vessels entering internal ring: inguinal exploration and nubbin excision.
19. Fowler–Stephens testis survives on deferential artery collaterals.
20. Second stage of Fowler–Stephens at about 6 months.
21. Preserve peritoneum between vas and vessels to keep Fowler–Stephens possible.
22. Prentiss manoeuvre: pass cord medial to inferior epigastric vessels.
23. Shehata technique: staged traction preserving testicular vessels.
24. Gastroschisis-associated UDT: observe; 50–55% descend.
25. Congenital diaphragmatic hernia-associated UDT is always ipsilateral.
26. Paternity after unilateral UDT ≈ controls; bilateral markedly reduced.
27. TGCT risk 2–5×; uncorrected UDT tumour is usually seminoma.
28. Orchidopexy after puberty: TGCT risk roughly double that of prepubertal surgery.
29. Postpubertal UDT with normal contralateral testis: orchidectomy.
30. Microlithiasis is not an independent TGCT risk; no routine US surveillance.
31. PMDS: AMH or AMHR2 mutation; transverse testicular ectopia.
32. Agenesis is associated with ipsilateral Müllerian duct persistence.

# 61 Urologic Considerations in Pediatric Renal Transplantation

*Reading time ≈ 18 min (slow pace)*

## 1 · CAUSES OF END-STAGE KIDNEY DISEASE (ESKD) IN CHILDREN

- Children needing kidney replacement therapy (KRT) more often have **congenital anomalies of the kidney and urinary tract (CAKUT)**: hypoplasia/aplasia/dysplasia, then obstructive uropathy; acquired: focal segmental glomerulosclerosis. Commonest age at transplant **13–17 years**.
- Younger: dysplasia or obstructive uropathy — mainly **posterior urethral valves (PUV)** in boys. Older: nephrological causes, neuropathic bladder.
- The urologist's task: the condition that destroyed the native kidneys must not destroy the graft. Join the team **before ESKD**, not reactively.

> **KEY POINT** **Five goals**: unimpeded drainage · capacious low-pressure reservoir with social continence · volitional emptying · no infection · fewest operations. Chronic rejection is unavoidable, but every added insult accelerates it.

## 2 · PRE-TRANSPLANT UROLOGICAL EVALUATION

- **Basic screen (all)**: history + renal ultrasound (US) + uroflow with post-void residual (PVR) if toilet-trained.
- **Voiding cystourethrogram (VCUG)** not routine; do it with neuropathic bladder, obstructive uropathy, vesicoureteral reflux (VUR), prior lower-tract surgery, recurrent febrile urinary tract infection (UTI), worsening hydroureteronephrosis or refractory voiding problems.
- **Urodynamics**: neuropathic bladder, PUV, large ureterocele, refractory dysfunction, high-grade hydronephrosis, recurrent UTI. Measures capacity, compliance, overactivity, emptying, sphincter; **repeat serially**.
- Anuric child with no urological history: no further testing.

**ALGORITHM**

Positive history → US + VCUG → abnormal dynamics → urodynamics → drugs + clean intermittent catheterisation (CIC) ± bladder cycling → reassess → still unsafe → augmentation → clear for transplant

## 3 · VALVE BLADDER AND BLADDER MANAGEMENT

- Commonest ESKD pattern: **low capacity, detrusor overactivity, poor compliance** — dangerous because high pressure reaches the kidney even without reflux.
- Valve bladder evolves: overactive small → larger, less contractile → **myogenic failure** (poor emptying).
- **Target**: expected capacity for age at detrusor pressure **< 40 cm $H_2O$**, socially continent.
- Medical first: oxybutynin (titrate), solifenacin/darifenacin, mirabegron, intradetrusor botulinum toxin; α-blockers aid emptying. Better storage can worsen emptying → CIC. Prove emptying with repeated low PVRs.
- Adherence to pre-transplant CIC tests adherence to the transplant regimen.

### Defunctionalised bladder

Does not cycle (anuria or diversion), so response to post-transplant volumes is unpredictable.
**Bladder cycling**: instil saline in rising volumes via catheter with set dwell → add medication → target expected capacity → confirm by urodynamics → clear, or augment. **A renal allograft must never be used to test whether a bladder works.**

**Augmentation**

- Indication: cannot reach > **75%** expected capacity at < 40 cm $H_2O$ with CIC no more often than every **3–4 h**, despite maximal medical therapy. No compelling evidence it increases transplant complications.
- PUV bladder → **postpone** (heads to myogenic failure); neuropathic bladder → **earlier** reconstruction (deteriorates under high graft output).

| Segment | Point |
|---|---|
| Dilated ureter (ureterocystoplasty) | Urothelial, no mucus/stones/metabolic problems — always explore first |
| Ileum, sigmoid | Most used; mucus, stones, infection, malignancy (worse with immunosuppression) → lifelong surveillance |
| Stomach | Haematuria–dysuria syndrome, especially in anuric patients; rarely used |

- **Mitrofanoff/continent catheterisable stoma**: decided by spontaneous emptying; sensate or difficult urethra favours a stoma. Site it **medial to the planned transplant incision**; preserve native ureter for stoma, augment or graft drainage.
- **Timing**: major reconstruction **before** transplant, allowing **6–12 weeks** healing. Exception: transplant before 2 years of age, before continence is expected.
- Peritoneal dialysis (PD): intraperitoneal surgery forces temporary haemodialysis — protect the peritoneum. Avoid concurrent procedures in infants.
- Long-term stoma or transplant ureterostomy only for severe neurodevelopmental impairment or CIC intolerance.
- Circumcision in boys at risk (especially valves) — discuss early; reduces post-transplant UTI.

## 4 · NATIVE NEPHRECTOMY

| Remove (named indication) | Keep (default) |
|---|---|
| Refractory hypertension; nephrotic syndrome with malnutrition/hypercoagulability; massive polyuria; recurrent upper-tract infection; stones; malignancy risk; very large kidneys; massive VUR (relative) | Water excretion (less fluid restriction); erythropoietin and vitamin D; avoids morbidity; graft usually fits |

- On PD: retroperitoneal (bilateral prone) so PD restarts at once. High risk: embolisation.
- Normal ureter: transect high, keep as reserve for salvage ureteroureterostomy. Dilated refluxing/obstructed ureter: remove totally (infection).
- Combined with transplant only for room (small child, huge kidneys) — ideally ipsilateral via transplant incision.

## 5 · TRANSPLANT INTO THE ABNORMAL BLADDER

- Graft retroperitoneal via Gibson incision; child < **15 kg** may need aorta/vena cava anastomosis and transperitoneal approach. Know augment pedicle position.
- **Extravesical non-refluxing ureteroneocystostomy** (preferred): partially fill bladder → 3–4 cm detrusorotomy → mucosal disc → spatulate 4–5 mm → interrupted mucosa-to-mucosa → anchor stitches → close flaps loosely at entry.
- Alternatives: Barry tunnel; **ureteroureterostomy** to native ureter (small bladder, short donor ureter, hostile pelvis; always the salvage option).
- Stent if difficult reimplant or poor ureter: short double-J out at **4–6 weeks**; early removal at 5–7 days may cut infection. Use the largest urethral catheter tolerated.

## 6 · UROLOGICAL COMPLICATIONS

| Complication | Rate | Timing / management |
|---|---|---|
| Ureteric stenosis | **6%** | Most within 100 days; usually distal (ischaemic donor ureter); focal → balloon + stent 4–6 weeks; otherwise open repair with native ureter or bladder flap — do not delay |
| Urine leak | **3%** | Early; drain-fluid creatinine; check catheter → US → mercaptoacetyltriglycine (MAG3) scan or cystogram → small: observe; large: drain/explore; short necrosis → reimplant; long → native ureter, psoas hitch, flap |
| Significant VUR | 2% | See below |
| Stones | 2% | Treat if > 1 cm and growing, obstructing, infected; one-stage modality |

MAG3 localises a leak only if graft function is adequate. Pre-existing obstructive or reflux uropathy raises all complications — lower threshold to intervene.

### Rising creatinine with hydronephrosis

**ALGORITHM**
Indwelling catheter 1–2 weeks → creatinine falls = bladder cause → if not, stent + catheter; signs of rejection → biopsy; failed retrograde stent → nephrostomy + antegrade study

- Graft is unusually sensitive to obstruction; grade of hydronephrosis does not track impairment. Obstruction and rejection can coexist — low threshold for diagnostic stenting.
- Extrinsic causes: **lymphocele**; **post-transplant lymphoproliferative disease (PTLD)** (allograft mass = PTLD until proven otherwise).
- Bladder dysfunction: CIC, medication; augmentation rarely and late.

### UTI and graft VUR

- Early UTI: catheter/stent; late UTI reflects bladder function — check emptying, hydronephrosis, selective VCUG. Distinguish colonisation (universal with CIC) from infection.
- Transplant VUR costs more than native VUR (solitary immunosuppressed kidney). Febrile UTI → cystogram.
- Recurrent febrile UTI + optimised bladder + graft VUR → **open/robotic reimplant** (intravesical, transtrigonal or Politano–Leadbetter; stent). Asymptomatic reflux → observe. Endoscopic injection: ≈ one-third success, obstruction risk.

## 7 · OUTCOMES

- Urological complications are uncommon and rarely lose the graft; thrombosis is a leading cause of early graft loss.
- **≥ 2 UTIs** → lower graft glomerular filtration rate at 1 and 4 years, gap widening with time.
- PUV history → lower obstruction-free survival; link with graft dysfunction inconsistent.

## 8 · MCQ ONE-LINERS

1. Commonest cause of paediatric ESKD: CAKUT (hypoplasia/dysplasia), then obstructive uropathy.
2. Commonest obstructive cause in boys: posterior urethral valves.
3. Routine pre-transplant VCUG: not required.
4. Safe storage pressure target: < 40 cm $H_2O$.
5. Augment if < 75% expected capacity despite CIC every 3–4 h and drugs.
6. Valve bladder end-stage: myogenic failure.
7. PUV bladder: postpone augmentation; neuropathic: earlier.
8. Gastrocystoplasty problem: haematuria–dysuria syndrome.
9. Preferred augment material if available: dilated ureter.
10. Never use the allograft to test a bladder.
11. Reconstruction heals 6–12 weeks before transplant.
12. Catheterisable stoma sited medial to transplant incision.
13. Native kidneys: keep by default.
14. Child < 15 kg: graft to aorta/vena cava.

5. Preferred ureteric anastomosis: extravesical non-refluxing ureteroneocystostomy.
6. Salvage for graft ureteric stenosis: ureteroureterostomy to native ureter.
7. Commonest urological complication: ureteric stenosis (≈ 6%).
8. Most obstructions occur within 100 days.

19. Leak fluid identified by high creatinine.
20. First step in creatinine rise + hydronephrosis: indwelling catheter.
21. Allograft mass: PTLD until proven otherwise.
22. Endoscopic injection for graft VUR: ≈ one-third success.
23. Asymptomatic graft VUR: observe.

# 63 Paediatric Urologic Oncology: Renal and Adrenal

*Reading time ≈ 35 min (slow pace)*

## 1 · WILMS TUMOUR (WT): EPIDEMIOLOGY AND GENETICS

- Nephroblastoma: **6–7%** of childhood cancers, **95%** of kidney cancers under 15 y; embryonal tumour from immature renal remnants. Median age **3.5 y**, >80% before 5 y; younger in bilateral and syndromic disease.
- Mostly somatic and multi-hit (Knudson two-hit does not explain most). **10–15%** germline/early epigenetic (11p15); familial 1–2%.
- **Refer for genetics**: bilateral/multifocal, age <2 y, nephrogenic rests, overgrowth/syndromic features, family history.

| Gene (locus) | Frequency | Note |
|---|---|---|
| IGF2 (11p15) — putative "WT2" locus | 70% | Loss of imprinting/LOH (loss of heterozygosity); Beckwith–Wiedemann syndrome |
| WT1 (11p13) | 20% | First WT gene (WAGR deletions); germline or somatic |
| WTX (Xq11), CTNNB1 (β-catenin) | 20%, 15% | Somatic only |
| TP53 (17p13) | 4% | Majority of **anaplastic** tumours |

| | WAGR | Denys–Drash (DDS) |
|---|---|---|
| Features | WT, aniridia, genitourinary anomalies, intellectual disability | 46,XY disorder of sex development (DSD) + mesangial sclerosis + WT |
| WT1 lesion | Deletion (one allele lost) | Zinc-finger point mutation, **dominant-negative** → worse phenotype |
| Kidney | Renal-failure risk into puberty | Early nephropathy → end-stage renal disease (ESRD) |

- **Ideal type I** (WT1 ± CTNNB1): stromal-predominant, intralobar rests, early age, WAGR/DDS. **Type II** (11p15 epigenetic): blastemal/epithelial, perilobar rests, later age, BWS.
- **Beckwith–Wiedemann syndrome (BWS)**/hemihypertrophy: WT risk 4–10%, 21% bilateral; also hepatoblastoma, adrenocortical tumours. Nephromegaly (≥95th centile) = highest risk.
- Prognostic markers (favourable histology): **1p + 16q LOH** (~5%) adverse at all stages → intensified **regimen M**; 1q gain (25–30%) = next stratifier.

| Syndrome risk | Examples (WT risk) |
|---|---|
| High (>20%) | WAGR, DDS (**50%**); Fanconi D1 (BRCA2); Perlman (DIS3L2); mosaic variegated aneuploidy |
| Moderate (5–10%) | BWS, isolated hemihypertrophy, Frasier (WT1), Simpson–Golabi–Behmel (GPC3) |
| Low | Li–Fraumeni, neurofibromatosis 1, Sotos, trisomy 18, Bloom |

> **KEY POINT** Trap: BWS is the **commonest** predisposing syndrome but only moderate risk; WAGR/DDS are rarer and highest.

### Screening protocol

> **ALGORITHM**
> WT risk >1% → renal ultrasound (US) every **3–4 months** until **7 y** → suspicious lesion → computed tomography (CT)/magnetic resonance imaging (MRI)

- Interval does not lengthen with age (growth rate assumed constant).
  BWS/hemihypertrophy/trisomy 18/SGB: abdominal US + α-fetoprotein (AFP) 3-monthly to 4 y (hepatoblastoma).

- Contralateral kidney after nephrectomy: US 3–4-monthly to 5 y (highest risk: infants <12 months with rests).
- Benefit = lower stage → **nephron-sparing surgery (NSS)**, not proven survival. Benign cysts are common in BWS — avoid needless nephrectomy.

## 2 · PATHOLOGY

- **Favourable histology**: classic **triphasic** — blastema + epithelium (tubules, glomeruloid bodies) + stroma; pseudocapsule; soft, friable → spill risk. 12% multicentric unilateral.
- **Anaplasia** (3 criteria): nuclei ≥**3×** neighbours, hyperchromasia, abnormal mitoses. ~2% under 2 y, ~13% over 5 y. Chemoresistant; **diffuse** worse than focal. Acquired (later age, absent from rests); TP53-driven.
- International Society of Paediatric Oncology (SIOP) post-chemotherapy risk: **low** = completely necrotic; **high** = diffuse anaplasia, **blastemal-predominant**; intermediate = rest. Blastema shrinks but predicts relapse if it survives; differentiated stroma/epithelium barely shrinks yet is cured by excision.

### Nephrogenic rests (NRs)

Present in >1/3 of WT kidneys and **>90%** of bilateral WT; most regress. Nephroblastomatosis = multiple rests; diffuse hyperplastic perilobar nephroblastomatosis gives a thick rind, almost always bilateral; ~half develop WT.

| | **Perilobar** | **Intralobar** |
|---|---|---|
| Site/timing | Lobar periphery; late embryogenesis | Anywhere in lobe, sinus; early |
| Syndrome | BWS (11p15) | WAGR, DDS (WT1) |

## 3 · CLINICAL FEATURES AND WORK-UP

- >85% symptomatic: mass, pain; haematuria ~20% (gross → cystoscopy + retrograde pyelogram at surgery for collecting-system extension); hypertension ~25% (renin). Examine for aniridia, hemihypertrophy, genital anomalies.
- Varicocele persisting supine or hepatomegaly → suspect inferior vena cava (IVC) thrombus.
- **Acquired von Willebrand disease** up to 8% — clotting times may be normal; correct with desmopressin (DDAVP).
- Imaging: US → contrast CT or MRI abdomen–pelvis → **chest CT** (lung = commonest metastatic site; do before MRI anaesthesia to avoid atelectasis). No contralateral exploration if imaging normal. Rupture: ascites beyond cul-de-sac is best predictor. Nodes cannot be judged on imaging.
- Imaging cannot separate WT from renal cell carcinoma (RCC) or clear cell sarcoma (CCSK). Clear cell sarcoma and rhabdoid tumour → add bone scan + brain imaging.
- Emergency surgery only for active bleeding/rupture.

## 4 · STAGING AND COG VS SIOP

| **Children's Oncology Group (COG) stage** | **Definition (surgical–pathological)** |
|---|---|
| I | Confined, completely resected; capsule intact; no rupture; no renal sinus extension |
| II | Extracapsular but completely resected; renal sinus or extrarenal vessel involvement |
| III | Residual abdominal disease: **positive nodes, any spill**, peritoneal implants, positive margin, piecemeal/incomplete removal |
| IV | Haematogenous metastases (lung, liver, bone, brain) |
| V | Bilateral at diagnosis |

| | COG (North America) | SIOP (Europe) |
|---|---|---|
| Unilateral | Upfront nephrectomy | Pre-op vincristine (VCR) + dactinomycin (AMD) ~6 weeks, no biopsy |
| Advantage | Untreated histology and biology (LOH) for risk; very-low-risk cured by surgery alone | Smaller, firmer tumour; fewer complications; spill near zero |
| Cost | More spill/complications | Loses pre-treatment stage/biology; a minority not WT |
| Risk basis | Stage + histology + biology | Post-chemotherapy histology |
| Follow-up | CT-based | US + chest radiograph |

**KEY POINT** Because COG stage is surgical-pathological, the operating surgeon — not the scan — decides chemotherapy intensity and irradiation: one spill converts stage I/II to III.

## 5 · SURGICAL PRINCIPLES

- **Transperitoneal radical nephrectomy**; inspect liver, peritoneum, nodes. Palpate renal vein/IVC before ligation (intra-op US if equivocal). Spare adrenal if uninvolved.
- **Lymph node sampling mandatory** (formal dissection not needed): hilar, paracaval/para-aortic, interaortocaval (skip metastases). COG requires **≥5 nodes**. Omission is the commonest protocol violation (10–20%) → under-staging.
- **Avoid spill** (~10%; right side, large tumours) — causes local abdominal relapse. Local recurrence risks: spill, unfavourable histology, incomplete removal, no node sampling.
- Complications ~11% (haemorrhage, small-bowel obstruction); higher with >10 cm, caval extension, thoracoabdominal incision, wrong diagnosis.

### Chemotherapy

- Backbone VCR + AMD; doxorubicin (DOX, cardiotoxic) for higher stage; cyclophosphamide + etoposide in regimen M and CCSK; carboplatin for anaplastic/relapsed.
- † COG favourable histology: stage I–II → VCR + AMD; stage III–IV → VCR + AMD + DOX with flank/abdominal (and lung) radiotherapy.
- CT-only lung nodules: AREN0533 — complete response at week 6 without 1p/16q LOH → lung irradiation omitted safely.

### Bilateral WT (stage V) and NSS

**ALGORITHM**
Synchronous bilateral (5–7%) → **no upfront nephrectomy, no biopsy** → VCR + AMD + DOX 6 weeks → imaging → NSS if feasible → else more chemo but **surgery by week 12** → non-responder: **bilateral open biopsy**

- Why 12 weeks: in National Wilms Tumor Study (NWTS)-4 prolonged chemo delayed surgery to ~7 months and some tumours became anaplastic.
- Open (not needle) biopsy detects anaplasia better; bilateral because anaplasia is discordant between kidneys in ~4/5.
- AREN0534: 4-year event-free survival (EFS) 82%, overall survival (OS) 95%. Commonest cause of renal failure = bilateral nephrectomy for persistent tumour. Enucleation margins are often positive — acceptable with flank radiotherapy. Transplant after 1–2 y, removing residual renal tissue first.
- Unilateral NSS indications: syndromic/predisposed or solitary kidney, bilateral risk; SIOP UMBRELLA criteria: **<300 mL**, substantial remnant, negative nodes. Use frozen section and a margin of normal kidney.

### IVC thrombus

- ~6–10% (renal vein/IVC/atrium); more than half asymptomatic.

- Below hepatic veins → COG upfront nephrectomy with **en bloc** thrombectomy (piecemeal = stage III → radiotherapy). **At/above hepatic veins → pre-op chemotherapy** to avoid cardiopulmonary bypass. Chemo beyond 6–12 weeks adds nothing. Viable cells in thrombus do not affect survival.

## 6 · RELAPSE AND LATE EFFECTS

| Relapse group | Definition | Salvage |
|---|---|---|
| Standard | Non-anaplastic, prior VCR/AMD only | 70–80% |
| High | Non-anaplastic, after ≥3 drugs | 40–50% |
| Very high | Anaplastic/blastemal | ~10% |

- Late effects track **radiotherapy and anthracycline**: heart failure 4.4% after DOX (rare below **150 mg/m²**); second cancers 1% at 10 y (in-field); gonadal failure (alkylators, pelvic/whole-abdominal RT). Renal failure after unilateral nephrectomy only **0.6%** at 20 y — higher in WT1 syndromes.

## 7 · OTHER RENAL TUMOURS

| Tumour | Key facts | Management |
|---|---|---|
| Congenital mesoblastic nephroma (CMN) | Commonest **neonatal/antenatal** renal tumour (mean 3½ months); ETV6–NTRK3 fusion (cellular type, shared with congenital fibrosarcoma); classic vs cellular | Radical nephrectomy alone; cellular type can recur — adjuvant only if incompletely resected |
| Clear cell sarcoma (CCSK) | 3%; BCOR duplication; **bone**/brain metastasis; late relapse (up to 10 y); never bilateral | DOX-containing chemo; radiotherapy even at stage I |
| Rhabdoid tumour (RTK) | 2%; **most lethal**; median ~16 months; SMARCB1/INI1 loss (22q), INI1 stain negative; brain tumours/metastases | No standard; EZH2 inhibitor (tazemetostat) under trial |
| Cystic nephroma / cystic partially differentiated nephroblastoma (CPDN) | Radiologically identical; CPDN has blastema in septa (first 2 y); DICER1 | Excision cures (100%); frozen section if partial nephrectomy |
| RCC | Commonest renal malignancy of 2nd decade (RCC > WT above 13–14 y); **translocation RCC (TFE3/TFEB, Xp11.2)** commonest; renal medullary carcinoma — sickle trait, lethal | Complete resection; nodes less ominous than adults; chemo/radioresistant |
| Angiomyolipoma | Tuberous sclerosis (50–80%); bleeding risk >3 cm and growing | mTOR inhibitor first-line; embolise for bleeding |

> **KEY POINT** Age points to diagnosis: neonate → CMN; toddler → WT/rhabdoid; adolescent → RCC. Metastatic site sets staging tests: lung (WT), bone (CCSK), brain (rhabdoid/CCSK).

## 8 · NEUROBLASTOMA

- Commonest **extracranial solid tumour** of childhood and commonest malignancy of infancy; neural-crest origin (adrenal medulla, sympathetic chain); median 19 months. 50% adrenal; metastases in ~70% at diagnosis.
- Familial <1% (PHOX2B, ALK). Behaviour ranges from regression to lethal disease.
- Spectrum: neuroblastoma → ganglioneuroblastoma → **ganglioneuroma** (benign; operate only for symptoms). Shimada classification: stroma-poor vs stroma-rich, graded against age.
- Presentation: hard fixed mass, bone pain, periorbital ecchymosis; cord compression; pelvic (organ of Zuckerkandl) → retention. Paraneoplastic: catecholamine excess, VIP (vasoactive intestinal peptide) → watery diarrhoea, **opsoclonus–myoclonus–ataxia**.

## Diagnosis

- Urinary **VMA** (vanillylmandelic acid) and **HVA** (homovanillic acid) raised in **90–95%**; used to monitor response/relapse.
- Marrow: two aspirates + two biopsies. CT/MRI (MRI best for intraspinal extension). Calcification and vessel **encasement** favour neuroblastoma over WT.
- **$^{123}$I-MIBG** (meta-iodobenzylguanidine) scan required for staging (replaces bone scan); non-avid → positron emission tomography (PET). $^{131}$I-MIBG is therapeutic.
- Screening (Japan, Quebec, Germany) increased favourable infant tumours without reducing advanced disease or mortality — classic **overdiagnosis**.

## Staging and risk

| International Neuroblastoma Risk Group (INRG) stage | Definition |
|---|---|
| L1 | Localised, **no** image-defined risk factor (IDRF) |
| L2 | Localised, ≥1 IDRF (mainly vessel/nerve encasement) |
| M | Distant metastases |
| MS | Age **<18 months**; metastases to skin, liver, marrow (<10%) only; no bone (old INSS 4S: <12 months) |

- INRG is imaging-based, pre-treatment; INSS (International Neuroblastoma Staging System, stages 1–4S) depends on resection and nodes, so surgeon-dependent.
- Risk groups (very low/low/intermediate/high) combine stage, age, histology, **MYCN**, 11q, ploidy. **MYCN amplification** (>10 copies, 2p24; ~20%) = high risk at any stage (except maturing histology); M at ≥18 months always high risk; MS with 11q aberration = high risk.
- Other adverse: 1p36 deletion, unbalanced 11q LOH, 17q gain (strongest on multivariate), **diploidy**. Favourable: **hyperdiploidy**, TrkA expression, age <18 months, thoracic site.

> **KEY POINT** Trap: in neuroblastoma diploid is bad and hyperdiploid good. Stage 4S/MS: OS 80–88% with frequent spontaneous regression; deaths occur in infants <2 months from massive hepatomegaly with respiratory compromise or coagulopathy.

## Management by risk

- **Low risk (stage 1, 2, 4S)**: resection alone (stage 1 >90% disease-free); chemo only for symptoms, progression or adverse biology. COG P9641: surgery alone, 5-y OS ~97%. Never remove kidney or vital structures; RT to residual stage 2 does not help.
- **4S**: resection of primary not mandatory (equal survival after biopsy); favourable and asymptomatic → supportive care ± limited chemo.
- **Intermediate**: chemo then resection; pre-op chemo for stage 3/peri-hilar tumours reduces nephrectomy.
- **High risk**: induction (cyclophosphamide, vincristine, doxorubicin alternating with cisplatin–etoposide) → **delayed surgery** after 3–4 courses → myeloablative consolidation with autologous stem-cell rescue (tandem better) → radiotherapy (15–30 Gy) → **13-cis-retinoic acid + anti-GD2 antibody** (with granulocyte-macrophage colony-stimulating factor and interleukin-2) for minimal residual disease.
- Surgical role in stage 4 is debated: no survival benefit from radical resection, but better local control. Delayed surgery gives firmer tumours, fewer ruptures and nephrectomies.
- Surgical complications: diarrhoea (autonomic nerve division at coeliac/superior mesenteric origins), chylous ascites (self-limiting).
- **Cord compression**: chemotherapy first; laminectomy only for progressive deficit (late scoliosis); avoid RT (spinal growth).

**KEY POINT** Surgical value runs inversely with biology: favourable tumours are cured by little or no resection; unfavourable ones are not rescued by radical resection — so vital structures are never traded for margin.

## 9 · ADRENOCORTICAL TUMOURS

- BWS predisposes to adrenocortical tumours (and hepatoblastoma).
- † **Genetics**: germline **TP53 (Li–Fraumeni)** in a large proportion of children → offer TP53 testing and genetic counselling to every child with an adrenocortical tumour.
- † **Presentation by hormone**: most are functional, usually under 5 y. Androgens → **virilisation** (pubarche, clitoromegaly, accelerated growth) — commonest; cortisol → Cushing syndrome (often mixed with virilisation); oestrogen → feminisation; aldosterone → hypertension, hypokalaemia (rare).
- † **Work-up**: serum dehydroepiandrosterone sulphate (DHEA-S), testosterone, androstenedione, 17-hydroxyprogesterone; cortisol (24-h urinary free cortisol, dexamethasone suppression); **urinary steroid profile**; plasma metanephrines to exclude phaeochromocytoma. Contrast CT/MRI abdomen (size, invasion, IVC thrombus) + chest CT.
- † **Adverse features**: tumour weight >200 g or large volume, incomplete resection or rupture/spill, metastases, Cushing features; adenoma versus carcinoma is hard to separate histologically in children.
- † **Surgery**: **open en bloc adrenalectomy** — **avoid capsular rupture** (spill worsens outcome); avoid laparoscopy if carcinoma suspected; regional retroperitoneal node dissection for larger/advanced tumours; Give **perioperative steroid cover** for cortisol-secreting tumours (contralateral adrenal suppressed).
- † **Advanced/residual disease**: **mitotane** (adrenolytic; requires glucocorticoid replacement) with cisplatin, etoposide and doxorubicin.

**KEY POINT** Why the open approach? Cure depends on complete removal of an intact tumour; capsular rupture seeds the retroperitoneum, with poor salvage.

## 10 · LANDMARK TRIALS

| Trial | Design | Result | Take-home |
|---|---|---|---|
| AREN0532/0533 | Combined 1p/16q LOH → regimen M | Stage III–IV EFS ~90% vs 61% | Biology-driven intensification |
| AREN0533 (lung) | Lung nodule response at week 6 | RT omitted, EFS/OS preserved | Response-adapted RT omission |
| AREN0534 | Bilateral WT: 3-drug, surgery by week 12 | EFS 82%, OS 95% (vs 56%/81% NWTS-5) | Fixed clock, maximise NSS |
| COG P9641 | Low-risk neuroblastoma, surgery alone | EFS ~89%, OS ~97% | Chemo only if symptomatic |
| COG anti-GD2 | Anti-GD2 + cytokines + retinoid vs retinoid | Antibody superior | Immunotherapy standard in high risk |
| German NB97 | Extent of resection, stage 4 >18 months | No EFS/OS benefit | Radical resection not essential |

## 11 · MCQ ONE-LINERS

1. WT1 is at 11p13; WT2 putative locus at 11p15 (IGF2).
2. Denys–Drash: dominant-negative WT1 point mutation, 46,XY DSD, mesangial sclerosis.
3. WAGR and Denys–Drash carry ~50% WT risk; BWS 5–10%.
4. Screening: renal US every 3–4 months until age 7 if risk >1%.

5. Anaplasia: nuclei ≥3× normal, hyperchromasia, atypical mitoses; TP53 mutated.
6. Nephrogenic rests in >90% of bilateral WT.
7. Any tumour spill upstages WT to COG stage III.
8. Minimum lymph nodes sampled at Wilms nephrectomy (COG): five.
9. Acquired von Willebrand disease in up to 8% of WT.
0. Bilateral WT: definitive surgery by week 12 of chemotherapy.
1. Thrombus above hepatic veins → preoperative chemotherapy.
2. Doxorubicin cardiotoxicity rare below 150 mg/m² cumulative.
3. Adrenocortical tumour: open adrenalectomy, avoid rupture; test TP53.
4. Commonest neonatal renal tumour: congenital mesoblastic nephroma (ETV6–NTRK3).
15. Clear cell sarcoma metastasises to bone; rhabdoid tumour associated with brain tumours.
16. Rhabdoid tumour: SMARCB1/INI1 loss, most lethal paediatric renal tumour.
17. Commonest paediatric RCC: Xp11.2 translocation (TFE3).
18. Renal medullary carcinoma: sickle cell trait.
19. Neuroblastoma: urinary VMA/HVA raised in 90–95%.
20. 123I-MIBG diagnostic; 131I-MIBG therapeutic.
21. MYCN amplification: >10 copies at 2p24, high risk.
22. Hyperdiploidy favourable in neuroblastoma; diploidy adverse.
23. INRG MS: age <18 months; skin, liver, marrow <10%; no bone.
24. Opsoclonus–myoclonus–ataxia: half have neuroblastoma.
25. Calcification and vessel encasement favour neuroblastoma over Wilms tumour.

# 64 Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia

*Reading time ≈ 26 min (slow pace)*

## 1 · RHABDOMYOSARCOMA (RMS): BIOLOGY

- Aggressive sarcoma of rhabdomyoblasts; **commonest childhood sarcoma** (~3% of paediatric cancers); 20–30% genitourinary (GU). Bimodal: peak in first 2 y, smaller adolescent peak.
- GU sites: bladder/prostate (BP), paratesticular (PT), vulvovaginal (VV). GU-RMS 5-y survival **>80%**.
- Distant metastases in 16% at diagnosis (lung, bone, marrow).
- Most sporadic; ≥5% syndromic (usually embryonal): **Li–Fraumeni** (TP53), Beckwith–Wiedemann, **DICER1**, neurofibromatosis type 1 (pelvic RMS), Costello (bladder tumours). Diagnosis <3 y → look for germline predisposition.
- Surveillance follows the dominant tumour: Li–Fraumeni → abdominal US 3–4-monthly + annual whole-body and brain MRI; DICER1 → chest CT (pleuropulmonary blastoma) + abdominal US; Costello → abdominopelvic US 3-monthly to 8–10 y.

| | **Embryonal (ERMS)** | **Alveolar (ARMS)** |
|---|---|---|
| Site | GU tract (commonest GU histology); **botryoid** variant protrudes from bladder/vagina | Trunk, extremities, perineum |
| Age | Younger (~70% <10 y) | Older children, adolescents |
| Fusion | Usually fusion-negative | ~80% **PAX3/7–FOXO1** [t(2;13) or t(1;13)] |
| Prognosis | Better | Fusion-positive worse; fusion-negative ARMS behaves like ERMS |

> **KEY POINT** Fusion status predicts outcome better than histology, so current trials assign risk by **fusion status**, not microscopy. MYOD1 mutation (fusion-negative) and circulating tumour DNA at diagnosis are emerging adverse markers.

## 2 · PRESENTATION AND WORK-UP

- BP: haematuria, stranguria, retention; botryoid mass at meatus. A 3–5-year-old boy with haematuria + new stranguria needs urgent assessment.
- PT: painless scrotal mass, separate from testis. VV: bloody discharge, prolapsing mass.

**ALGORITHM**

Ultrasound (US) → CT or MRI of primary + nodal basin (BP: pelvic and retroperitoneal to renal vessels; PT: retroperitoneal; VV: inguinal, pelvic, retroperitoneal) → **chest CT** (lung commonest metastatic site) → bone scan + bilateral marrow (omit in fusion-negative ERMS without nodal/lung disease)

CT = computed tomography; MRI = magnetic resonance imaging. Role of positron emission tomography–CT is unsettled.

## 3 · STAGE, GROUP AND RISK

Three layers: **stage** (pretreatment tumour–node–metastasis, TNM — where the tumour began) → **clinical group** (what the surgeon left behind) → **risk group** (adds fusion status and age).

| **COG TNM stage** | **Site** | **Size / nodes** |
|---|---|---|
| 1 | Favourable: **paratesticular, female GU**, orbit, head–neck | Any, M0 |
| 2 | Unfavourable: **bladder/prostate**, extremity, parameningeal, trunk | ≤5 cm, N0/NX |

| COG TNM stage | Site | Size / nodes |
|---|---|---|
| 3 | Unfavourable | ≤5 cm N1, or >5 cm any N |
| 4 | Any | M1 |

COG = Children's Oncology Group. N1 = node ≥1 cm short axis. BP-RMS can never be below stage 2.

| Clinical group (Intergroup Rhabdomyosarcoma Study Group/COG) | Definition |
|---|---|
| I | Localised, completely resected, nodes negative |
| IIa / IIb / IIc | Microscopic residual / involved nodes resected, most distal node negative / microscopic residual or most distal node positive |
| III | Gross residual or **biopsy only** |
| IV | Distant metastases at onset |

| Risk | Criteria |
|---|---|
| Low | Fusion-negative; stage 1 group I–II (III orbit only), or stage 2 group I–II |
| Intermediate | Fusion-negative stage 1–2 group III (non-orbit), stage 3 group I–II; any fusion-positive non-metastatic; stage 4 fusion-negative <10 y |
| High | Stage 4 fusion-positive, or ≥10 y |

**KEY POINT** Organ-preserving **biopsy** deliberately places BP-RMS in group III → **intermediate risk**; PT-RMS (resected by orchidectomy) is almost always low risk. Age <1 y or >10 y is independently adverse. Trap: resected nodes with most distal node positive = IIc, not IIb.

## 4 · TREATMENT PRINCIPLES

- **Chemotherapy is universal**; local control by radiotherapy (RT) and/or surgery; clinical group sets radiotherapy dose. Complete resection (except PT) only if the organ can be preserved.
- Primary exenteration fell from almost all (IRS-I) to **13%** (IRS-IV) with unchanged OS (78–83%). IRS = Intergroup Rhabdomyosarcoma Study; OS = overall survival.

| | Pretreatment re-excision (PRE) | Delayed primary excision (DPE) |
|---|---|---|
| Timing | Before chemotherapy | After chemotherapy starts |
| GU survival | **Improves** | Not shown |
| Group | Can change group | Cannot |
| Indication | Resectable without organ loss; positive/uncertain margins | Only if R0 (negative-margin) resection likely; may lower RT dose |

### Bladder/prostate RMS

- Biopsy: **cystoscopic** (preferred; 9–11.5 French resectoscope, undermine with cutting current, remove with forceps, minimal cautery) → percutaneous → open transvesical (last resort; spill).
- Relieve obstruction: **urethral catheter** first (usually removed within weeks as chemo works) → JJ stent → percutaneous nephrostomy if stent impossible. Avoid suprapubic tube (seeding); vesicostomy/ureterostomy delays chemotherapy.
- **End-of-therapy residual mass**: usually stroma or mature rhabdomyoblasts (misread as tumour) → observe if stable and asymptomatic; resection does not improve survival.
- Viable tumour after chemoradiation without metastasis → partial cystectomy (if adequate margin and capacity) or cystoprostatectomy/exenteration can cure.

- Bladder function: only ~40% "normal" after preservation (detrusor overactivity, small capacity). Follow-up: **urodynamics at 6 months**, renal US annually ≥10 y.

### Paratesticular RMS

- **Radical inguinal orchidectomy**. Prior scrotal approach → excise remaining cord inguinally as PRE; hemiscrotectomy/scrotal RT only for positive cord margin or scrotal skin invasion. Tumour abutting but not invading the tunica vaginalis is **not** a positive margin.
- **Ipsilateral nerve-sparing retroperitoneal lymph node dissection (RPLND)** before chemotherapy if **age ≥10 y** or radiological N1 at any age; ≥7–13 nodes, up to renal vessels. Open, laparoscopic or robotic — none proven superior.

> **KEY POINT** Why age, not CT? Older boys have far more retroperitoneal disease and CT has a high false-negative rate for nodes.

### Female GU, perineal, infants

- Work-up: MRI → examination under anaesthesia → cystoscopy → vaginoscopy + biopsy. Chemotherapy ± brachytherapy controls most; no node dissection; hysterectomy/exenteration only for unresponsive tumours. Cervical RMS (small pedunculated) → polypectomy or loop excision biopsy; needs repeat biopsy after induction even with no visible disease; residual → brachytherapy or trachelectomy.
- Vulvovaginal (two-thirds of female GU-RMS): MRI and vaginoscopy after 3 and/or 6 cycles; residual positive biopsy → brachytherapy. Uterine: younger, worse OS; hysterectomy only if R0, preserving ovaries.
- Perineal/perianal: mostly alveolar, fusion-positive; 5-y OS ~50%.
- Infants: more treatment morbidity and predisposition syndromes; radiotherapy often withheld (bony pelvis, hips) → more local failure; protocols allow RT deviation in fusion-negative children <24 months.

### Late effects

Radiotherapy causes almost tenfold more sequelae. Bladder dysfunction >60% despite preservation; sexual dysfunction, infertility, ovarian failure, vaginal stenosis; second cancers ~6× (alkylators, anthracyclines). Acute: vincristine neuropathy, doxorubicin cardiotoxicity, cyclophosphamide haemorrhagic cystitis.

## 5 · OTHER BLADDER TUMOURS

- **Urothelial**: mostly papillary urothelial neoplasm of low malignant potential (PUNLMP)/low-grade; haematuria; transurethral resection of bladder tumour (TURBT) with paediatric resectoscope, fulgurate base; renal–bladder US surveillance. **Recurrence should prompt search for a syndrome** (e.g. Lynch).
- **Spina bifida/exstrophy**: baseline lifetime bladder cancer risk **2–4%**, aggressive, **not increased by intestinal augmentation**; screening asymptomatic patients is neither effective nor cost-effective (cancers arise within months of normal surveillance). Evaluate symptoms (haematuria) promptly instead.

| Benign lesion | Key point | Treatment |
|---|---|---|
| Inflammatory myofibroblastic tumour | ALK rearrangement; intermediate malignancy | Complete resection; excellent outcome |
| Nephrogenic adenoma | After trauma, infection, surgery, transplant; biopsy needed | TURBT/fulguration + treat cause; recurs |
| Eosinophilic cystitis | Biopsy essential; eosinophilia often absent | Steroids, antihistamines |
| Haemangioma | Severe haematuria; Klippel–Trenaunay, Sturge–Weber | Propranolol, laser; partial cystectomy |

| Benign lesion | Key point | Treatment |
|---|---|---|
| Fibroepithelial polyp | Mimics pelviureteric obstruction (Dietl crisis) or urethral valves | — |

## 6 · TESTICULAR TUMOURS

- ~1/100,000 under 15 y; peak 2–4 y and puberty. Benign far commoner than in adults (38–74%).
- Prepubertal: **teratoma** = commonest tumour (benign); **yolk sac tumour (YST)** = commonest malignancy.
- Painless mass; scrotal Doppler US + markers — lactate dehydrogenase (LDH), α-fetoprotein (**AFP**), β-human chorionic gonadotrophin (β-hCG); hormones if precocious puberty.
- US cannot reliably separate benign from malignant: anechoic/cystic with echogenic foci → teratoma; **"onion-skin" rings → epidermoid cyst**.

| | AFP | β-hCG |
|---|---|---|
| Source | Yolk sac, fetal liver, gut — **YST invariably** | Embryonal carcinoma, mixed tumours; rare prepubertal |
| Half-life | **5 days** | **24 h** |
| Caveat | Physiologically high in infancy; adult level (<10 ng/mL) by **6–9 months** | — |

### Surgery

**ALGORITHM**
Inguinal approach → markers normal + benign-looking US → **testis-sparing partial orchidectomy** with frozen section → benign: close; germ cell malignancy or raised AFP → radical inguinal orchidectomy

- Prepubertal teratoma: partial orchidectomy; **no follow-up** needed. † Postpubertal teratoma is malignant (adult-type) → radical orchidectomy.
- **YST**: radical inguinal orchidectomy; **90%** stage I. CT chest/abdomen/pelvis; follow AFP decline by half-life. Stage I → **surveillance, no RPLND or chemotherapy**. Scrotal violation → stage II (chemo, no hemiscrotectomy). Lymphadenopathy with normal markers → node biopsy to confirm; nodes + raised markers = stage III; stage III–IV → cisplatin, etoposide, bleomycin. Persistent mass → biopsy/RPLND.
- **Stromal tumours**: Leydig (commonest; precocious puberty; Reinke crystals ~40%), Sertoli, juvenile granulosa (benign in infants) → partial orchidectomy if markers normal; radical after puberty (Tanner 5). **Testicular adrenal rests** (TART) in poorly controlled congenital adrenal hyperplasia (high adrenocorticotrophic hormone) → increase steroid dose first; partial orchidectomy if endocrine function threatened.
- Why testis-sparing? US cannot separate stromal from germ cell lesions, yet prepubertal tumours are overwhelmingly benign; systematic reviews find no justification for radical orchidectomy with normal markers. Chemotherapy is rarely needed for stromal tumours.
- Leukaemia/lymphoma: commonest secondary testicular tumours; testicular relapse now ≤2%.
- Microlithiasis: no childhood surveillance; self-examination from puberty only if risk factors.

### Disorders of sex development (DSD)

Risk rises with Y material (SRY/GBY regions) in dysgenetic gonads: complete gonadal dysgenesis ~35%, partial ~24%, Frasier, Denys–Drash, partial androgen insensitivity, Turner with Y = high; complete androgen insensitivity, mixed gonadal dysgenesis = intermediate; no Y = low. Gonadoblastoma is premalignant (analogous to germ cell neoplasia in situ), often bilateral → invasive dysgerminoma (seminoma-like). Gonadectomy is individualised (risk, gender identity, fertility).

**KEY POINT** Trap: **complete** androgen insensitivity is intermediate (not high) risk; partial is high.

## 7 · OTHER PELVIC AND GENITAL TUMOURS

- **Sacrococcygeal teratoma**: ~1/40,000, female; neurogenic bladder 36–51%, worse with Altman II–IV (internal). Annual renal–bladder US until toilet training; urodynamics for higher types.
  **Currarino triad** (anterior sacral mass, sacral defect, anorectal malformation) → neurogenic bladder.
- Paratesticular masses other than RMS (epididymal cyst, adenomatoid tumour, cystadenoma — von Hippel–Lindau): testis-sparing excision, but inguinal approach + frozen section mandatory.
- Ovary: neonatal cysts reflect maternal hormones and regress; operate only if persistent, symptomatic or suspicious, preserving the ovary. Malignancy rare (germ cell commonest; markers AFP, β-hCG, LDH). Sertoli–Leydig tumour → **screen DICER1**.
- **Urethral prolapse** mimics RMS (doughnut swelling, prepubertal Black girls) → topical oestrogen/steroid; no follow-up.

## 8 · MCQ ONE-LINERS

1. Commonest paediatric soft-tissue sarcoma: rhabdomyosarcoma.
2. Commonest GU-RMS histology: embryonal; botryoid variant in bladder/vagina.
3. Alveolar RMS fusion: PAX3/7–FOXO1 (chromosome 13).
4. Risk now assigned by fusion status, not histology.
5. Paratesticular and vulvovaginal are favourable sites; bladder/prostate unfavourable.
6. BP-RMS is at least stage 2.
7. Biopsy-only BP-RMS = group III → intermediate risk.
8. Resected nodes, most distal positive = group IIc.
9. Pretreatment re-excision improves GU-RMS survival; delayed excision does not.
0. First drainage in obstructing BP-RMS: urethral catheter.
1. Stable end-of-therapy bladder mass: observe.
2. Paratesticular RMS RPLND: age ≥10 years or N1.
3. Paratesticular RMS: radical inguinal orchidectomy.
4. Exenteration fell to 13% by IRS-IV without survival loss.
15. Recurrent paediatric urothelial tumour: consider Lynch or other syndrome.
16. Neurogenic bladder cancer risk 2–4%, not raised by augmentation.
17. Commonest prepubertal testicular tumour: teratoma; malignant: yolk sac tumour.
18. AFP half-life 5 days; β-hCG 24 hours.
19. AFP reaches adult levels by 6–9 months of age.
20. Onion-skin appearance on US: epidermoid cyst.
21. Stage I prepubertal YST: surveillance after orchidectomy.
22. Reinke crystals: Leydig cell tumour.
23. Testicular adrenal rests: optimise steroid replacement first.
24. Complete androgen insensitivity: intermediate gonadal tumour risk.
25. Sertoli–Leydig ovarian tumour: test for DICER1.
26. Li–Fraumeni surveillance: annual whole-body MRI plus 3–4-monthly abdominal US.
27. Gonadoblastoma progresses to dysgerminoma.
28. Currarino triad: sacral mass, sacral defect, anorectal malformation.

# ★ Putting it together: common paediatric decisions

*Reading time ≈ 8 min (slow pace) · Integrates the paediatric chapters · use these as the skeleton of long-essay answers*

## A · VESICOURETERAL REFLUX (VUR) AFTER A FEBRILE URINARY TRACT INFECTION (UTI)

**ALGORITHM**

**1.** Renal and bladder ultrasound (US) → voiding cystourethrogram (VCUG) if febrile UTIs are multiple, need admission, or US is abnormal.

**2.** Everyone: treat bladder and bowel dysfunction (BBD), starting with constipation.

**3.** Low risk (grade I–II, no BBD, boy, older) → observation ± continuous antibiotic prophylaxis (CAP). High risk (grade ≥III, girl, BBD, scars, infant) → CAP.

**4.** Breakthrough febrile UTI, new scars, or persistent high grade with poor function → correction: endoscopic injection or ureteric reimplantation (→ *Ch 41, 53*).

## B · DILATED UPPER TRACT: WHICH DIAGNOSIS?

| Condition | Key test | Operate when | Operation |
|---|---|---|---|
| Pelviureteric junction (PUJ) obstruction | Mercaptoacetyltriglycine (MAG3) diuretic renogram | Split function falling or <40%, worsening dilatation, symptoms or infection | Anderson–Hynes dismembered pyeloplasty (open, laparoscopic or robotic) |
| Primary obstructive megaureter | US + MAG3 (VCUG to exclude reflux) | Falling function, infection, progressive dilatation; most resolve | Tapered or folded reimplant; temporary diversion or balloon dilatation in infants |
| Duplex with ureterocele / ectopic ureter | US, VCUG, functional scan | Obstruction, infection, incontinence (ectopic ureter in girls) | Endoscopic incision, upper-pole heminephrectomy, or ureteroureterostomy (→ *Ch 52*) |
| Multicystic dysplastic kidney | US + dimercaptosuccinic acid (DMSA) scan (non-functioning) | Rarely; most involute | Observation; check the contralateral kidney for VUR or PUJ obstruction |

## C · UNDESCENDED TESTIS (UDT)

**ALGORITHM**

**Palpable** → orchidopexy at **6–18 months** (ideally by 12). Imaging is not recommended.

**Non-palpable** → examination under anaesthesia → still impalpable → diagnostic laparoscopy: blind-ending vessels = vanished testis → stop · vessels and vas entering the ring → groin exploration, excise the nubbin · intra-abdominal testis → laparoscopic orchidopexy (primary or Fowler–Stephens).

**Bilateral non-palpable or UDT + hypospadias** → work up for a difference of sexual development (DSD) first (→ *Ch 59, in the Oncology, Genitalia & DSD book*).

## D · ACUTE SCROTUM

| | Testicular torsion | Torsion of an appendage | Epididymitis |
|---|---|---|---|
| Typical | Sudden severe pain, vomiting, high-riding hard testis; neonates and adolescents | Prepubertal; localised upper-pole tenderness, "blue dot" | Gradual onset; urinary symptoms; consider underlying anomaly in young boys |

| | Testicular torsion | Torsion of an appendage | Epididymitis |
|---|---|---|---|
| Action | **Explore without delay** if suspicion is high (Doppler can miss partial torsion); detorse, fix both sides, remove if non-viable | Analgesia, rest | Urinalysis/culture; antibiotics if infected |

## E · NEWBORN WITH ATYPICAL GENITALIA

> **KEY POINT** Bilateral impalpable gonads + atypical genitalia = salt-wasting congenital adrenal hyperplasia (CAH) until proven otherwise. Check electrolytes and 17-hydroxyprogesterone (after 36 h) and karyotype; the salt-wasting crisis usually comes at days 5–7. Sex designation and surgery are not emergencies and need a multidisciplinary team.

## F · WILMS TUMOUR VS NEUROBLASTOMA

| | Wilms tumour | Neuroblastoma |
|---|---|---|
| Age | Median 3.5 years | † Younger (median about 18 months) |
| Imaging | Arises within the kidney († claw sign); may extend into the inferior vena cava (IVC) | Calcification and vessel encasement; † crosses the midline, displaces the kidney |
| Markers / genetics | WT1, 11p15; syndromes (WAGR — Wilms, aniridia, genitourinary anomalies, intellectual disability; Denys–Drash; Beckwith–Wiedemann) | Urinary catecholamine metabolites — vanillylmandelic acid (VMA), homovanillic acid (HVA) — raised in 90–95%; MYCN amplification = high risk |
| Strategy | Children's Oncology Group (COG): upfront nephrectomy · International Society of Paediatric Oncology (SIOP): neoadjuvant chemotherapy then surgery | By International Neuroblastoma Risk Group: observe (some infants) → surgery → multimodal therapy |